

OEIS A290268 at Unbounded Logarithmic Deficit

Regular saddle phases, the Airy fold, two-dimensional cosine laws,
superseded lower bounds, and arithmetic families

ProveIt research report, built from eight research notes
prepared for Vladimir Reshetnikov with OpenAI (batch 72A)

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Abstract

OEIS A290268 counts the nonzero terms $a(N)$ in the N th derivative of x^{x^2} . In the depth coordinates of the source report, each term is an integer $H_{d,k,q}$, and the depth d is the logarithmic deficit. This report collects eight manuscripts on the regime in which that deficit grows with the derivative order, merged into four parts. Part I determines a regular saddle branch $\kappa = \sin t/t - 2 \cos t - 2$, $t_0 \leq t < t_c$, ending at a curvature-zero fold $\kappa_c \approx 0.0281461$. On it, a uniform fixed-offset phase law proves eventual nonvanishing at every algebraic limiting ratio $k/d \rightarrow \kappa < \kappa_c$, and uniformly for $|b| \leq 8$ and $0 \leq k \leq d/36$. Possible phase zeros lie only on explicit transcendental rays. An $O(1/d)$ rate confines possible extra zeros near those rays to bounded strips, and the phase law persists for offsets $|b| \leq \sqrt{d}$. The fixed-logarithmic-degree phase laws of the arithmetic manuscript are the case $\kappa = 0$. Part II proves an additive Airy law in the critical window $|k - \kappa_c d| = O(d^{1/3})$, and hence eventual nonvanishing at the nearest integer to $\kappa_c d$. Part III proves uniform two-dimensional cosine laws, explicit phase curvatures, density one with $O(N^{5/3})$ exceptions on tail patches, and uniform positivity on the triangle $40k + 420q \leq 91d$. It also records the lower bounds these gave: $a(N) \geq (3/8 + c)N^2$, $a(N) \geq 3N^2/8 + CN^{3/2}$, and $\liminf a(N)/N^2 \geq 28176/57247$. All of them are ineffective, and all are superseded by $a(N) = N^2/2 + o(N^2)$, proved in the sibling report *power-tower-exponent-supports*. Part IV proves block congruences modulo primes, prime-unit families with exact valuations at unbounded depth, all-depth central strips at $k = 0$, and an exact example showing that root geometry alone cannot prove the support conjecture. The full support conjecture remains open, no numerical depth threshold is asserted, and nothing here is formally verified.

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1 Introduction

Write

$$\frac{d^N}{dx^N} x^{x^2} = x^{x^2} \sum_{j \in \mathbb{Z}, k \geq 0} c_N(j, k) x^j (\log x)^k,$$

and let $a(N)$ be the number of pairs (j, k) with $c_N(j, k) \neq 0$; this is OEIS A290268. The ProveIt research report *power-tower-derivative-term-counts*, cited here as [1], studies this count in its Part 3. Its Section 3.15 reduces every coefficient with $j \leq -1$ to an integer $H_{d,k,q}$ in *depth coordinates* (Section 2 below). It proves that the reflection holes $q = 2d$, $k + d$ even, vanish, classifies depths one through four completely, and conjectures that there are no other zeros. That conjecture is its complete support classification, label `xxb:conj:support`. Its Questions 4 and 5 (Section 3.24) ask for results uniform in the depth and for arithmetic separation of real zeros.

The manuscripts collected here study the coefficients when the depth d grows with N . In the manuscripts' words, this is the regime of *unbounded logarithmic deficit*: the depth is the deficit $m = k - j$ of the source's Part 3 proper. Every manuscript was written against the source at commit `ca81647a9`.

1.1 How the report is organized

- **Part I**, the regular saddle branch and fixed-offset phases: manuscript 58 (the base of the merge, Sections 2–8); manuscript 64's fixed-degree phase laws as the case $\kappa = 0$ (Sections 9–10); manuscript 45's absolute rate and bounded strips (Section 11); and manuscript 53's phase law through square-root offsets (Section 12).
- **Part II**, the Airy fold at κ_c : manuscript 54.
- **Part III**, the two-dimensional tail and the lower bounds: manuscripts 49, 45 (its second part), 40, 37 and the count of 53, followed by the table of superseded headline bounds (Section 25).
- **Part IV**, arithmetic at unbounded deficit: manuscript 64's congruences, valuations, central strips and root-geometry example.
- Questions (Section 32) and provenance (Section 33).

Manuscript numbers are those of batch 72A (Section 33). A ninth manuscript, 61, is wholly implied by 58 and is not printed; Section 33 says what it contained.

1.2 What the merge printed once

The members overlap heavily, and Vladimir asked for duplicates to be watched for. Each item below is printed once, credited to every manuscript that proved it, and the other copies are replaced by references.

- The coefficient model, derivative bridge and bulk count, restated by all eight manuscripts: Section 2.
- The regular-branch geometry (the functions K, F, D_0 , the gap identity and the limiting curvature), proved in 58 and restated in 49, 53, 54 and 40: Proposition 4.1.
- The exact-rational certificate $\kappa_c > 1/36$, delivered byte for byte by 49, 53 and 58: Lemma 4.2. One copy of the script and its record is shipped.

- The closed-walk trace estimate: in full as Lemma 14.1 (manuscript 54) and in its offset-dependent form as Lemma 12.1 (manuscript 53). Manuscripts 45 and 49 use it with a sketch, kept as citations.
- The integer-triangle (Jarník-type) lemma, proved in 45 and repeated in 40 and 37: Lemma 21.2.
- The single-contour representation, derived in both 40 and 37: equation (22.7).
- The two-saddle cosine law of 49, which 45 restates as its input: Theorem 20.5.
- The fixed-degree phase laws of 64 are the case $\kappa = 0$ of 58’s phase theorem. They are printed as corollaries, with 64’s proofs kept as second proofs (Section 9).

The headline count bounds of 37, 49, 53, 45 and 40 are all implied by the leading asymptotic $a(N) = N^2/2 + o(N^2)$ of manuscript 34. That theorem is printed in the sibling report *power-tower-exponent-supports* [2], not here. The bounds are kept, each with a dated note, because the cosine laws, phase structure, positivity and explicit constants behind them are not consequences of a count asymptotic. Section 25 states the relation exactly.

1.3 Status

This is an AI-assisted research report. It is unrefereed and not formalized. Each manuscript’s statement that its proofs “received independent mathematical review” is the authors’ statement, kept as delivered; the intake checked selected constants and identities and reran the shipped scripts on copies (Section 33). It did not re-derive every proof. No statement here has a Lean or Rocq counterpart. The A290268 Lean development at `Combinatorics/DerivativeExpansions/A290268/Lean` formalizes the coefficient recurrence, the support window, the count of the conjectured support, the values $a(n)$ for $n \leq 53$, and the reduction of the conjectured closed form to a support statement (conditional). It contains none of the theorems of this report. Placement beside the source report confers no formal status. All thresholds in this report are existential unless a number is displayed.

Text added in the merge is marked [Merge note, batch 72A.] or [Added 1 October 2026, batch 72A: ...]. Everything else is the manuscripts’ text, with the label prefixes, symbol renames and cross-references listed below.

1.4 Conventions and notation

All parts use the depth coordinates of [1, Section 3.15]: depth d , logarithmic degree k , tail index q , and the *signed* offset $b = q - 2d$. The following table fixes the remaining letters. It lists, for each, the tempting false reading from the source report or from another part. A letter marked “local” is defined where it is used and means nothing outside that section.

Table 1: Symbols of this report and their false readings.

Symbol	Meaning here	Not to be confused with
d, k, q	depth, power of $\log x$, tail index; $N = 2k + 2d + 1 + q$	the source’s Part 3 proper, where $d = n - 2k - 1$ and the depth is $m = k - j$
b	signed offset $q - 2d$ (Parts I–III, and Part IV except one section)	manuscript 64’s $b = 2d - q $ in its central-strip section; that quantity is written s here (Section 29)

Symbol	Meaning here	Not to be confused with
$s, \sigma, \varepsilon, h$	$s = b $, $\sigma = \pm 1$, parity $\varepsilon = (d - k - 1) \bmod 2$, $h = (d - k - 1 - \varepsilon)/2$	the block residue s of Theorem 27.2; the source's $h_d(\tau)$
κ, δ, η	ratios k/d , b/d , and $q/d = \delta + 2$ (37, 40)	the source's $\delta = q - p$; manuscript 53's small constant, written η_*
r	$2d + q + 1 = M - k$ (Section 2, Parts I, II, IV); in Part III (37, 40, 45) the centered ratio parameter $4 + \delta$	the source's r in $n = 2r$, $n = 2r + 1$
R	$4d + b + 1$, equal to $2d + q + 1$ when $q = 2d + b$; $(4 + \delta)d + 1$	—
M	coefficient index $k + r = N - k$	the limiting mean, written $\mathcal{M}$; the source's moduli M_1, M_2
$\mathcal{M}(\kappa, y)$	limiting saddle mean (4.4)	renamed from M in 58, 49, 53 and 54
m	$\min(2d, q)$	the source's deficit $m = k - j$
ω	dummy variable of the coefficient extraction $[\omega^M]$	the source and the manuscripts write w ; renamed because w is the saddle radius
$w(\kappa), a(\kappa)$	$w = 1/t(\kappa) = \sqrt{y(\kappa)}$, $a = w/2$, equation (2.8); $a_0 = a(0)$	the count $a(N)$; manuscript 64's a is a_0
t, T	branch parameter $t \in [t_0, t_c)$; written T in 37 and 40, where t is the contour variable ($T_0 = t_0$, $T_c = t_c$)	in 45 and 54, local T and T_* are third logarithmic derivatives
K, F, D_0	branch functions (4.7), (4.8)	the source's cutoff constants K_d , falling factorial F_n and Mellin transform $\mathcal{F}_{d,k}$
$D(v) = A(v)B(v)$	Laurent product (4.1); 49's D_R	the source's D_δ ; 40's angular deficit, written Δ_{ang}
ρ	finite-depth saddle radius, $\rho \approx d^2 y$	the source's count index ρ in $\beta(2\rho)$
ζ	outer variable $u = d\zeta$ (49, Part III)	the source's $\Pi_d(\zeta)$; 45's phase-crossing ratios, written κ^*
$\mathcal{P}$	analytic prefactor (49, 45); P (40), P_h (37); $P_d(u) = (2d + u)_{4d+b+1}$	the source's falling product $\mathcal{P}_M$; 45's patch, written $\mathcal{C}_N$
L	action $L(t)$ (37, 40); monic Jacobi polynomials L_j (58); Airy constants $L_\varepsilon(b)$ (54); local $L^2 = w^2 + r^2/4$ (45)	the source's $L = \log x$
U_b, Q_b	unpaired product and shift quotient (49, 53)	the source's $U(n)$; 37's kernel, written $\mathcal{U}(s, y)$
$G_k(c, r)$	centered recurrence (2.2); 64 also writes $Q_k^{(R)}(u - b/2)$ and $C_j(c; R)$	$G_k(c, r) = \mathcal{Q}_k^{(r)}(c/2)$ in the source's Part 3 notation
$\mathcal{J}, J_{d,k}$	integral (4.5); Jacobi matrix	the source's zero budget $J(d, k)$
E	$E(\kappa, y) = 2\mathcal{M} + \kappa - 1$ (58, 54); local elsewhere; $E_{m,r}$ elementary symmetric (64)	the source's E_q ; 53's error, written $\text{Err}_{d,k,b}$
$\beta(N), \beta_b$	bulk count (2.5); $\beta_b = b + 1$ (53)	—
p	path length (trace lemmas); odd prime (Part IV); integer in the exceptional ratios (58, 45)	local in each
z	inner generating variable (49); local in 37 and 40	the source's $z = x^2$

Renamed symbols. No normalization was changed. The renames are: the extraction variable $w \mapsto \omega$ (all manuscripts); the limiting mean $M(\kappa, y) \mapsto \mathcal{M}(\kappa, y)$ (58, 49, 53, 54); 54's Airy scale $c \mapsto c_A$; 45's phase-crossing ratios $\zeta \mapsto \kappa^*$, its patch $\mathcal{P}_N \mapsto \mathcal{C}_N$, its lattice-lemma curvature constants $m, M \mapsto \gamma_-, \gamma_+$ and its lattice determinant $D \mapsto \Delta_3$; 49's chain lengths $M, M_d \mapsto n, n_d$, its harmonic remainder $\mathcal{R}(v, w) \mapsto \mathcal{R}_h(v, w)$ and its rotated exponent $E(Z) \mapsto E_{\text{in}}(Z)$; 53's small constant $\eta \mapsto \eta_*$, its depth threshold $D \mapsto d_*$ and its error $E_{d,k,b} \mapsto \text{Err}_{d,k,b}$; 40's angular

deficit $D(T, \kappa) \mapsto \Delta_{\text{ang}}(T, \kappa)$; 37's kernel $U(s, y) \mapsto \mathcal{U}(s, y)$, its regions $U_{\pm} \mapsto \mathcal{W}_{\pm}$, its local $D = s^2 + y^2 \mapsto \Sigma$, $B(s) \mapsto B_h(s)$, $A(s) \mapsto A_h(s)$, $U(z), B(z), R(z), h(z) \mapsto S_A(z), S_B(z), \varrho(z), h_B(z)$ and $E = e^3 \mapsto E_3$; 64's recurrence variable $a \mapsto c$ in Part IV, its auxiliary polynomial $F(z) \mapsto Y_p(z)$, its example polynomials $F, Q \mapsto F_{\sharp}, Q_{\sharp}$, its evaluation parameter $h \mapsto h'$, and, in its central-strip section, $b = |2d - q| \mapsto s$ with the summation index $s \mapsto \ell$.

Every label carries the prefix `aud:`. Labels from a manuscript carry its own prefix as a sub-prefix: `aud:fr:` (58), `aud:af:` (54), `aud:sc:` (45), `aud:qg:` (49), `aud:go:` (53), `aud:gh:` (37), `aud:dc:` (40, distinct from the source's `xxb:dc:`) and `aud:ap:` (64).

2 The coefficient model and the explicit region

For integers $d \geq 1$ and $k, q \geq 0$, put $r = 2d + q + 1$ and $M = k + r$. The integer model from [1, Section 3.15] is

$$H_{d,k,q} = \frac{M!}{d!} [\omega^M] (2 + \omega)^k (1 + \omega)^{2d} \log^d(1 + \omega). \quad (2.1)$$

Define

$$G_0(c, r) = 1, \quad G_1(c, r) = c, \quad G_{j+1}(c, r) = cG_j(c, r) + j(j + r)G_{j-1}(c, r). \quad (2.2)$$

The source's centered factorization is

$$H_{d,k,q} = [u^d](2d + u)_{\underline{r}} G_k(2u + 2d - q, r). \quad (2.3)$$

These are the centered Meixner–Pollaczek polynomials $Q_j^{(r)}(c/2)$; the classical generating-function normalization is recorded in [3]. The source proves the reflection zeros $q = 2d$, $k + d$ even. Its full conjecture asserts that no other zeros occur.

In the derivative expansion of x^{x^2} , with coefficient $c_N(j, k)$ of $x^j(\log x)^k$ after division by x^{x^2} , the exact bridge is

$$N = 2k + 2d + 1 + q, \quad j = -q - 1, \quad c_N(j, k) = \binom{N}{k} H_{d,k,q}. \quad (2.4)$$

Consequently every nonvanishing result below proves the presence of its corresponding derivative term. The source [1, Section 3.23] proves the positive-bulk count

$$\beta(N) = \begin{cases} (3N^2 + 10N + 8)/8, & N \text{ even}, \\ (3N^2 + 12N + 1)/8, & N \text{ odd}. \end{cases} \quad (2.5)$$

This bulk consists of the nonnegative x -power columns and the positive $j = -1$ boundary. Cells with $q \geq 1$, equivalently $j \leq -2$, therefore add terms outside it. At fixed N , distinct pairs (d, k) give distinct monomials (j, k) .

[Merge note, batch 72A.] The model (2.1), the recurrence (2.2), the factorization (2.3), the bridge (2.4) and the bulk count (2.5) are restated in every member manuscript (37, 40, 45, 49, 53, 54, 64); they are printed once, here. In the source they are Section 3.15 (the factorization is its Proposition `xxb:dc:prop:factor`), Theorem `xxb:dc:thm:holes` (the reflection holes) and Section 3.23 (the bulk, first proved as Theorem `xxb:thm:main-bounds`). The polynomials $G_k(c, r)$ are the source's $Q_k^{(r)}(c/2)$.

Fix an integer b and set

$$q = 2d + b, \quad R = 4d + b + 1, \quad m = \min(2d, q), \quad s = |b|, \quad P_d(u) = (2d + u)_{\underline{4d+b+1}},$$

$$\sigma = \begin{cases} 1, & b \leq 0, \\ -1, & b > 0, \end{cases} \quad \varepsilon = (d - k - 1) \bmod 2, \quad h = \frac{d - k - 1 - \varepsilon}{2}.$$

All statements concern sufficiently large d , so that $q \geq 0$.

The explicit region uses

$$K(t) = \frac{\sin t}{t} - 2 \cos t - 2, \quad F(t) = (2t^2 - 1) \sin t + t \cos t. \quad (2.6)$$

Let t_0 be the nonzero solution of $\tan(t_0/2) = 2t_0$ in $(0, \pi)$, and let t_c be the unique zero of F in (t_0, π) . We prove below that K increases strictly from zero to

$$\kappa_c = K(t_c), \quad t_0 \leq t < t_c.$$

Thus each $0 \leq \kappa < \kappa_c$ has a unique inverse $t = t(\kappa)$ on this branch. Put

$$y(\kappa) = t(\kappa)^{-2}, \quad \psi(\kappa) = \frac{\pi - t(\kappa)}{2}. \quad (2.7)$$

We also write

$$w(\kappa) = \frac{1}{t(\kappa)} = \sqrt{y(\kappa)}, \quad a(\kappa) = \frac{w(\kappa)}{2}, \quad a_0 = a(0) = \frac{1}{2t_0}. \quad (2.8)$$

Manuscripts 45 and 49 use w , and 58 and 53 use a . The constant a_0 is manuscript 64's a (Section 9).

For orientation only, $\kappa_c \approx 0.0281461097171$.

Theorem 2.1 (An explicit finite-ratio region). *The following conclusions hold.*

1. *If $k/d \rightarrow \kappa$ for an algebraic real number $0 \leq \kappa < \kappa_c$, then for every fixed integer b , $H_{d,k,2d+b}$ is nonzero for all sufficiently large d , except for the exact reflection holes $b = 0$, $k + d$ even.*
2. *If $I \subset [0, \kappa_c)$ is compact, then for all sufficiently large d , the same nonvanishing holds simultaneously for every integer k with $k/d \in I$ and every integer $|b| \leq 8$. In particular, this holds throughout $0 \leq k \leq d/36$.*
3. *For any other fixed b , a uniform parity-dependent phase law holds on every such compact interval. Its possible vanishing phases occur only at the explicit ratios*

$$\kappa = K((1 - p/s)\pi), \quad s = |b| > 0,$$

whose arguments lie in $[t_0, t_c)$, with p odd for $\varepsilon = 0$ and p even for $\varepsilon = 1$. These ratios are transcendental. The phase law alone makes no claim of exact cancellation on those rays.

No explicit depth threshold is asserted. The offset b is fixed, and the curvature-zero endpoint κ_c is excluded. The proof gives the full phase and its positive amplitude, not merely the stated nonvanishing corollaries.

Part I

The regular saddle branch and fixed-offset phases

Sections 2–8 are manuscript 58, the base of this report. The three sections after them refine its phase law along fixed-offset lines. Section 9 gives the case $\kappa = 0$ with manuscript 64's proofs. Section 11 gives manuscript 45's absolute $O(1/d)$ rate, and Section 12 manuscript 53's extension to offsets $|b| \leq \sqrt{d}$.

3 The limiting Jacobi roots

For each fixed d , the monic polynomials

$$L_j(x) = i^{-j} G_j(ix, R)$$

obey

$$L_{j+1}(x) = xL_j(x) - j(j+R)L_{j-1}(x).$$

Thus L_k is the characteristic polynomial of the $k \times k$ real symmetric Jacobi matrix $J_{d,k}$ with zero diagonal and off-diagonal entries $\sqrt{j(j+R)}$, $1 \leq j < k$. Its roots λ are real, simple, and symmetric. For every positive root put $\alpha = \lambda^2/4$. Then

$$G_k(2u, R) = (2u)^k \prod_{\lambda > 0} (1 + \alpha/u^2). \quad (3.1)$$

Empty products are one, and empty root maxima are zero. The elementary matrix norm bound gives, for $k \geq 2$,

$$\max \alpha \leq (k-1)(k-1+R). \quad (3.2)$$

Lemma 3.1 (A finite-mass arcsine mixture). *If $k/d \rightarrow \kappa \geq 0$, then the finite measures*

$$\nu_{d,k} = \frac{1}{d} \sum_{L_k(\lambda)=0} \delta_{\lambda/d}$$

converge weakly to the measure characterized by

$$\int f d\nu_\kappa = \frac{1}{\pi} \int_0^\kappa \int_0^\pi f(2\sqrt{x(x+4)} \cos \theta) d\theta dx. \quad (3.3)$$

The measure has total mass κ . For each continuous f on a common compact spectral interval, convergence to the right-hand side with $\kappa = k/d$ is uniform when k/d ranges over a compact interval.

Proof. The norm bound gives a common compact spectral interval whenever k/d is bounded. Odd trace moments vanish. For a fixed even power $2p$, expand the diagonal entries of $(J_{d,k}/d)^{2p}$ as sums over nearest-neighbor returning paths. Away from the $O(p)$ boundary rows, there are $\binom{2p}{p}$ such paths. At a row with $j/d \rightarrow x$, every path product tends to $[x(x+4)]^p$. The convergence is uniform on compact x intervals, because $\sqrt{x(x+4)}$ is continuous there. The boundary rows contribute $o(1)$ after dividing the trace by d . Riemann sums therefore give

$$\frac{1}{d} \text{tr}(J_{d,k}/d)^{2p} \longrightarrow \binom{2p}{p} \int_0^\kappa [x(x+4)]^p dx.$$

These are exactly the moments of (3.3); the zeroth moment is $k/d \rightarrow \kappa$. Polynomial approximation on the common compact support proves weak convergence. The case $\kappa = 0$ also follows directly from the vanishing total mass. Uniformity follows by the same estimates, or by extracting a convergent subsequence from any hypothetical sequence violating it. $\square$

Lemma 3.1 is an elementary specialization of the classical varying-recurrence zero-distribution theorem of Kuijlaars and Van Assche [4, Theorem 1.4]. The proof above is included so that no general zero-distribution theorem is required for the argument.

4 The exterior saddle and its regular branch

Define

$$D(v) = A(v)B(v), \quad A(v) = \prod_{j=1}^m (1 + v/j^2), \quad B(v) = \prod_{\lambda>0} (1 - \alpha/v). \quad (4.1)$$

For $\rho > \max \alpha$, the value $D(\rho)$ is positive, and

$$\mu(\rho) = \frac{\rho D'(\rho)}{D(\rho)} = \sum_{j=1}^m \frac{\rho}{\rho + j^2} + \sum_{\lambda>0} \frac{\alpha}{\rho - \alpha}, \quad (4.2)$$

$$V(\rho) = \frac{d\mu}{d \log \rho} = V_A - V_B, \quad (4.3)$$

$$V_A = \sum_{j=1}^m \frac{\rho j^2}{(\rho + j^2)^2}, \quad V_B = \sum_{\lambda>0} \frac{\alpha \rho}{(\rho - \alpha)^2}.$$

Write $\rho = d^2 y$. If $k/d \rightarrow \kappa$ and $y > \kappa(\kappa + 4)$, the norm bound separates this radius from all roots. Lemma 3.1 gives

$$\frac{\mu(d^2 y)}{d} \rightarrow \mathcal{M}(\kappa, y) = \sqrt{y} \arctan \frac{2}{\sqrt{y}} + \frac{1}{2} \int_0^\kappa \left(\sqrt{\frac{y}{y - x(x+4)}} - 1 \right) dx. \quad (4.4)$$

The factor $1/2$ counts one root from each symmetric pair; a possible zero root contributes zero. The convergence, including derivatives in y , is uniform on compact subsets of the strict exterior region. Indeed all the rational test functions and their derivatives are uniformly bounded there.

Put

$$\mathcal{J}(\kappa, y) = \int_0^\kappa \frac{dx}{\sqrt{y - x(x+4)}} = \arcsin \frac{\kappa + 2}{\sqrt{y+4}} - \arcsin \frac{2}{\sqrt{y+4}}. \quad (4.5)$$

Since $h/d \rightarrow (1 - \kappa)/2$, the limiting equation $\mu(\rho) = h$ becomes

$$\sqrt{y} \left(\arctan \frac{2}{\sqrt{y}} + \arcsin \frac{\kappa + 2}{\sqrt{y+4}} \right) = 1. \quad (4.6)$$

Equivalently, $2\sqrt{y} \arctan(2/\sqrt{y}) + \sqrt{y} \mathcal{J}(\kappa, y) = 1$.

To describe the relevant branch explicitly, define

$$K(t) = \frac{\sin t}{t} - 2 \cos t - 2, \quad (4.7)$$

$$F(t) = (2t^2 - 1) \sin t + t \cos t, \quad D_0(t) = \cos t + 2t \sin t. \quad (4.8)$$

Let t_0 be the unique nonzero solution of $\tan(t_0/2) = 2t_0$ in $(0, \pi)$. The derivative of $\tan(t/2) - 2t$ increases strictly on this interval, begins negative, and tends to infinity; this proves uniqueness. Evaluation at $3\pi/4$ gives $t_0 > 3\pi/4$. The earlier fixed-degree parameter satisfies $t_0 = 1/(2a_0)$, where $a_0 \arctan(1/a_0) = 1/4$.

[Merge note, batch 72A.] The “earlier fixed-degree parameter” is the constant a of manuscript 64, written a_0 in (2.8). Manuscript 58 does not cite 64. That manuscript’s fixed-degree phase laws are printed as the case $\kappa = 0$ in Sections 9 and 10.

Proposition 4.1 (The positive-curvature branch). *There is a unique $t_c \in (t_0, \pi)$ with $F(t_c) = 0$. Set*

$$\kappa_c = K(t_c).$$

The function K increases strictly from zero to κ_c on $[t_0, t_c]$. For every $0 \leq \kappa < \kappa_c$, let $t = t(\kappa)$ be its inverse on this interval. Then

$$y(\kappa) = \frac{1}{t(\kappa)^2} \quad (4.9)$$

solves (4.6), lies strictly outside the root support, and has limiting curvature

$$yM_y(\kappa, y) = \frac{F(t)}{4tD_0(t)} > 0. \quad (4.10)$$

At κ_c the curvature is zero while the root gap is still positive. No global uniqueness assertion for all exterior saddles is made.

Proof. One has $K(t_0) = 0$, $K'(t) = F(t)/t^2$, and

$$F(t_0) = \frac{t_0(4t_0^2 - 3)}{1 + 4t_0^2} > 0, \quad F(\pi) = -\pi.$$

For $3\pi/4 < t < \pi$,

$$F'(t) = t(3\sin t + 2t\cos t) < 0,$$

because $\sin t \leq -\cos t$ and $2t > 3$. Thus F has exactly one zero in (t_0, π) , proving the asserted monotonicity of K .

On $[t_0, t_c]$, D_0 is positive: $D_0(t_0) = 1$, its derivative $\sin t + 2t\cos t$ is negative, and $F(t_c) = 0$ gives $D_0(t_c) = \sin(t_c)/t_c > 0$. Direct substitution into (4.7) gives the exact gap identity

$$\frac{1}{t^2} - K(t)(K(t) + 4) = \frac{D_0(t)^2}{t^2} > 0. \quad (4.11)$$

Furthermore, $t - \arctan(2t)$ is positive and less than $\pi/2$: it lies in $(0, \pi)$, and its cosine has the positive sign of $D_0(t)$. Therefore

$$\arcsin \frac{K(t) + 2}{\sqrt{t^{-2} + 4}} = t - \arctan(2t),$$

with the principal inverse-sine branch. This proves (4.6).

Finally set $E(\kappa, y) = 2\mathcal{M}(\kappa, y) + \kappa - 1$. From (4.6),

$$E_\kappa(\kappa, y) = \frac{\sqrt{y}}{\sqrt{y - \kappa(\kappa + 4)}}.$$

Differentiate $E(K(t), t^{-2}) = 0$ and use (4.11). The result is

$$yM_y = \frac{K'(t)}{4\sqrt{y - \kappa(\kappa + 4)}} = \frac{F(t)}{4tD_0(t)}.$$

It is positive before t_c and zero there, whereas (4.11) remains strictly positive at t_c . □

[Merge note, batch 72A.] Shared statement. Proposition 4.1, the gap identity (4.11) and the curvature (4.10) are restated, with their elementary proofs, in manuscripts 49 (its Section 2), 53 (Section 2), 54 (Section 3) and 40 (Section 3, in the variable T). Manuscripts 34 and 37 use the point $T = 14/5$ of the same branch. These facts are printed once, here, and the members' sections refer to them.

For orientation only, the implicit constants are approximately

$$t_0 \approx 2.78649815065118, \quad t_c \approx 2.96463500776812, \quad \kappa_c \approx 0.0281461097171248.$$

These decimals are not used to prove the branch statements.

Lemma 4.2 (An explicit rational subinterval). *One has $\kappa_c > 1/36$.*

Proof. At $x = 74/25$, let

$$S_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{(2j+1)!}, \quad C_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j}}{(2j)!}.$$

The alternating Taylor bounds give $S_5 < \sin x < S_6$ and $C_5 < \cos x < C_6$. Exact rational arithmetic gives

$$\frac{S_5(x)}{x} - 2C_6(x) - 2 > \frac{1}{36}, \quad (2x^2 - 1)S_5(x) + xC_5(x) > 0.$$

The inequalities can also be verified directly by clearing factorials and powers of 25; an accompanying exact-rational script records both positive differences. The elementary bounds $31/10 < \pi < 22/7$ place x in $(3\pi/4, \pi)$. Hence $K(x) > 1/36$ and $F(x) > 0$, so Proposition 4.1 gives $t_0 < x < t_c$ and $\kappa_c > 1/36$. $\square$

[Merge note, batch 72A.] Manuscripts 49 and 53 delivered the same exact-rational certificate, with the same script and output, byte for byte. They use it to place their base intervals $[1/200, 1/100]$ and $[1/100, 1/50]$ inside $(0, \kappa_c)$ (Sections 20.1 and 12). It is shipped once, as `code/58-finite-ratio-verify_rational_band.py` with its record `data/58-finite-ratio-rational_band_certificate.json`.

5 Uniform positive saddle coefficients

Lemma 5.1. *Fix a compact interval $I \subset [0, \kappa_c)$. For all sufficiently large d and integers k with $k/d \in I$, there is a solution ρ of $\mu(\rho) = h$ in a neighborhood of $d^2 y(k/d)$, unique in that neighborhood. Uniformly over these pairs,*

$$\frac{\rho}{d^2} - y(k/d) \longrightarrow 0, \quad \frac{V(\rho)}{d} - \frac{F(t(k/d))}{4t(k/d)D_0(t(k/d))} \longrightarrow 0. \quad (5.1)$$

Let $C_h = [v^h]D(v)$. Then

$$C_h = \frac{D(\rho)\rho^{-h}}{\sqrt{2\pi V(\rho)}}(1 + o(1)) > 0. \quad (5.2)$$

For every fixed integer j ,

$$\frac{[v^{h-j}]D(v)}{\rho^j C_h} \longrightarrow 1 \quad (5.3)$$

uniformly on the same interval of ratios.

Proof. The limiting root gap and limiting curvature have positive uniform lower bounds on I . The norm estimate (3.2) and uniform convergence in (4.4), including its derivative, therefore provide a common neighborhood of the branch on which the finite mean is strictly increasing and its values surround h . The intermediate value theorem gives the saddle and (5.1).

Write $x_\lambda = \alpha/\rho$. For every real θ there is the exact bound

$$\left| \frac{D(\rho e^{i\theta})}{D(\rho)} \right| \leq \exp(-2(V_A - V_B) \sin^2(\theta/2)). \quad (5.4)$$

Indeed, for an A factor the squared normalized modulus is $1 - 4p(1-p)\sin^2(\theta/2)$, with $p = \rho/(\rho + j^2)$. For a B factor it is

$$\left| \frac{1 - x_\lambda e^{-i\theta}}{1 - x_\lambda} \right|^2 = 1 + \frac{4x_\lambda \sin^2(\theta/2)}{(1 - x_\lambda)^2}.$$

Apply $\log(1+z) \leq z$ to both products. This proves (5.4); if an A factor vanishes, the desired bound follows immediately.

The quantities x_λ are uniformly bounded away from one. There are $O(d)$ factors, and every third logarithmic derivative near $\theta = 0$ has a common bound after summation. Consequently

$$\log \frac{D(\rho e^{i\theta})}{D(\rho)} - ih\theta = -\frac{V(\rho)}{2}\theta^2 + O(d|\theta|^3)$$

uniformly. On $|\theta| \leq d^{-2/5}$ the remainder tends to zero; outside this arc, (5.4) and $V(\rho) \geq c_I d$ give exponential decay. Cauchy's coefficient formula, valid for Laurent polynomials on every circle of positive radius, now gives (5.2) by the Gaussian integral. Inserting $e^{ij\theta}$ gives (5.3). The leading expression in (5.2) is positive, proving eventual positivity of C_h . $\square$

6 The fixed shift and the two-arc phase

We use the positive saddle coefficient $C_h = [v^h]D(v)$ obtained above. The finite- d saddle satisfies $\rho/d^2 \rightarrow y(\kappa)$ uniformly for k/d in each compact regular-ratio interval. Write $w = \sqrt{y}$ and $a = w/2$.

Lemma 6.1 (The fixed recurrence shift). *On the upper Gaussian arc $u \approx i\sqrt{\rho}$, uniformly on compact regular-ratio intervals,*

$$\frac{G_k(2u - b, R)}{G_k(2u, R)} \rightarrow \exp\left(\frac{ib}{2}\mathcal{J}(\kappa, y)\right), \quad \mathcal{J}(\kappa, y) = \int_0^\kappa \frac{dt}{\sqrt{y - t(t+4)}}. \quad (6.1)$$

The lower-arc limit is its complex conjugate. The ratio is uniformly bounded on the entire circle $|u| = \sqrt{\rho}$.

Proof. The roots of $G_k(c, R)$ are $i\lambda$, where λ runs over the real Jacobi spectrum. The contour stays a positive multiple of d away from all these roots. Hence, on the upper arc,

$$\begin{aligned} \log \frac{G_k(2u - b, R)}{G_k(2u, R)} &= -b \sum_\lambda \frac{1}{2u - i\lambda} + O(k/d^2) \\ &= ib \sum_\lambda \frac{1}{2\sqrt{\rho} - \lambda} + o(1). \end{aligned}$$

The second line is uniform on a shrinking Gaussian arc, because its angular variation is $o(1)$. The limiting root measure gives

$$\sum_\lambda \frac{1}{2\sqrt{\rho} - \lambda} \rightarrow \int_0^\kappa \frac{dt}{\sqrt{4y - 4t(t+4)}} = \frac{1}{2}\mathcal{J}(\kappa, y).$$

This proves (6.1). Real coefficients give the conjugate lower limit. On the whole circle, the root-product estimate is

$$\left| \frac{G_k(2u - b, R)}{G_k(2u, R)} \right| \leq \exp\left(\frac{k|b|}{2\sqrt{\rho} - \max|\lambda|}\right) = O_b(1).$$

Empty maxima are zero. The estimate also covers $k = 0$, when the ratio is one. $\square$

Pairing the falling-factorial roots gives

$$P_d(u) = (-1)^q (2d)! q! u A(-u^2) \prod_{j=1}^s \left(1 + \frac{\sigma u}{m+j}\right). \quad (6.2)$$

At the upper arc, the finite extra product tends to

$$(1 + \sigma ia)^s = (1 + a^2)^{s/2} \exp(-ib \arctan a).$$

Combining this with Lemma 6.1, the total upper-arc factor is

$$Z = (1 + a^2)^{s/2} e^{-ib\psi}, \quad \psi = \arctan a - \frac{1}{2} \mathcal{J}(\kappa, y). \quad (6.3)$$

The lower-arc factor is $\bar{Z}$. The saddle equation simplifies the angle exactly:

$$2w \arctan(2/w) + w\mathcal{J} = 1 \implies \psi = \frac{\pi}{2} - \frac{1}{2w} = \frac{\pi - t}{2}. \quad (6.4)$$

Theorem 6.2 (Uniform finite-ratio phase). *Fix $b \in \mathbb{Z}$ and a compact $I \subset [0, \kappa_c)$. Put $\kappa_d = k/d$ and define $y = y(\kappa_d)$, $a = \sqrt{y}/2$, and $\psi = \psi(\kappa_d)$. With the signed normalization*

$$\mathcal{N}_{d,k} = 2^k (-1)^{q+h} \sigma^\varepsilon (2d)! q! C_h m^{-\varepsilon}, \quad (6.5)$$

one has, uniformly over all integers k with $\kappa_d \in I$,

$$\frac{H_{d,k,2d+b}}{\mathcal{N}_{d,k}} = \begin{cases} (1 + a^2)^{s/2} \cos(s\psi) + o(1), & \varepsilon = 0, \\ (1 + a^2)^{s/2} \sin(s\psi)/a + o(1), & \varepsilon = 1. \end{cases} \quad (6.6)$$

Here $C_h > 0$ for all sufficiently large d . When $b = 0$, $\varepsilon = 1$, the coefficient is exactly zero.

Proof. The centered root product and (6.2) express the integrand as the real prefactor $2^k (-1)^q (2d)! q!$ times

$$u^{k+1} D(-u^2) \prod_{j=1}^s \left(1 + \frac{\sigma u}{m+j} \right) \frac{G_k(2u-b, R)}{G_k(2u, R)}.$$

The exact angular estimate for D gives two Gaussian arcs near $u = \pm i\sqrt{\rho}$. The other factors are uniformly bounded on the full contour, so all remaining arcs contribute an exponentially small amount relative to the positive saddle scale.

In the Cauchy integral, the residual radius and parity factor is $u^{k+1-d} = u^{-2h-\varepsilon}$. On the two arcs its phases are respectively $(-1)^h i^{-\varepsilon}$ and $(-1)^h (-i)^{-\varepsilon}$. Each arc contributes half the positive saddle coefficient. Their sum therefore gives $\operatorname{Re} Z$ for $\varepsilon = 0$, and $\operatorname{Im} Z$ for $\varepsilon = 1$, multiplied by

$$2^k (-1)^{q+h} (2d)! q! C_h \rho^{-\varepsilon/2}.$$

For $\varepsilon = 0$, this gives the cosine in (6.6). For $\varepsilon = 1$, $\operatorname{Im} Z = -(1 + a^2)^{s/2} \sin(b\psi)$, and division by σ changes this to $(1 + a^2)^{s/2} \sin(s\psi)$ for $b \neq 0$. Also $m/\sqrt{\rho} \rightarrow 1/a$ uniformly, giving the stated factor $1/a$. If $b = 0$, the odd-parity term is zero by the exact reflection identity.

The integral of the absolute value of the Cauchy integrand is bounded by a constant times

$$2^k (2d)! q! D(\rho) \rho^{-h-\varepsilon/2} / \sqrt{d} = O_b(|\mathcal{N}_{d,k}|).$$

Consequently the uniform $o(1)$ errors in the arc factors are relative errors on the required coefficient scale. This remains valid if the amplitude is exponentially small, and completes the uniform argument. $\square$

7 Arithmetic ratios, uniform strips, and exceptional rays

Corollary 7.1 (Algebraic limiting ratios). *If $0 \leq \kappa < \kappa_c$ is algebraic, then $\psi(\kappa)/\pi$ is irrational. For each fixed b , every sequence $k/d \rightarrow \kappa$ therefore has eventual nonvanishing apart from the reflection holes.*

Proof. If ψ/π were rational, then $t = \pi - 2\psi$ would be a nonzero rational multiple of π . Both $\sin t$ and $\cos t$ would be algebraic, and the branch equation gives

$$[\kappa + 2 \cos t + 2]t = \sin t.$$

Since $0 < t < \pi$, its right side is nonzero. The bracket is nonzero and algebraic, so t would be algebraic, contradicting transcendence of π . Thus no applicable sine or cosine phase in (6.6) vanishes. Continuity and the phase theorem prove eventual nonvanishing. $\square$

Corollary 7.2 (The first eight offsets). *For every compact $I \subset [0, \kappa_c)$, the nonvanishing conclusion holds uniformly for $k/d \in I$ and $|b| \leq 8$, apart from reflection holes. In particular, for each $0 < \kappa_1 < \kappa_c$, this is a full eventual band $0 \leq k \leq \kappa_1 d$ in those offsets. Lemma 4.2 permits the explicit choice $\kappa_1 = 1/36$.*

Proof. The effective angle decreases from $\psi(0) = \arctan a_0$ to a positive endpoint, where $a_0 \arctan(1/a_0) = 1/4$. The estimate $a_0 < 3/16$ follows from

$$\arctan(16/3) > \pi/2 - 3/16 > 4/3,$$

using $\pi > 31/10$. Monotonicity of $a \arctan(1/a)$ then proves the claim. Hence

$$0 < \psi \leq \arctan a_0 < a_0 < 3/16 < \pi/16.$$

For $1 \leq s \leq 8$, both $\sin(s\psi)$ and $\cos(s\psi)$ are positive on the branch and have positive minima on each compact ratio interval. The phase theorem gives uniform nonvanishing. There are only finitely many offsets to combine, and $b = 0$ is handled by the exact parity distinction. $\square$

For a general fixed $s = |b| > 0$, the possible phase zeros are exactly

$$t = (1 - p/s)\pi \in [t_0, t_c), \quad \begin{cases} p \text{ odd}, & \varepsilon = 0, \\ p \text{ even}, & \varepsilon = 1. \end{cases} \quad (7.1)$$

This is a finite set. Every corresponding $\kappa = K(t)$ is transcendental by the proof of Corollary 7.1. Away from these ratios, the phase theorem gives uniform nonvanishing on compact sets. At an exceptional ratio, it gives a zero leading phase, not an exact zero of H .

Proposition 7.3 (The first phase crossing). *For $s = 9$ and $\varepsilon = 0$, the point $t_9 = 8\pi/9$ lies strictly between t_0 and t_c . It is the first offset admitting a phase crossing within the regular branch.*

Proof. Put $\delta = \pi/9$ and $x = \delta/2$. Since $\pi < 22/7$,

$$\frac{65}{2}x^2 < \frac{7865}{7938} < 1.$$

Therefore

$$\tan x < \frac{x}{1 - x^2/2} < \frac{1}{32x}.$$

This is equivalent to $K(8\pi/9) > 0$. Also $\tan \delta > \delta$ and $\pi > 3$ imply

$$\begin{aligned} F(8\pi/9) &= \cos \delta [(128\delta^2 - 1) \tan \delta - 8\delta] \\ &> \cos \delta \delta (128\delta^2 - 9) > 0. \end{aligned}$$

Since $t_9 \in (3\pi/4, \pi)$, strict decrease of F gives $t_9 < t_c$. On this interval K increases, so $K(t_9) > 0 = K(t_0)$ gives $t_9 > t_0$. At that point $\psi = \pi/18$, and $\cos(9\psi) = 0$. Corollary 7.2 excludes all smaller nonzero offsets. $\square$

For orientation, the first exceptional ratio is

$$\kappa_9 = K(8\pi/9) \approx 0.00186218357819.$$

This decimal is not used in the certification. No claim of an exact coefficient zero is made at κ_9 or any other phase-crossing ratio.

8 Scope of the finite-ratio theorem (manuscript 58)

The root-distribution limit is a special case of the classical varying-recurrence theory of Kuijlaars and Van Assche [4]; the proof here derives its required instance directly from trace moments. The Meixner–Pollaczek identification and the coefficient bridge come from the cited source. The explicit saddle parametrization, shifted phase, arithmetic-ratio consequence, and stated uniform offset region are the conclusions established here for this coefficient model. No broader priority claim is made.

The branch ends when its curvature vanishes while the contour still lies strictly outside the recurrence-root support. The present Gaussian argument does not cover that endpoint, later branches, growing offsets, or the exact coefficients on exceptional limiting rays. The full A290268 noncancellation conjecture remains open. [Added 1 October 2026, batch 72A: the endpoint κ_c is treated in Part II (manuscript 54), growing offsets $|b| \leq \sqrt{d}$ in Section 12 (manuscript 53), and the rate near exceptional rays in Section 11 (manuscript 45); the exact coefficients on exceptional rays and later branches remain open.]

The proofs received independent mathematical review in both directions, including the spectral and phase components. They have not been formally verified in Lean or externally refereed. The supplied numerical script checks finite Jacobi spectra and limiting transforms for orientation; it is not a certificate of any infinite theorem. All displayed decimal values are likewise illustrative, while the branch definitions and the first-crossing certification are exact.

9 The case $\kappa = 0$ at logarithmic degree zero (manuscript 64)

Manuscript 64 (its Sections 5 and 6) proves eventual nonvanishing at $k = 0$, and at every fixed k , for every fixed offset, with an explicit phase. These are the case $\kappa = 0$ of Theorems 2.1 and 6.2. At $k = 0$ the Laurent product is $D(v) = A(v)$, so $C_h = [v^h]A(v)$ is the elementary symmetric value $E_{m,h}$ below; $a(0) = a_0$ is 64’s a , and $\psi(0) = \arctan a_0$ (proof of Corollary 7.2) is 64’s ϕ . Vladimir asked for duplicates to be printed once, so the statements appear as corollaries. Manuscript 64’s proofs, which use a local limit theorem for sums of independent Bernoulli variables instead of the Jacobi spectral limit, are kept as second proofs. Manuscript 64 was written before 58, and 58 cites neither it nor any sibling.

This section concerns $k = 0$ only. The offset is fixed before the depth tends to infinity; no uniform claim over offsets growing with the depth is made.

Let $a > 0$ be the unique solution of

$$a \arctan(1/a) = \frac{1}{4}, \quad \phi = \arctan a \in (0, \pi/2). \quad (9.1)$$

Numerically, $a \approx 0.1794366883$ and $\phi \approx 0.17755$; these approximations are not used in the proof. This a is the constant $a_0 = a(0)$ of (2.8), and $\phi = \psi(0)$; inside Sections 9 and 10 manuscript 64's letters a and ϕ are kept.

Lemma 9.1 (Fixed-distance coefficient ratios). *Put $E_{m,r} = e_r(1, 1/2^2, \dots, 1/m^2)$. Suppose $h = h(m)$ is integral and $h = m/4 + O(1)$. For every fixed integer $j \geq 0$,*

$$\frac{E_{m,h-j}}{m^{2j} E_{m,h}} \longrightarrow a^{2j} \quad (m \rightarrow \infty).$$

Proof. Write $T_{m,r} = m^{2r} E_{m,r}$, so

$$G_m(z) = \prod_{\ell=1}^m \left(1 + \frac{m^2 z}{\ell^2}\right) = \sum_{r=0}^m T_{m,r} z^r.$$

For all sufficiently large m , $0 < h < m$. There is a unique $z_m > 0$ such that

$$\sum_{\ell=1}^m \frac{z_m}{z_m + (\ell/m)^2} = h.$$

For fixed $z > 0$, the sum divided by m tends to

$$\int_0^1 \frac{z}{z + x^2} dx = \sqrt{z} \arctan(1/\sqrt{z}).$$

This integral is strictly increasing from zero to one. Monotonicity therefore shows that $z_m \rightarrow a^2$.

Let S_m be the sum of independent Bernoulli variables with success probabilities $p_{m,\ell} = z_m/[z_m + (\ell/m)^2]$. Its mean is exactly h , and its probability generating function gives

$$\mathbb{P}(S_m = r) = \frac{T_{m,r} z_m^r}{G_m(z_m)}.$$

Its variance V_m satisfies

$$\frac{V_m}{m} \longrightarrow \int_0^1 \frac{a^2 x^2}{(a^2 + x^2)^2} dx = \frac{1}{8} - \frac{a^2}{2(1+a^2)} > 0.$$

The positivity is also immediate from the integral.

The local limit theorem for sums of independent Bernoulli variables gives, uniformly for integers $0 \leq r \leq m$,

$$\mathbb{P}(S_m = r) = \frac{1}{\sqrt{2\pi V_m}} \exp\left(-\frac{(r-h)^2}{2V_m}\right) + O(V_m^{-1}).$$

For example this is a consequence of Auld–Neammanee's Theorem 1.1 [12]; its hypotheses require only independent Bernoulli variables and variance at least one. Therefore $\mathbb{P}(S_m = h-j)/\mathbb{P}(S_m = h) \rightarrow 1$ for each fixed j . The exact probability identity now yields

$$\frac{E_{m,h-j}}{m^{2j} E_{m,h}} = \frac{T_{m,h-j}}{T_{m,h}} = z_m^j \frac{\mathbb{P}(S_m = h-j)}{\mathbb{P}(S_m = h)} \longrightarrow a^{2j}.$$

□

Corollary 9.2 (The case $k = 0$: a fixed-offset phase limit). *Fix an integer b , put $q = 2d + b$, and consider sufficiently large d so that $q \geq 0$. Set*

$$s = |b|, \quad m = \min(2d, q), \quad h = \lfloor (d-1)/2 \rfloor, \quad \varepsilon = d - 1 - 2h,$$

and let $\sigma = 1$ for $b \leq 0$ and $\sigma = -1$ for $b > 0$. Along either parity of d ,

$$\frac{H_{d,0,2d+b}}{(-1)^{q+h}\sigma^\varepsilon(2d)!q!E_{m,h}m^{-\varepsilon}} \longrightarrow \begin{cases} (1+a^2)^{s/2}\cos(s\phi), & d \text{ odd}, \\ (1+a^2)^{s/2}\sin(s\phi)/a, & d \text{ even}. \end{cases} \quad (9.2)$$

Consequently, for every fixed integer b , there is a depth threshold D_b such that $H_{d,0,2d+b}$ is nonzero for all $d \geq D_b$, apart from $b = 0$ with d even, when it is identically zero.

First proof. Apply Theorem 6.2 with $I = \{0\}$. For $k = 0$ the Laurent product is $D(v) = A(v)$, so $C_h = E_{m,h}$; also $\varepsilon = (d-1) \bmod 2$, $h = \lfloor (d-1)/2 \rfloor$, $a(0) = a$ and $\psi(0) = \phi$, so (6.6) is (9.2). Eventual nonvanishing is the case $\kappa = 0$ of Corollary 7.1. $\square$

Second proof (manuscript 64). Pair opposite factors as in Proposition 29.1. With $B_r = e_r(1/(m+1), \dots, 1/(m+s))$, the quotient on the left of (9.2) equals

$$\sum_{j=0}^{\lfloor (s-\varepsilon)/2 \rfloor} (-1)^j \frac{E_{m,h-j}}{m^{2j}E_{m,h}} m^{\varepsilon+2j} B_{\varepsilon+2j}.$$

An empty sum is zero. Here $h = m/4 + O(1)$, and s is fixed. For each fixed r , $m^r B_r \rightarrow \binom{s}{r}$. Lemma 9.1 therefore gives the limit

$$\sum_j (-1)^j a^{2j} \binom{s}{\varepsilon + 2j}.$$

For $\varepsilon = 0$ this is $\operatorname{Re}(1+ia)^s$, and for $\varepsilon = 1$ it is $\operatorname{Im}(1+ia)^s/a$, proving the displayed phase limits.

It remains to show that the applicable limits do not vanish. Let $\theta = \arctan(1/a) = \pi/2 - \phi$. Equation (9.1) gives $\tan \theta = 4\theta$ and $0 < \theta < \pi/2$. If ϕ/π were rational, then θ would be a nonzero rational multiple of π , hence transcendental. But $\tan \theta$ would be algebraic, since $e^{2i\theta}$ would be a root of unity. This contradicts $\tan \theta = 4\theta$. Thus ϕ/π is irrational.

For every positive integer s , neither $\cos(s\phi)$ nor $\sin(s\phi)$ vanishes. For $s = 0$, the odd-depth limit is one, while the even-depth coefficient is exactly zero by symmetry. Every other limiting constant is nonzero, so convergence establishes eventual nonvanishing. $\square$

Scope and effectiveness. This is a fixed-offset theorem, not a classification of the entire $k = 0$ row. The argument gives a finite threshold for each offset but does not supply its numerical value here. The finite central strips proved in Part IV (Theorem 29.2) give all-depth results at their specified offsets, whereas this theorem gives an eventual result at every fixed offset.

10 The case $\kappa = 0$ at every fixed logarithmic degree (manuscript 64)

The logarithmic degree k can also be held fixed. This section does not allow k or the offset to grow with the depth.

Corollary 10.1 (Fixed logarithmic degree and fixed offset). *Fix integers $k \geq 0$ and b . For all sufficiently large depths d , the coefficient $H_{d,k,2d+b}$ is nonzero except when $b = 0$ and $k + d$ is even; the latter coefficients vanish identically by symmetry.*

More precisely, put $q = 2d + b$, $m = \min(2d, q)$, $h = \lfloor (d - k - 1)/2 \rfloor$, and $\varepsilon = d - k - 1 - 2h$. With a , ϕ , and σ as in Corollary 9.2, along either parity of $d - k$ one has

$$\frac{H_{d,k,2d+b}}{2^k(-1)^{q+h}\sigma^\varepsilon(2d)!q!E_{m,h}m^{-\varepsilon}} \longrightarrow \begin{cases} (1+a^2)^{|b|/2} \cos(|b|\phi), & d-k \text{ odd}, \\ (1+a^2)^{|b|/2} \sin(|b|\phi)/a, & d-k \text{ even}. \end{cases}$$

First proof of the nonvanishing. For fixed k , $k/d \rightarrow 0$ and 0 is algebraic, so Theorem 2.1(1) gives eventual nonvanishing apart from the reflection holes. Theorem 6.2 with $I = [0, \kappa_1]$ gives the phase limit with the Laurent-product amplitude C_h in place of $E_{m,h}$. $\square$

Second proof (manuscript 64). Write

$$P_d(u) = (2d+u)_{4d+b+1}, \quad F_d(\ell) = [u^\ell]P_d(u), \quad R = 4d + b + 1.$$

The established centered factorization is

$$H_{d,k,2d+b} = [u^d]P_d(u)Q_k^{(R)}(u - b/2).$$

Let $C_j(c; R) = Q_j^{(R)}(c/2)$. Its recurrence is

$$C_0 = 1, \quad C_1 = c, \quad C_{j+1} = cC_j + j(j+R)C_{j-1}.$$

Induction shows that the coefficient of c^{k-2v} in C_k has degree at most v in R , and its leading coefficient in c is one. After substituting $c = 2u - b$, it follows that

$$Q_k^{(R)}(u - b/2) = 2^k u^k + \sum_{\delta=1}^k A_{d,\delta} u^{k-\delta}, \quad A_{d,\delta} = O_{k,b}(d^{\lfloor \delta/2 \rfloor}).$$

The proof of Corollary 9.2 applies verbatim to $F_d(d-k)$: its paired coefficient index is now $h = \lfloor (d-k-1)/2 \rfloor = m/4 + O(1)$, so Lemma 9.1 remains applicable and gives precisely the phase limit displayed in this theorem, without its factor 2^k .

For completeness, the same ratio lemma gives, for each fixed integer v ,

$$E_{m,h+v}/E_{m,h} = O(m^{-2v}).$$

For positive v this follows by applying the lemma at $h+v$ and taking reciprocals; negative v is the lemma itself. If the shifted index has parameters $h+v, \varepsilon_\delta$, then $\delta = 2v + \varepsilon_\delta - \varepsilon$. Thus its amplitude ratio is $O(m^{-2v+\varepsilon-\varepsilon_\delta}) = O(m^{-\delta})$, including odd shifts. The finite paired-factor expansion therefore gives this same order for the natural coefficient amplitudes. On any nonhole parity, the leading phase limit for $F_d(d-k)$ is nonzero, so

$$\frac{F_d(d-k+\delta)}{F_d(d-k)} = O_{k,b}(m^{-\delta}).$$

A numerator whose phase limit vanishes causes no difficulty for this upper bound. In particular, when $b = 0$ and $d-k$ is odd, every opposite-parity numerator is exactly zero because P_d is odd. Since $m \sim 2d$,

$$\begin{aligned} H_{d,k,2d+b} &= 2^k F_d(d-k) + \sum_{\delta=1}^k A_{d,\delta} F_d(d-k+\delta) \\ &= 2^k F_d(d-k)(1 + O_{k,b}(d^{-1})). \end{aligned}$$

This proves the displayed phase limit and eventual nonvanishing on the nonhole parities. If $b = 0$ and $d+k$ is even, the symmetry identity already proves exact vanishing; alternatively P_d is odd and $Q_k^{(4d+1)}$ has parity k . $\square$

The depth threshold depends on both fixed parameters. Neither uniformity in growing k or growing $|b|$ nor an explicit numerical threshold is asserted here.

11 Absolute rate and bounded strips (manuscript 45, part 1)

Manuscript 45 sharpens the $o(1)$ of Theorem 6.2 to $O(1/d)$ at each fixed offset. Its main theorem has two parts. The first is proved in this section. The second is a density statement on a two-dimensional patch and is proved in Part III (Section 21).

Theorem 11.1 (Two forms of sparse cancellation). *The following statements hold.*

1. *For each fixed integer b and compact interval $I \subset [0, \kappa_c)$, every sufficiently deep extra zero with $k/d \in I$ lies in one of finitely many strips $|k - \kappa^*d| \leq C_{b,I}$, whose ratios κ^* are explicit phase crossings. There are only $O_{b,I}(1)$ such candidate zeros per depth.*
2. *There is a fixed nondegenerate rectangle $\mathcal{R}$ of positive ratios $(\kappa, \delta) = (k/d, b/d)$ such that, for every derivative order N and every sufficiently large depth d , at most $O_{\mathcal{R}}(d^{2/3})$ coefficients can vanish among the integer k with*

$$(k/d, (N - 1 - 4d - 2k)/d) \in \mathcal{R}.$$

In the associated positive-area patch at order N , the candidate count is $C_{\text{patch}}N^2 + O(N)$ with $C_{\text{patch}} > 0$, and at most $O(N^{5/3})$ candidates vanish. Their nonzero proportion therefore tends to one.

These are statements about specified regions. They neither prove global density one for the entire support nor decide every candidate inside a bounded-width strip.

11.1 A fixed-offset phase estimate with an absolute rate

Fix $b \in \mathbb{Z}$. The quantities $q, R, m, s, \sigma, \varepsilon$ and h are those of Section 2. The roots λ , the numbers $\alpha = \lambda^2/4$ and the Laurent product $D(v)$ are those of (3.1) and (4.1).

On each compact $I \subset [0, \kappa_c)$ there is a positive-curvature saddle ρ satisfying $\rho D'(\rho)/D(\rho) = h$, uniformly for $k/d \in I$. The corresponding coefficient $C_h = [v^h]D(v)$ is positive for all sufficiently large d (Lemma 5.1).

We need the stronger rate in the saddle location. The closed-walk trace argument (Lemma 14.1, proved in full in Part II; compare Lemma 12.1) gives, for a suitable fixed W strictly inside the root-free exterior radius,

$$\left| \text{tr}((J_{d,k}/d)^{2p}) - d \binom{2p}{p} \int_0^{k/d} [x(x+4)]^p dx \right| \leq C_{b,I}(p+1)^2(4W)^p. \quad (11.1)$$

Here $J_{d,k}$ is the symmetric Jacobi matrix with entries $\sqrt{j(j+R)}$. A returning path has p squared edge weights; each differs from its base value by $O_{b,I}((p+1)/d)$. Product telescoping, at most 4^p paths per row, and $O(p)$ boundary rows give (11.1). The increasing-function Riemann-sum error is bounded by W^p . This also covers $p \geq k$, when all rows can be treated as boundary rows, and hence remains uniform as $k/d \rightarrow 0$.

Summing the exterior resolvent series and its fixed-order derivatives gives an $O_{b,I}(1)$ error in the mean and its logarithmic derivatives. Therefore, with $\kappa = k/d$,

$$\rho/d^2 = w(\kappa)^2 + O_{b,I}(d^{-1}), \quad V \geq c_I d > 0, \quad (11.2)$$

and every fixed higher logarithmic derivative is $O_{b,I}(d)$.

Lemma 11.2 (A Gaussian multiplier with a second-order rate). *At the exact saddle, let*

$$Z_d(\theta) = D(\rho e^{i\theta}) D(\rho)^{-1} e^{-ih\theta}.$$

If W_d has uniformly bounded derivatives through the needed fixed orders near zero and is uniformly bounded on the full circle, then

$$\frac{\int_{-\pi}^{\pi} Z_d(\theta) W_d(\theta) d\theta}{\int_{-\pi}^{\pi} Z_d(\theta) d\theta} = W_d(0) + O_{b,I}(d^{-1}). \quad (11.3)$$

Proof. The exact angular bound is $|Z_d(\theta)| \leq e^{-c_I d \theta^2}$. Near zero,

$$\log Z_d(\theta) = -V\theta^2/2 - iT\theta^3/6 + O(d\theta^4), \quad T = O(d).$$

On $|\theta| \leq d^{-2/5}$, expand

$$Z_d(\theta) = e^{-V\theta^2/2} (1 - iT\theta^3/6 + O(d\theta^4 + d^2\theta^6))$$

and $W_d(\theta) = W_d(0) + W'_d(0)\theta + O(\theta^2)$. The terms linear in θ and cubic in θ integrate to zero against the even Gaussian. The remaining normalized integrals are bounded by moments of θ^2 , $d\theta^4$, and $d^2\theta^6$, all of order d^{-1} . The other arcs are exponentially small. The same calculation with $W_d = 1$ gives the positive denominator asymptotic, so division proves (11.3). No vanishing phase is divided out. $\square$

Theorem 11.3 (Fixed-offset absolute phase rate). *Let $\Lambda = (1 + a(\kappa)^2)^{s/2}$ and*

$$\mathcal{N}_{d,k,b} = 2^k (-1)^{q+h} \sigma^\varepsilon (2d)! q! C_h \rho^{-\varepsilon/2} \Lambda.$$

Uniformly over $k/d \in I$,

$$\frac{H_{d,k,2d+b}}{\mathcal{N}_{d,k,b}} = \begin{cases} \cos(s\psi(\kappa)) + O_{b,I}(d^{-1}), & \varepsilon = 0, \\ \sin(s\psi(\kappa)) + O_{b,I}(d^{-1}), & \varepsilon = 1. \end{cases} \quad (11.4)$$

The reflection case remains exactly zero.

Proof. On the upper u -arc, the multiplier consists of the finitely many unpaired falling factors, the quotient $G_k(2u - b, R)/G_k(2u, R)$, and $e^{-i\varepsilon\theta/2}$. Root separation gives uniform fixed-order derivative bounds because b is fixed. The exact mean estimate and (11.2) show that its center value, divided by Λ , is $e^{-ib\psi(\kappa)} + O_{b,I}(d^{-1})$. The recurrence-shift logarithmic remainder is $O(b^2/d)$, the resolvent error is $O(d^{-1})$, and each of the finitely many unpaired factors has an $O(d^{-1})$ error. The lower center is the conjugate.

Apply Lemma 11.2 on both arcs. Their parity factors combine into the cosine in even parity and the imaginary part in odd parity. Division by σ changes $-\sin(b\psi)$ to $\sin(s\psi)$. The exact radius factor $\rho^{-\varepsilon/2}$ in the normalization removes the otherwise present factor $1/a$. This proves (11.4). $\square$

11.2 Bounded-width strips at each fixed offset

For $b \neq 0$, the phase zeros are the finitely many ratios

$$\kappa^* = K((1 - p/s)\pi), \quad (1 - p/s)\pi \in [t_0, t_c), \quad \begin{cases} p \text{ odd}, & \varepsilon = 0, \\ p \text{ even}, & \varepsilon = 1. \end{cases} \quad (11.5)$$

They are transcendental. If one were algebraic, then $t = (1 - p/s)\pi$ and $[\kappa^* + 2 \cos t + 2]t = \sin t$ would force the nonzero rational multiple t of π to be algebraic; here $\sin t \neq 0$ and both trigonometric values are algebraic.

Every phase zero is simple, because

$$\psi'(\kappa) = -1/[2K'(t(\kappa))] \neq 0$$

on a compact regular-ratio interval. Near each zero the relevant phase has absolute value bounded below by a positive constant times its distance from that zero; away from the zero set, compactness supplies a positive lower bound. The error in (11.4) therefore implies

$$H_{d,k,2d+b} = 0 \implies |k - \kappa^*d| \leq C_{b,I} \quad (11.6)$$

for some ratio in (11.5), once d is sufficiently large and reflection holes are excluded. If the relevant phase-zero set is empty, there are no extra zeros in the interval for large d . Each strip contains only a bounded number of integers k per depth, proving the first assertion of Theorem 11.1.

At a fixed order N , the admissible fixed- b segment is empty if $N - 1 - b$ is odd. Otherwise

$$k = (N - 1 - b)/2 - 2d,$$

and (11.6) becomes

$$|(N - 1 - b)/2 - (2 + \kappa^*)d| \leq C_{b,I}.$$

Each strip contains $O_{b,I}(1)$ integer depths. Thus only $O(1)$ extra cancellations can occur along the entire admissible fixed- b regular-ratio segment at each sufficiently large order.

Transcendence of the center ratio does not eliminate the remaining candidates: a rational k/d can lie within $O(d^{-1})$ of a transcendental number. The result is a reduction to bounded-width strips, not a proof of full noncancellation.

12 Phases through square-root offsets (manuscript 53)

Manuscript 53 lets the offset grow with the depth, $|b| \leq \sqrt{d}$, keeping the positive saddle amplitude computed with the actual offset. Its count consequence, $a(N) \geq \beta(N) + CN^{3/2}$, is in Part III (Section 24).

We allow the integer offset $b = q - 2d$ to grow. The quantities q , R , m , s , σ , ε and h are those of Section 2, and

$$\beta_b = s + 1.$$

The subscript on β_b distinguishes this offset size from the bulk count $\beta(N)$. We use the fixed interval

$$I = [\kappa_-, \kappa_+] = [1/100, 1/50] \subset (0, \kappa_c). \quad (12.1)$$

The containment is Lemma 4.2: $\kappa_c > 1/36 > 1/50$. Manuscript 53 re-derives it from the same certificate. The branch quantities $y(\kappa)$, $a(\kappa)$ and $\psi(\kappa)$ are those of Part I. So are the limiting mean (4.4), the gap (4.11), the curvature (4.10) and the saddle equation (4.6), with $\arctan a - \mathcal{J}/2 = \psi$ by (6.3) and (6.4). Manuscript 53's Section 2 restates them.

12.1 Trace bounds uniform in the offset

The Jacobi matrix $J_{d,k}$, its roots λ , the numbers $\alpha = \lambda^2/4$ for positive roots, the product $D(v) = A(v)B(v)$ of (4.1), the root bound (3.2) and the product formula (3.1) are those of Part I, with $R = 4d + b + 1$ for the actual offset.

Throughout this section $k/d \in I$ and $|b| \leq \sqrt{d}$. All constants are uniform in these parameters.

Lemma 12.1 (Offset-dependent trace and saddle errors). *Let $\mu_d(v) = vD'(v)/D(v)$. Uniformly near the compact regular branch,*

$$\mu_d(d^2y) = dM(k/d, y) + O_I(\beta_b), \quad (12.2)$$

including every fixed number of derivatives with respect to $\log y$. The actual saddle $\mu_d(\rho) = h$ exists locally, is unique there, and satisfies

$$\rho/d^2 = y(k/d) + O_I(\beta_b/d), \quad V(\rho) := \frac{d\mu_d}{d \log \rho} \geq c_I d > 0. \quad (12.3)$$

For all sufficiently large d , $C_h = [v^h]D(v)$ is positive and

$$C_h = \frac{D(\rho)\rho^{-h}}{\sqrt{2\pi V(\rho)}}(1 + o(1)) \geq c'_I D(\rho)\rho^{-h}/\sqrt{d}, \quad (12.4)$$

uniformly over the stated growing-offset band.

Proof. Choose a fixed W between the largest squared scaled root bound and the smallest exterior y on a neighborhood of the branch. This is possible by (4.11); on the increasing- κ branch both the root bound and the minimum radius have their closest gap at the upper endpoint. The perturbation $b/d = o(1)$ preserves the gap uniformly.

The scaled squared edge weight is

$$w_{d,j} = (j/d)(j/d + 4 + (b+1)/d).$$

A closed nearest-neighbor path of length $2p$ traverses each edge equally often in both directions. Its product therefore contains p squared edge weights. For a path starting at row i , each edge index differs from i by at most p , giving

$$|w_{d,j} - (i/d)(i/d + 4)| \leq C_I(p + \beta_b)/d.$$

Telescoping the product, summing at most $\binom{2p}{p} \leq 4^p$ paths per row, and treating the $O(p)$ boundary rows yield

$$\left| \text{tr}((J_{d,k}/d)^{2p}) - d \binom{2p}{p} \int_0^{k/d} [x(x+4)]^p dx \right| \leq C_I(p+1)(p+\beta_b)(4W)^p. \quad (12.5)$$

The increasing-function Riemann-sum error is at most W^p before the binomial factor. The estimate remains valid when p exceeds the matrix size. Odd moments vanish, and the zeroth trace moment is k .

Expand the exterior resolvent in even powers. Since W/y is uniformly below one, summing (12.5) gives an $O_I(\beta_b)$ error. Every fixed logarithmic derivative adds only a polynomial factor in p , preserving this bound. The central-binomial generating function gives the B contribution in (4.4), with a factor $1/2$ for positive roots. The A contribution is a smooth Riemann sum; its endpoint displacement $m - 2d = \min(0, b)$ costs $O_I(\beta_b)$. This proves (12.2).

The limiting curvature has a positive minimum on I , and the exact target h differs from $d(1 - k/d)/2$ by a bounded amount. The mean-value theorem gives (12.3). All fixed-order logarithmic derivatives are $O_I(d)$, and the exact angular estimate is

$$\left| D(\rho e^{i\theta})/D(\rho) \right| \leq \exp(-2V(\rho) \sin^2(\theta/2)) \leq e^{-c_I d \theta^2}, \quad -\pi \leq \theta \leq \pi. \quad (12.6)$$

For an A factor, put $p = \rho/(\rho + j^2)$; its squared normalized modulus is $1 - 4p(1-p) \sin^2(\theta/2)$. For a B factor, put $x = \alpha/\rho$; its squared normalized modulus is $1 + 4x \sin^2(\theta/2)/(1-x)^2$. The linear logarithmic bounds combine into $V = V_A - V_B$.

At the saddle the local logarithm is $-V\theta^2/2 + O_I(d|\theta|^3)$ after removing its linear term. On $|\theta| \leq d^{-2/5}$ the remainder tends uniformly to zero; (12.6) suppresses the remaining arcs. The Gaussian integral proves (12.4), including its positive uniform lower bound. $\square$

[Merge note, batch 72A.] The moment estimate (12.5) in this proof is the closed-walk argument of Lemma 14.1 (Part II) with the offset dependence tracked. Manuscripts 45 (equation (11.1)) and 49 (proof of Lemma 20.4) use the same estimate with a sketch.

12.2 A phase estimate through square-root offsets

Let $\kappa = k/d$, $a = a(\kappa)$, $\psi = \psi(\kappa)$, and

$$\Lambda = (1 + a^2)^{s/2}, \quad \mathcal{N}_{d,k,b} = 2^k (-1)^{q+h} \sigma^\varepsilon (2d)! q! C_h \rho^{-\varepsilon/2} \Lambda.$$

The saddle ρ and C_h use the actual finite b parameters.

Theorem 12.2 (Uniform growing-offset phase). *There is a constant C_I such that, for all sufficiently large d , uniformly over $k/d \in I$ and integer $|b| \leq \sqrt{d}$,*

$$\frac{H_{d,k,2d+b}}{\mathcal{N}_{d,k,b}} = \begin{cases} \cos(s\psi) + \text{Err}_{d,k,b}, & \varepsilon = 0, \\ \sin(s\psi) + \text{Err}_{d,k,b}, & \varepsilon = 1, \end{cases} \quad |\text{Err}_{d,k,b}| \leq C_I \left(\frac{\beta_b}{\sqrt{d}} + \frac{\beta_b^2}{d} \right). \quad (12.7)$$

The normalization is nonzero. The reflection case $b = 0$, $\varepsilon = 1$ remains exactly zero.

Proof. Put $r_\rho = \sqrt{\rho}$ and

$$Q_b(u) = \frac{G_k(2u - b, R)}{G_k(2u, R)}, \quad U_b(u) = \prod_{j=1}^s \left(1 + \frac{\sigma u}{m + j} \right).$$

All recurrence-root denominators on $|u| = r_\rho$ have magnitude at least a fixed multiple of d . At the upper center $u = ir_\rho$, the root product gives

$$\log Q_b(ir_\rho) = ibS_d + O_I(b^2/d), \quad S_d = \sum_{\lambda} \frac{1}{2r_\rho - \lambda}.$$

The exact identity

$$S_d = \frac{k}{2r_\rho} + \frac{\mu_B(\rho)}{r_\rho}, \quad \mu_B(\rho) = \sum_{\lambda > 0} \frac{\alpha}{\rho - \alpha},$$

and Lemma 12.1 give $S_d = \mathcal{J}(\kappa, y(\kappa))/2 + O_I(\beta_b/d)$. Also (12.3) gives

$$\log U_b(ir_\rho) = s \log(1 + \sigma ia) + O_I(\beta_b^2/d).$$

Indeed each ratio $r_\rho/(m + j)$ differs from a by $O_I((\beta_b + j)/d)$, and the sum of these errors is $O_I(\beta_b^2/d)$. Using (6.4), the combined center multiplier therefore satisfies

$$U_b(ir_\rho)Q_b(ir_\rho)/\Lambda = e^{-ib\psi} + O_I(\beta_b^2/d). \quad (12.8)$$

The constants absorb the bounded exponential of the error, since β_b^2/d is bounded. The lower-center value is the complex conjugate.

On the upper half-circle write $u = ir_\rho e^{i\theta/2}$ and set

$$W_b(\theta) = e^{-i\varepsilon\theta/2} U_b(u) Q_b(u), \quad -\pi \leq \theta \leq \pi.$$

The lower half is treated identically. Each unpaired factor is separated from zero on the full circle because $r_\rho/(m + j)$ has limit $a(\kappa) < 1$. The shifted and unshifted recurrence roots also remain inside a smaller radius. Their logarithmic angular derivatives satisfy

$$\begin{aligned} \frac{d}{d\theta} \log Q_b(u) &= \sum_{\lambda} \left(\frac{2u'}{2u - b - i\lambda} - \frac{2u'}{2u - i\lambda} \right) \\ &= \sum_{\lambda} \frac{2u'b}{(2u - b - i\lambda)(2u - i\lambda)} = O_I(|b|). \end{aligned}$$

The derivative from U_b is $O_I(s)$, and the residual exponential adds at most $1/2$. Thus W_b has a continuous logarithm along the half-circle with derivative bounded by $C_I\beta_b$. Equation (12.8) implies

$$\frac{|W_b(\theta) - W_b(0)|}{\Lambda} \leq C_I(e^{C_I\beta_b|\theta|} - 1). \quad (12.9)$$

Combining this with the exact Gaussian bound (12.6), and substituting $x = \sqrt{d}\theta$, gives

$$\int_{-\pi}^{\pi} e^{-c_I d \theta^2} (e^{C_I\beta_b|\theta|} - 1) d\theta \leq C_I\beta_b/d. \quad (12.10)$$

Here $\beta_b/\sqrt{d}$ is uniformly bounded. After division by the positive lower bound in (12.4), the multiplier average differs from its center by $O_I(\Lambda\beta_b/\sqrt{d})$.

Finally, exact factor pairing gives the original integrand as

$$2^k(-1)^q(2d)!q! u^{k+1}D(-u^2)U_b(u)Q_b(u).$$

The two half-circle center phases of $u^{k+1-d} = u^{-2h-\varepsilon}$ are $(-1)^h i^{-\varepsilon}$ and $(-1)^h (-i)^{-\varepsilon}$. Their sum gives the cosine for $\varepsilon = 0$ and the imaginary part for $\varepsilon = 1$. Division by σ in the latter case changes $-\sin(b\psi)$ to $\sin(s\psi)$. The exact radius factor $\rho^{-\varepsilon/2}$ is already in the normalization. Equations (12.8) and (12.10) prove the claimed error bound. $\square$

Corollary 12.3 (Phase-safe cells). *There exist $\eta_* > 0$ and d_* such that, for all $d \geq d_*$, $k/d \in I$, and $|b| \leq \eta_*\sqrt{d}$, the coefficient is nonzero whenever the applicable sine or cosine in (12.7) has absolute value at least $1/2$.*

Proof. Take $\eta_* \leq 1$ small enough that the bound in (12.7) is at most $1/4$ for all sufficiently large d . This is possible because it is $O_I(\eta_* + \eta_*^2) + o(1)$. A phase of absolute value at least $1/2$ cannot then be cancelled by the error. $\square$

Part II

The Airy fold at the critical ratio

Part II is manuscript 54. It describes the $d^{1/3}$ transition window at the fold κ_c , which Theorem 2.1 excludes. It also contains the complete proof of the closed-walk trace estimate (Lemma 14.1) that Sections 11 and 12 and Part III use.

13 The critical constants and the critical-window theorem

Throughout Part II, the integer offset $b = q - 2d$ is fixed, and $q, R, m, s, \sigma, \varepsilon, h$ are those of Section 2. The functions K, F, D_0 are (4.7) and (4.8), and t_0, t_c, κ_c are those of Section 2 and Proposition 4.1. The exact critical constants are

$$\kappa_c = K(t_c), \quad y_c = t_c^{-2}, \quad T_* = -\frac{F'(t_c)}{8D_0(t_c)} > 0, \quad c_A = (2/T_*)^{1/3}, \quad \gamma = \frac{c_A}{2D_0(t_c)}. \quad (13.1)$$

Existence, uniqueness, and positivity are checked below. Numerically, for orientation only, $\kappa_c \approx 0.0281461097171$.

Theorem 13.1 (Noncancellation at and near the critical degree). *For every fixed integer b , if*

$$k - \kappa_c d = o(d^{1/3}),$$

then $H_{d,k,2d+b}$ is nonzero for all sufficiently large d , except when $b = 0$ and $k + d$ is even. In particular, the conclusion holds when k is any nearest integer to $\kappa_c d$.

The stronger result is an additive Airy expansion throughout every bounded window of

$$\tau = \frac{k - \kappa_c d}{d^{1/3}}.$$

It proves nonvanishing on compact sets avoiding zeros of $\text{Ai}(-\gamma\tau)$. The normalization and precise statement are given in Section 18; no relative-error assertion is made at an Airy zero. No numerical depth threshold is claimed.

14 A quantitative spectral estimate

The Jacobi matrix $J_{d,k}$ has zero diagonal and off-diagonal entries $\sqrt{j(j+R)}$. Its real roots λ are simple and symmetric, and $\alpha = \lambda^2/4$ for positive roots. The Laurent product $D(v) = A(v)B(v)$ of (4.1), the product formula (3.1) and the norm bound (3.2) are those of Part I. Empty products are one and empty maxima are zero.

Lemma 14.1 (A uniform analytic trace error). *Fix compact neighborhoods I of κ_c and Y of y_c small enough that*

$$\max_{\kappa \in I} \kappa(\kappa + 4) < \min_{y \in Y} y.$$

Uniformly for $\kappa = k/d \in I$ and $y \in Y$, if $\mu_d(v) = vD'(v)/D(v)$, then

$$\mu_d(d^2 y) = dM(\kappa, y) + O_b(1), \quad (14.1)$$

where M is the limiting mean (4.4). The same $O_b(1)$ error holds after any fixed number of derivatives with respect to $\log y$.

Proof. Choose W strictly between the two sides of the displayed separation inequality. Let $T = J_{d,k}/d$. For every integer $p \geq 1$, we first prove

$$\left| \text{tr}(T^{2p}) - d \binom{2p}{p} \int_0^\kappa [x(x+4)]^p dx \right| \leq C_b(p+1)^2 (4W)^p. \quad (14.2)$$

The constant is independent of d, k, p for sufficiently large d .

A closed nearest-neighbor path crosses each edge equally often in both directions. Its length- $2p$ product therefore contains p squared edge weights, with repetitions. The squared weight at edge j is

$$w_{d,j} = \frac{j}{d} \left(\frac{j}{d} + 4 + \frac{b+1}{d} \right).$$

For a path starting at row i , every edge index differs from i by at most p , so

$$|w_{d,j} - (i/d)(i/d + 4)| \leq C_b(p+1)/d.$$

All relevant weights are below W . Telescoping the product bounds its difference from $[(i/d)(i/d + 4)]^p$ by $C_b p(p+1)W^{p-1}/d$. There are at most $\binom{2p}{p} \leq 4^p$ returning paths per row. Summing over $O(d)$ rows gives the principal error in (14.2). The $O(p)$ boundary rows with missing paths cost at most $Cp4^p W^p$. This remains valid when p exceeds the matrix size. Finally, the Riemann-sum error for the increasing function $[x(x+4)]^p$ is at most W^p , before multiplication by $\binom{2p}{p}$. Absorb the fixed factor $1/W$ into the constant to obtain (14.2).

Odd trace moments vanish exactly, and the zeroth trace moment is $k = d\kappa$. Expand the even resolvent in its convergent power series. For $z > 4W$, (14.2) gives an $O_b(1)$ error bounded by

$$C_b \sum_{p \geq 1} (p+1)^2 (4W/z)^p.$$

Each fixed derivative adds only a polynomial factor in p , preserving convergence uniformly away from $z = 4W$. The generating identity for central binomial coefficients gives

$$\sum_{\lambda} \frac{(\lambda/d)^2}{4y - (\lambda/d)^2} = d \int_0^{\kappa} \left(\sqrt{\frac{y}{y - x(x+4)}} - 1 \right) dx + O_b(1).$$

The contribution of B to μ_d is half this sum, because it includes one root from each symmetric pair. A zero root contributes zero. The contribution of A is a smooth Riemann sum; its endpoint $m = 2d + \min(0, b)$ differs from $2d$ by a bounded amount. It equals $d\sqrt{y} \arctan(2/\sqrt{y}) + O_b(1)$, with the same bound for every fixed logarithmic derivative. This proves the lemma. $\square$

15 Critical constants and exact discrete centering

We record the critical geometry to make the argument independent of a weak-convergence approximation. Let $E(\kappa, y) = 2\mathcal{M}(\kappa, y) + \kappa - 1$. Direct evaluation of the integral in (4.4) gives

$$E(\kappa, y) = \sqrt{y} \left(\arctan \frac{2}{\sqrt{y}} + \arcsin \frac{\kappa + 2}{\sqrt{y+4}} \right) - 1. \quad (15.1)$$

By Proposition 4.1, its regular zero branch is $\kappa = K(t)$, $y = t^{-2}$ for $t_0 \leq t \leq t_c$. There $t_0 > 3\pi/4$, $K(t_0) = 0$ and $K' = F/t^2$, and F decreases on $(3\pi/4, \pi)$, which gives the unique critical root t_c . On $[t_0, t_c]$ the gap identity (4.11) holds with $D_0 > 0$, and the curvature is (4.10). The curvature vanishes at t_c while the gap stays positive. Thus the neighborhoods required in Lemma 14.1 exist.

[Merge note, batch 72A.] Manuscript 54's Section 3 re-derives these branch facts in full. They are printed once, as Proposition 4.1.

Put

$$\rho = d^2 y_c, \quad \Delta = k - \kappa_c d, \quad \tau = \Delta/d^{1/3}. \quad (15.2)$$

At this fixed radius, write

$$\mu = \mu_d(\rho), \quad V = (v\partial_v)\mu_d(v)|_{v=\rho}, \quad T = (v\partial_v)^2 \mu_d(v)|_{v=\rho}.$$

Lemma 14.1 and $E_{\kappa}(\kappa_c, y_c) = 1/D_0(t_c)$ give

$$\mu - h = \frac{\Delta}{2D_0(t_c)} + O_b(1 + \Delta^2/d). \quad (15.3)$$

The exact parity term $(1 + \varepsilon)/2$ is retained in the bounded error. Moreover,

$$V = O_b(1 + |\Delta|), \quad T = dT_* + O_b(1 + |\Delta|), \quad T_* = -\frac{F'(t_c)}{8D_0(t_c)} > 0. \quad (15.4)$$

To check the last constant, differentiate (4.10) along the branch. At t_c , $K'(t_c) = 0$, so its first-order variation is the same as variation in y at fixed κ . The logarithmic derivative is $-(t/2)d/dt$, giving exactly T_* . All higher logarithmic derivatives of any fixed order are $O_b(d)$ by Lemma 14.1.

For exact finite-depth centering, define

$$a_d = (2/T)^{1/3}, \quad \xi_d = a_d(\mu - h). \quad (15.5)$$

The quantity T is positive for large d . Uniformly for τ in a prescribed bounded interval,

$$a_d = c_A d^{-1/3}(1 + o(1)), \quad \xi_d - \gamma\tau \longrightarrow 0. \quad (15.6)$$

These definitions preserve all rounding, parity, and bounded parameter shifts. They are formed directly from the finite Laurent product; numerical root finding is not part of the theorem. In particular,

$$B(v) = \frac{G_k(2i\sqrt{v}, R)}{(2i\sqrt{v})^k}$$

is a polynomial in $1/v$, so its logarithmic derivatives can be computed from the recurrence itself.

16 Quartic localization and oscillatory tails

Throughout this section, $|\tau| \leq S$ for a fixed constant S . All estimates are uniform over the associated integer pairs (d, k) , with b fixed. The quantitative input proved above gives

$$\mu - h = \frac{k - \kappa_c d}{2D_0(t_c)} + O\left(1 + \frac{(k - \kappa_c d)^2}{d}\right), \quad V = O(d^{1/3}), \quad T/d \longrightarrow T_* > 0. \quad (16.1)$$

The logarithmic derivatives of any fixed order are $O(d)$, and the fixed radius $\rho = d^2 y_c$ stays strictly beyond all positive roots of the Laurent product.

Write

$$p_j = \frac{\rho}{\rho + j^2}, \quad x_\ell = \frac{\alpha_\ell}{\rho}, \quad A_j = 4p_j(1 - p_j), \quad B_\ell = \frac{4x_\ell}{(1 - x_\ell)^2}.$$

For $z = \sin^2(\theta/2)$, the exact modulus identity is

$$\log \left| \frac{D(\rho e^{i\theta})}{D(\rho)} \right| = \frac{1}{2} \sum_j \log(1 - A_j z) + \frac{1}{2} \sum_\ell \log(1 + B_\ell z). \quad (16.2)$$

At a zero of an A factor the inequalities below follow by continuity.

Lemma 16.1 (Quartic decay and bounded variation). *There is a constant $q_0 > 0$ such that*

$$\log \left| \frac{D(\rho e^{i\theta})}{D(\rho)} \right| \leq -2V \sin^2(\theta/2) - Q_d \sin^4(\theta/2), \quad Q_d = 4 \sum_j [p_j(1 - p_j)]^2 \geq q_0 d. \quad (16.3)$$

The normalized modulus is uniformly bounded on the full circle. For every fixed $0 < \delta < \pi$, its integral over $\delta \leq |\theta| \leq \pi$ is exponentially small. On each of $[0, \delta]$ and $[-\delta, 0]$, its total variation is uniformly bounded.

Proof. Apply $\log(1 - u) \leq -u - u^2/2$ for $0 \leq u < 1$ and $\log(1 + v) \leq v$ for $v \geq 0$ in (16.2). Since $\sum A_j - \sum B_\ell = 4V$, this gives (16.3). The lower bound for Q_d uses only $d \leq j \leq 3d/2$: these indices belong to the product for large d , and $p_j(1 - p_j)$ is bounded below by a positive constant because $\rho/d^2 = y_c > 0$.

Completing the square in z gives

$$-2Vz - Q_d z^2 \leq V^2/Q_d = O(d^{-1/3}).$$

This proves the uniform bound. For $|\theta| \geq \delta$, the negative quartic term is a negative multiple of d , whereas the positive part of the quadratic term is only $O(d^{1/3})$. This proves exponential decay outside the fixed central arc.

Finally, the right-hand side of (16.2), as a function of z , has second derivative

$$-\frac{1}{2} \sum_j \frac{A_j^2}{(1 - A_j z)^2} - \frac{1}{2} \sum_\ell \frac{B_\ell^2}{(1 + B_\ell z)^2} < 0.$$

Thus its first derivative changes sign at most once. Since z increases with θ on $[0, \pi]$, the normalized modulus is unimodal on $[0, \delta]$. Its total variation is at most twice its uniformly bounded supremum. Symmetry gives the same conclusion on the negative interval. $\square$

Choose a sufficiently small fixed $\delta > 0$ on which a logarithm is analytic, and write

$$\frac{D(\rho e^{i\theta})}{D(\rho)} e^{-ih\theta} = R_d(\theta) e^{i\Phi_d(\theta)}, \quad \Phi_d(0) = 0.$$

Real coefficients make the real part of the logarithm even and the imaginary part odd. The derivative estimates therefore imply

$$\begin{aligned} \Phi'_d(\theta) &= (\mu - h) - \frac{T}{2} \theta^2 + O(d\theta^4), \\ \Phi''_d(\theta) &= -T\theta + O(d|\theta|^3). \end{aligned} \tag{16.4}$$

Take δ small enough using $T_* > 0$, then take L large enough in terms of S . Equations (16.1) and (16.4) show that, for $Ld^{-1/3} \leq |\theta| \leq \delta$,

$$|\Phi'_d(\theta)| \geq c_1 d \theta^2, \quad |\Phi''_d(\theta)| \leq c_2 d |\theta|. \tag{16.5}$$

All scaled stationary points lie in the excluded central interval once L is sufficiently large.

Lemma 16.2 (Oscillatory tails). *For all such L and sufficiently large d ,*

$$\left| \int_{Ld^{-1/3}}^{\delta} R_d(\theta) e^{i\Phi_d(\theta)} d\theta \right| \leq C d^{-1/3} L^{-2}. \tag{16.6}$$

The same estimate holds on the negative interval. It also holds after multiplying the integrand by a complex function with a uniform C^1 bound on $[-\delta, \delta]$.

Proof. Put $\theta_0 = Ld^{-1/3}$ and integrate by parts using $e^{i\Phi_d} = (i\Phi'_d)^{-1} (e^{i\Phi_d})'$. The boundary term is $O((d\theta_0^2)^{-1})$. The term involving R'_d has the same bound by Lemma 16.1, because the total variation of R_d is $O(1)$. The remaining term is bounded by

$$C \int_{\theta_0}^{\delta} \frac{d \cdot \theta}{d^2 \theta^4} d\theta \leq \frac{C}{d\theta_0^2}.$$

This proves (16.6). For a multiplier W_d , the product $R_d W_d$ has total variation at most

$$\|W_d\|_{\infty} \text{TV}(R_d) + \|R_d\|_{\infty} \int_{-\delta}^{\delta} |W'_d(\theta)| d\theta,$$

which is uniformly bounded. The same calculation applies to that product. $\square$

17 The additive Airy law

Recall the exact discrete quantities

$$a_d = (2/T)^{1/3}, \quad \xi_d = a_d(\mu - h).$$

For large d , $T > 0$, and the quantitative input gives

$$a_d \sim c_A d^{-1/3}, \quad \xi_d - \gamma\tau \longrightarrow 0 \quad (17.1)$$

uniformly for $|\tau| \leq S$.

Theorem 17.1 (Uniform additive Airy asymptotics). *For every fixed $S < \infty$, uniformly over integer pairs with $|\tau| \leq S$,*

$$[v^h]D(v) = D(\rho)\rho^{-h}a_d(\text{Ai}(-\xi_d) + o(1)) \quad (17.2)$$

$$= D(\rho)\rho^{-h}c_A d^{-1/3}(\text{Ai}(-\gamma\tau) + o(1)). \quad (17.3)$$

The error inside the brackets is additive. No positivity of this coefficient is assumed, and no division by the Airy function is made at its zeros.

Proof. Cauchy's formula reduces the assertion to the normalized integral

$$I_d = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{D(\rho e^{i\theta})}{D(\rho)} e^{-ih\theta} d\theta.$$

For fixed L , put $\theta = a_d r$ on $|r| \leq L$. The logarithmic expansion gives

$$\begin{aligned} \log \frac{D(\rho e^{i\theta})}{D(\rho)} - ih\theta &= i(\mu - h)\theta - \frac{V}{2}\theta^2 - \frac{iT}{6}\theta^3 + O(d\theta^4) \\ &= i\xi_d r - ir^3/3 + o(1) \end{aligned}$$

uniformly on this fixed r interval. Here $V a_d^2 = O(d^{-1/3})$, $d a_d^4 = O(d^{-1/3})$, and the cubic coefficient is exact because $T a_d^3 = 2$.

Thus the central part of I_d/a_d differs by $o(1)$ from

$$\frac{1}{2\pi} \int_{-L}^L e^{i\xi_d r - ir^3/3} dr.$$

The rest is uniformly $O(L^{-2}) + o(1)$ by Lemmas 16.1 and 16.2, since a_d and $d^{-1/3}$ are uniformly comparable. The tail of the limiting oscillatory integral is also $O(L^{-2})$ uniformly for bounded ξ_d , by one integration by parts using the derivative $\xi_d - r^2$.

Let d tend to infinity for fixed L , then let L tend to infinity. The standard real integral representation [5, Eq. 9.5.1],

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos(r^3/3 + xr) dr,$$

identifies the limit as $\text{Ai}(-\xi_d)$ and proves (17.2). Equation (17.1) and continuity of the Airy function on bounded intervals give (17.3). $\square$

Lemma 17.2 (A smooth multiplier). *Suppose $W_d(\theta)$ is uniformly bounded on the full circle and has a uniform C^1 bound on a fixed interval $[-\delta, \delta]$. Then, uniformly in the same bounded scaled windows,*

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{D(\rho e^{i\theta})}{D(\rho)} e^{-ih\theta} W_d(\theta) d\theta = a_d(W_d(0) \text{Ai}(-\xi_d) + o(1)). \quad (17.4)$$

Proof. On the central interval $|\theta| \leq a_d L$, one has $W_d(\theta) - W_d(0) = O(a_d L)$, so this replacement contributes $o(a_d)$ for each fixed L . Lemma 16.2 treats the remaining local arcs with the multiplier retained, and its full-circle bound combines with the exponential outer estimate. The limiting procedure from Theorem 17.1 therefore applies unchanged. $\square$

The multiplier statement is needed when passing back to the original coefficient. On the two halves of the u circle, write $u = \pm i\sqrt{\rho} e^{i\theta/2}$. The fixed unpaired falling factors, the recurrence-shift quotient, and the residual factor $e^{-i\varepsilon\theta/2}$ have the required bounds because the root gap is positive and b is fixed. Consequently the two-arc phase can be restored on the positive scale $D(\rho)\rho^{-h}a_d$, even at points where the leading Airy term is zero.

18 The fixed-offset phase and noncancellation

Let

$$a_c = \frac{\sqrt{y_c}}{2}, \quad \psi_c = \frac{\pi - t_c}{2}, \quad L_\varepsilon(b) = \begin{cases} (1 + a_c^2)^{s/2} \cos(s\psi_c), & \varepsilon = 0, \\ (1 + a_c^2)^{s/2} \sin(s\psi_c)/a_c, & \varepsilon = 1. \end{cases} \quad (18.1)$$

The normalization for the original coefficient is the nonzero signed quantity

$$\mathcal{B}_{d,k} = 2^k (-1)^{q+h} \sigma^\varepsilon (2d)! q! D(\rho) \rho^{-h} m^{-\varepsilon} a_d. \quad (18.2)$$

Theorem 18.1 (The critical-window law for the original coefficients). *For every fixed integer b and every fixed $S > 0$, uniformly over integer pairs with $|k - \kappa_c d| \leq Sd^{1/3}$,*

$$\frac{H_{d,k,2d+b}}{\mathcal{B}_{d,k}} = L_\varepsilon(b) \text{Ai}(-\xi_d) + o(1) = L_\varepsilon(b) \text{Ai}(-\gamma\tau) + o(1). \quad (18.3)$$

Both errors are additive. They remain valid at zeros of the displayed Airy factors. If $b = 0$ and $k + d$ is even, the original coefficient is exactly zero.

Proof. By the pairing identity (6.2) and the product formula (3.1), the Cauchy integrand is the real prefactor $2^k (-1)^q (2d)! q!$ times

$$u^{k+1} D(-u^2) \prod_{j=1}^s \left(1 + \frac{\sigma u}{m+j} \right) \frac{G_k(2u-b, R)}{G_k(2u, R)}.$$

Split the u -circle into its two halves and write $u = \pm i\sqrt{\rho} e^{i\theta/2}$. The strict root gap (4.11) bounds the fixed-shift quotient and its first angular derivative uniformly. Indeed each logarithmic factor has denominator of order d , and their number is $O(d)$; the fixed b shift cancels the potentially larger unshifted derivative. The finite extra product and $e^{-i\varepsilon\theta/2}$ also have uniform C^1 bounds. The multiplier Airy estimate proved above therefore applies to each half without dividing by $[v^h]D(v)$.

At the upper center $u = i\sqrt{\rho}$, the root product and Lemma 14.1 yield

$$\begin{aligned} \log \frac{G_k(2u-b, R)}{G_k(2u, R)} &= -b \sum_{\lambda} \frac{1}{2u - i\lambda} + O(1/d) \\ &\rightarrow \frac{ib}{2} \mathcal{J}_c, \quad \mathcal{J}_c = \int_0^{\kappa_c} \frac{dx}{\sqrt{y_c - x(x+4)}}. \end{aligned}$$

The resolvent limit also follows directly from the exact identity

$$\sum_{\lambda} \frac{1}{2\sqrt{\rho} - \lambda} = \frac{k}{2\sqrt{\rho}} + \frac{\mu_B(\rho)}{\sqrt{\rho}}, \quad \mu_B(\rho) = \sum_{\lambda > 0} \frac{\alpha}{\rho - \alpha},$$

and the mean estimate in Lemma 14.1. The finite extra product tends to $(1 + \sigma i a_c)^s$. Hence the combined upper-center factor is

$$Z_c = (1 + a_c^2)^{s/2} e^{-ib\psi_c}, \quad \psi_c = \arctan a_c - \mathcal{J}_c/2.$$

The equality with (18.1) follows from the exact saddle equation $2\sqrt{y_c} \arctan(2/\sqrt{y_c}) + \sqrt{y_c} \mathcal{J}_c = 1$. The lower-center factor is $\bar{Z}_c$.

The residual power is $u^{k+1-d} = u^{-2h-\varepsilon}$. At the upper and lower centers its phases are $(-1)^h i^{-\varepsilon}$ and $(-1)^h (-i)^{-\varepsilon}$. Summing the two half-integrals gives $\operatorname{Re} Z_c$ for $\varepsilon = 0$, and $\operatorname{Im} Z_c$ for $\varepsilon = 1$, times their common Airy factor. Division by σ^ε , and $m/\sqrt{\rho} \rightarrow 1/a_c$ in odd parity, give precisely $L_\varepsilon(b)$. The multiplier estimate is additive on the positive scale $D(\rho)\rho^{-h}a_d$; thus it proves (18.3) even when the leading Airy term vanishes. Finally (15.6) and continuity of Ai replace ξ_d by $\gamma\tau$ uniformly on bounded windows. $\square$

Lemma 18.2 (The critical phase cannot cancel). *The number ψ_c/π is irrational. Hence every $L_\varepsilon(b)$ is nonzero except for $b = 0$, $\varepsilon = 1$.*

Proof. If ψ_c/π were rational, then $t_c = \pi - 2\psi_c$ would be a nonzero rational multiple of π . The numbers $\sin t_c$ and $\cos t_c$ would be algebraic. But

$$(2t_c^2 - 1) \sin t_c + t_c \cos t_c = 0$$

would be a nonzero quadratic equation over the algebraic numbers for t_c , since $\sin t_c > 0$. This would make t_c algebraic, contradicting transcendence of π . The stated sine and cosine phases are therefore nonzero, with only the exact $s = 0$ odd-parity exception. $\square$

Proof of Theorem 13.1. If $k - \kappa_c d = o(d^{1/3})$, then $\xi_d \rightarrow 0$ by (15.3)–(15.6). The exact value [5]

$$\operatorname{Ai}(0) = \frac{1}{3^{2/3}\Gamma(2/3)} > 0$$

and Lemma 18.2 make the limiting right side of (18.3) nonzero on each nonhole parity. Since $\mathcal{B}_{d,k} \neq 0$, eventual nonvanishing follows. Any nearest integer satisfies $|k - \kappa_c d| \leq 1/2$, which is covered by the same argument. $\square$

Corollary 18.3 (Bounded windows away from Airy zeros). *Fix b , and let $J \subset \mathbb{R}$ be compact with $\operatorname{Ai}(-\gamma\tau) \neq 0$ for all $\tau \in J$. Then, for all sufficiently large d , every integer k with $(k - \kappa_c d)/d^{1/3} \in J$ gives a nonzero coefficient, apart from the reflection holes.*

Proof. The continuous Airy factor has a positive minimum in absolute value on J , and the applicable phase constants are nonzero. The uniform additive estimate proves the assertion. $\square$

19 Scope and relation to the regular saddle

The critical point is a curvature-zero fold strictly outside the recurrence-root spectrum. The preceding regular-saddle analysis covered compact ratio intervals below κ_c [Added 1 October 2026, batch 72A: this is manuscript 58, Part I here; manuscript 54 does not cite it]; the present result describes its $d^{1/3}$ transition window, including the critical ratio itself. The fixed offset remains bounded independently of d . Neither growing offsets nor the full A290268 noncancellation conjecture are settled here. [Added 1 October 2026, batch 72A: No manuscript of this report covers fixed b in the intermediate range $d^{-2/3} \ll \kappa_c - k/d \ll 1$, between the compact intervals of Theorem 6.2 and the window of Theorem 18.1; see Question 1.]

Zeros of $\text{Ai}(-\gamma\tau)$ are zeros of a leading asymptotic term. The additive theorem remains true there but does not decide whether any corresponding integer coefficient is zero. No inference of exact cancellation is made from an Airy zero or from an illustrative numerical calculation.

The coefficient factorization is credited to the source report. Airy integral representations and $\text{Ai}(0)$ are classical [5]. The quantitative trace bound, finite-depth centering, localization proof, and stated coefficient consequences are proved explicitly here. The arguments received independent mathematical review. They are not Lean formalizations or externally refereed results, and no worldwide priority claim is made.

Part III

The two-dimensional tail and the lower bounds

Part III concerns regions in which both k and the offset b are proportional to d , or the offset grows like $\sqrt{d}$. Five manuscripts convert asymptotic laws there into lower bounds for $a(N)$ beyond the bulk $\beta(N)$ of (2.5): 49, 45 (its second part), 40, 37 and 53. Each headline bound is ineffective, with existential thresholds. Each is implied by $a(N) = N^2/2 + o(N^2)$, which manuscript 34 proved later the same day; that theorem is printed in the sibling report [2]. Section 25 collects the bounds and says what each manuscript keeps. In this part, r is the centered ratio parameter $4 + \delta$ (37, 40, 45) and $R = (4 + \delta)d + 1$. The bulk, the bridge and the injectivity of $(d, k) \mapsto (j, k)$ at fixed N are those of Section 2.

20 Two saddles and a strict quadratic gain (manuscript 49)

Theorem 20.1 (Strict pointwise quadratic improvement). *There are constants $c > 0$ and N_0 such that*

$$a(N) \geq (3/8 + c)N^2 \tag{20.1}$$

for every integer $N \geq N_0$. In fact the proof gives

$$a(N) \geq \beta(N) + C_{\mathcal{R}}(N - 1)^2 - O(N)$$

for a positive rectangle constant $C_{\mathcal{R}}$ described below.

Neither a numerical gain nor a numerical threshold is claimed. This is a pointwise theorem at each sufficiently large order, rather than an average or summatory statement. [Added 1 October 2026, batch 72A: the gain is implied by $a(N) = N^2/2 + o(N^2)$ (manuscript 34, [2]) for every $c < 1/8$; see Section 25. The cosine law and the separated phases below are not implied by it.]

We write

$$b = q - 2d, \quad \kappa = k/d, \quad \delta = b/d, \quad R = (4 + \delta)d + 1, \quad u = d\zeta. \tag{20.2}$$

The proof uses small fixed positive κ and δ . In particular $b > 0$, so none of its selected cells is a reflection hole.

20.1 The regular base point

For the base $\delta = 0$, the regular branch of Proposition 4.1 is used, with $w(\kappa) = 1/t(\kappa)$ and $\psi(\kappa) = (\pi - t(\kappa))/2$ as in (2.8) and (2.7). On it $D_0(t) > 0$, and the gap identity (4.11) holds.

We may use the concrete interval

$$I = [1/200, 1/100] \subset (0, \kappa_c). \quad (20.3)$$

It lies in $(0, \kappa_c)$ by Lemma 4.2; manuscript 49 re-derives this from the same certificate. With the limiting mean $\mathcal{M}$ of (4.4) and $\mathcal{J}$ of (4.5), the saddle equation (4.6), the curvature (4.10) and (6.4) give, at $y = w(\kappa)^2 = t(\kappa)^{-2}$,

$$\mathcal{M} = (1 - \kappa)/2, \quad y\mathcal{M}_y = \frac{F(t)}{4tD_0(t)} > 0, \quad 2w \arctan(2/w) + w\mathcal{J} = 1. \quad (20.4)$$

The inverse-sine branch is verified by $0 < t - \arctan(2t) < \pi/2$, whose cosine has the positive sign of $D_0(t)$.

20.2 A globally dominant inner saddle

The recurrence (2.2) has the exact exponential generating function

$$\sum_{k \geq 0} \frac{G_k(c, R)}{k!} z^k = (1 + z)^{(c-R-1)/2} (1 - z)^{(-c-R-1)/2}. \quad (20.5)$$

Indeed the right side equals one at zero and satisfies $(1 - z^2)Y' = [c + (R + 1)z]Y$; differentiating at zero gives precisely the recurrence. Substitution of (20.2) yields

$$\frac{G_k(d(2\zeta - \delta), (4 + \delta)d + 1)}{k!} = [z^k](1 + z)^{d(\zeta - 2 - \delta) - 1} (1 - z)^{d(-\zeta - 2) - 1}. \quad (20.6)$$

The logarithms in $|z| < 1$ are their branches zero at zero. Define

$$g(z; \zeta, \kappa, \delta) = (\zeta - 2 - \delta) \log(1 + z) + (-\zeta - 2) \log(1 - z) - \kappa \log z. \quad (20.7)$$

Its stationary equation is

$$(\kappa + 4 + \delta)z^2 + (2\zeta - \delta)z - \kappa = 0. \quad (20.8)$$

At $\delta = 0$, $\zeta = iw(\kappa)$, the small root is $z = -iv$, where

$$v = \frac{w - \sqrt{w^2 - \kappa(\kappa + 4)}}{\kappa + 4} = \frac{\kappa}{w + \sqrt{w^2 - \kappa(\kappa + 4)}}. \quad (20.9)$$

For $\kappa \in I$, $w > 1/\pi > 7/22$ and $w^2 - \kappa(\kappa + 4) > 1/25$, so

$$0 < v < 11/570 < 1/50.$$

Lemma 20.2 (Global inner dominance). *There is a uniform neighborhood of the compact base graph $\{(iw(\kappa), \kappa, 0) : \kappa \in I\}$ in which the small saddle $z = z_s(\zeta, \kappa, \delta)$ is analytic, remains inside $|z| < 1$, and satisfies*

$$\operatorname{Re}(g(z_s e^{i\theta}) - g(z_s)) \leq -c_I(1 - \cos \theta), \quad -\pi \leq \theta \leq \pi. \quad (20.10)$$

Its angular curvature $V_{\text{in}} = z_s^2 g_{zz}(z_s)$ has positive real part.

Proof. At the base, rotate $z = -iZ$. The generating exponent becomes

$$E_{\text{in}}(Z) = 2w \arctan Z - 2 \log(1 + Z^2).$$

The term $-\kappa \log z$ has no real angular variation. Using $1 - \cos(n\theta) \leq n^2(1 - \cos \theta)$ on the power series for E_{in} gives

$$\text{Re}(E_{\text{in}}(ve^{i\theta}) - E_{\text{in}}(v)) \leq -[2wv - \mathcal{R}_h(v, w)](1 - \cos \theta), \quad (20.11)$$

where

$$\mathcal{R}_h(v, w) = \frac{8v^2 + 2wv^3(3 - v^2)}{(1 - v^2)^2}.$$

The even harmonics contribute $8v^2/(1 - v^2)^2$; the odd harmonics of degree at least three contribute the other term. Our rational bounds imply

$$\frac{\mathcal{R}_h(v, w)}{2wv} \leq \frac{4v/w + 3v^2}{(1 - v^2)^2} < \frac{44/175 + 3/2500}{(2499/2500)^2} < \frac{1}{3}.$$

Thus the base gap and its curvature are uniformly positive.

The small quadratic root is simple, so it continues analytically. On the circle $z_s e^{i\theta}$, both the exponent difference and its first angular derivative vanish at zero for every parameter. The change from the base real part is bounded by a constant times the parameter displacement times $1 - \cos \theta$: near zero this follows from bounded second derivatives, and away from zero it follows by compactness. Shrinking the neighborhood preserves (20.10) and the positive real part of the curvature. $\square$

Lemma 20.3 (Strong inner asymptotics). *Uniformly on a sufficiently small fixed neighborhood as above,*

$$G_k(d(2\zeta - \delta), R) = \frac{k! e^{dg(z_s; \zeta, \kappa, \delta)}}{\sqrt{2\pi d} (1 - z_s^2) \sqrt{V_{\text{in}}}} (1 + O(d^{-1})). \quad (20.12)$$

The square root continues the positive base value. The estimate is analytic in ζ , with uniform derivative estimates on smaller neighborhoods. The lower-half-plane version is the complex conjugate.

Proof. Use the circle $z = z_s e^{i\theta}$ in the Cauchy integral for (20.6); the prefactor is $(1 - z^2)^{-1}$. Lemma 20.2 suppresses all noncentral arcs. The central logarithm is $-V_{\text{in}}\theta^2/2 + O(\theta^3)$ before multiplication by d , with $\text{Re } V_{\text{in}} > c_I > 0$. The Gaussian integral gives the displayed leading term.

The error is $O(d^{-1})$, not merely $O(d^{-1/2})$: the first prefactor derivative and the cubic exponent term integrate to zero against the even Gaussian on a symmetric interval. The remaining normalized error is bounded by Gaussian integrals of constant multiples of $d\theta^4 + d^2\theta^6 + \theta^2$, each of order d^{-1} . Uniform analytic bounds justify the Taylor remainders and make the omitted arcs exponentially small. Cauchy's formula in the parameter ζ gives derivative estimates on a smaller neighborhood. $\square$

20.3 The outer action and its conjugate saddles

Let $P_d(u) = (2d + u)_{\underline{R}}$. In the upper-half-plane neighborhood, Euler–Maclaurin gives

$$P_d(d\zeta) = d^R e^{df(\zeta, \delta)} A_F(\zeta, \delta) (1 + O(d^{-1})), \quad (20.13)$$

where

$$f(\zeta, \delta) = \int_0^{4+\delta} \log(\zeta + 2 - x) dx, \quad (20.14)$$

$$A_F(\zeta, \delta) = \exp\left(\frac{\log(\zeta + 2) + \log(\zeta - 2 - \delta)}{2}\right). \quad (20.15)$$

The upper logarithms have arguments in $(0, \pi)$, and the lower versions are conjugate. To verify (20.13), apply the trapezoidal Euler–Maclaurin estimate to the mesh j/d , $0 \leq j \leq (4+\delta)d$. The endpoint half-sum gives (20.15); the remainder is $O(d^{-1})$ uniformly, since the logarithm and its derivatives are bounded on the root-free neighborhood. The estimate and its ζ derivatives are uniform.

Define

$$\Phi(\zeta; \kappa, \delta) = f(\zeta, \delta) + g(z_s(\zeta, \kappa, \delta); \zeta, \kappa, \delta) - \log \zeta. \quad (20.16)$$

At $\delta = 0$, the upper saddle is $\zeta = iw(\kappa)$. Indeed the half-angle formula and (20.9) give

$$\arctan v = \mathcal{J}(\kappa, w^2)/2.$$

Consequently direct differentiation of (20.16) gives

$$\Phi_\zeta(iw) = i(w^{-1} - 2 \arctan(2/w) - 2 \arctan v) = 0$$

by (20.4). Its angular curvature is

$$V_{\text{out}} := \zeta^2 \Phi_{\zeta\zeta} = 4yM_y(\kappa, y) > 0, \quad y = w^2. \quad (20.17)$$

This follows by differentiating the same identities, or by the logarithmic derivative of the centered root product.

The implicit-function theorem therefore gives an analytic upper saddle $\zeta_+(\kappa, \delta)$ for all sufficiently small real $|\delta|$, uniformly in $\kappa \in I$. The lower saddle is $\overline{\zeta_+}$. Their moduli agree, their angular curvatures retain positive real parts, and

$$\zeta_+(\kappa, \delta) = iw(\kappa) + O(\delta).$$

Thus a single circle $|u| = d|\zeta_+|$ passes through both saddles.

Lemma 20.4 (No additional dominant outer arc). *For sufficiently small fixed $\delta_0 > 0$, uniformly for $\kappa \in I$ and $0 \leq \delta \leq \delta_0$, the exact outer Cauchy integral is exponentially smaller away from fixed neighborhoods of the two conjugate saddles than on their leading saddle scale.*

Proof. This assertion needs a global estimate; the local inner asymptotic alone would not suffice. Because $b \geq 0$, pair the $2d$ common falling roots. Up to a real prefactor, the exact coefficient integrand is

$$u^{k+1-d} D_R(-u^2) U_b(u) Q_b(u), \quad (20.18)$$

where

$$D_R(v) = \prod_{j=1}^{2d} (1 + v/j^2) \prod_{\lambda > 0} (1 - \lambda^2/(4v)),$$

$$U_b(u) = \prod_{j=1}^b (1 - u/(2d + j)), \quad Q_b(u) = G_k(2u - b, R)/G_k(2u, R).$$

Here λ are the real roots of $i^{-k}G_k(ix, R)$ for the actual value of R .

On the chosen circle, put $\rho = d^2|\zeta_+|^2$ and $V_R = (\rho \partial_\rho)^2 \log D_R(\rho)$. This curvature is at least a fixed positive multiple of d . We justify this uniformly. The all-order closed-walk trace comparison (compare Lemmas 14.1 and 12.1) gives an $O(1+|b|)$ error, including fixed logarithmic derivatives, between the actual- R root sums and the $R = 4d + 1$ sums. To see the dependence, the scaled squared edge weight is $(j/d)(j/d + 4 + (b+1)/d)$; a length- $2p$ returning path differs from its base product by at most $Cp(p + |b| + 1)W^{p-1}/d$. Summing at most 4^p paths per row

and the boundary rows, then summing the exterior resolvent's geometric series, gives the stated error. At $b = 0$, the same closed-walk series sums the central-binomial generating function and yields the mean $\mathcal{M}$ in (4.4), with curvature yM_y . The root gap keeps the geometric ratio below one for small δ_0 .

It follows that V_R/d differs from the $\delta = 0$ curvature at the same radius by $O(\delta) + O(d^{-1})$. The radius itself changes by $O(\delta)$ after scaling by d . The limiting base curvature is positive by (20.4), so

$$V_R/d = V_{\text{base}}/d + O(\delta) + o(1) \geq c_I > 0 \quad (20.19)$$

after reducing δ_0 .

The exact squared-modulus products give

$$\left| D_R(\rho e^{i\theta}) / D_R(\rho) \right| \leq e^{-2V_R \sin^2(\theta/2)}.$$

This bounds the two-arc decay in the u variable around $\pm i|u|$. Both U_b and Q_b have logarithmic modulus bounded in absolute value by $Cb = C\delta d$ on the whole circle. For U_b , every factor has modulus uniformly between two positive constants. For Q_b , root separation gives $|b/(2u - i\lambda)| \leq C\delta < 1/2$, so the elementary logarithmic bound may be summed over its $O(d)$ factors.

The actual saddle angles differ from $\pm\pi/2$ by $O(\delta)$. The real logarithm of the modulus of D_R has zero first angular derivative and uniformly $O(d)$ second derivative at its center. Its modulus at a perturbed saddle angle is therefore at least its central modulus times $e^{-C\delta d^2}$. Including $U_b Q_b$, the actual saddle value is bounded below by the reference central value times $e^{-C\delta d}$.

Outside fixed small neighborhoods of the two centers, the reference angular gap is e^{-cd} , while the perturbations cost only $e^{C\delta d}$. Choose δ_0 small relative to this fixed gap. Those arcs are then exponentially smaller than the actual saddle values. Inside the neighborhoods, the analytic action and its positive real angular curvature give the usual quadratic decay around ζ_+ and its conjugate. These estimates cover the full contour. $\square$

20.4 The strong cosine law and separated adjacent phases

Shrink δ_0 if necessary and choose any nondegenerate closed rectangle

$$\mathcal{R} \subset \text{int}(I) \times (0, \delta_0).$$

All following asymptotics are uniform over integer parameters whose scaled pair (κ, δ) belongs to this rectangle or a fixed small neighborhood of it.

Let z_s denote the inner saddle evaluated at ζ_+ . Set

$$\mathcal{P}(\kappa, \delta) = \frac{A_F(\zeta_+, \delta)}{(1 - z_s^2) \sqrt{V_{\text{in}}} \sqrt{V_{\text{out}}}}. \quad (20.20)$$

All branches continue their base choices. The prefactor is analytic and nonzero. In particular its argument can be chosen smoothly on the rectangle.

Theorem 20.5 (Uniform additive cosine law). *There are positive amplitudes $\mathcal{A}_{d,k,b}$ and phases $\Theta_{d,k,b}$ such that*

$$H_{d,k,2d+b} = \mathcal{A}_{d,k,b} (\cos \Theta_{d,k,b} + O(d^{-1})), \quad (20.21)$$

where

$$\mathcal{A}_{d,k,b} = \frac{d^{R-d} k!}{\pi d} e^{d \text{Re } \Phi(\zeta_+; \kappa, \delta)} |\mathcal{P}(\kappa, \delta)|, \quad (20.22)$$

$$\Theta_{d,k,b} = d \text{Im } \Phi(\zeta_+; \kappa, \delta) + \arg \mathcal{P}(\kappa, \delta). \quad (20.23)$$

The error inside the parentheses is additive, including at zeros of the leading cosine.

Proof. The change of variables $u = d\zeta$ in the coefficient integral contributes d^{-d} . Equations (20.12) and (20.13), followed by the outer Gaussian calculation, give the upper-saddle contribution

$$\frac{d^{R-d}k!}{2\pi d} e^{d\Phi(\zeta_+; \kappa, \delta)} \mathcal{P}(\kappa, \delta) (1 + O(d^{-1})).$$

The local analytic estimates are uniform, and the same odd-term cancellation used in Lemma 20.3 gives the $O(d^{-1})$ outer remainder. Lemma 20.4 suppresses the remaining contour. The lower contribution is the complex conjugate. Their sum gives (20.21) on the positive scale (20.22), without dividing by the cosine.

As a branch check, at $\delta = 0$ the upper action has imaginary part $2\pi + (\kappa - 1)\pi/2$, while A_F contributes phase $\pi/2$ and the Gaussian factors are positive. The total phase is $(k - d + 1)\pi/2$ modulo 2π , reproducing the exact reflection parity. $\square$

Theorem 20.6 (Separated adjacent phases). *The rectangle $\mathcal{R}$ may be chosen so that there is a constant $\alpha > 0$ with*

$$\text{dist}(\Theta_{d,k+2,b-4} - \Theta_{d,k,b}, \pi\mathbb{Z}) \geq \alpha \quad (20.24)$$

for every sufficiently large d whenever both adjacent scaled cells lie in $\mathcal{R}$.

Proof. The saddle-envelope derivatives are

$$\Phi_\kappa = -\log z_s, \quad \Phi_\delta = \log(\zeta_+ - 2 - \delta) - \log(1 + z_s).$$

The derivatives through either stationary variable vanish. Therefore Taylor's theorem and the smooth nonzero prefactor give

$$\Theta_{d,k+2,b-4} - \Theta_{d,k,b} = \text{Im}[-2\log z_s - 4\log(\zeta_+ - 2 - \delta) + 4\log(1 + z_s)] + O(d^{-1}). \quad (20.25)$$

At $\delta = 0$, the literal upper branches give

$$4(\arctan(w/2) - \arctan v) - 3\pi = 4\psi(\kappa) - 3\pi \equiv 4\psi(\kappa) - \pi \pmod{2\pi}. \quad (20.26)$$

This is separated from $\pi\mathbb{Z}$ uniformly on I : ψ is positive there and is bounded strictly below $\pi/4$. Continuity preserves a positive separation for sufficiently small positive δ . Choose $\mathcal{R}$ in that region, then absorb the uniform $O(d^{-1})$ error by reducing the separation constant. This proves (20.24). $\square$

20.5 Adjacent phases and a two-dimensional lattice patch

We now convert the analytic results into a pointwise quadratic improvement. Choose a fixed nondegenerate rectangle

$$\mathcal{R} = [\kappa_-, \kappa_+] \times [\delta_-, \delta_+], \quad 0 < \kappa_- < \kappa_+, \quad 0 < \delta_- < \delta_+,$$

inside the region of Theorems 20.5 and 20.6. Their conclusions give a uniform additive cosine law and a constant $\alpha > 0$ such that

$$\text{dist}(\Theta_{d,k+2,b-4} - \Theta_{d,k,b}, \pi\mathbb{Z}) \geq \alpha \quad (20.27)$$

for all sufficiently large d , whenever both adjacent cells lie in $\mathcal{R}$. We may take $\alpha \leq \pi/2$. A global choice of phase argument is unnecessary for this separation condition.

Lemma 20.7 (One safe cell in each adjacent pair). *On a same-parity chain of n consecutive cells*

$$(k, b), (k+2, b-4), \dots$$

inside $\mathcal{R}$, at least $\lfloor n/2 \rfloor$ coefficients are nonzero for all sufficiently large d , uniformly over all such chains.

Proof. Choose $r = \sin(\alpha/4) > 0$, so $2 \arcsin r < \alpha$. If both phases of an adjacent pair had cosine of absolute value less than r , then, modulo π , both would lie in

$$(\pi/2 - \arcsin r, \pi/2 + \arcsin r).$$

Their difference would have distance less than $2 \arcsin r$ from $\pi\mathbb{Z}$, contradicting (20.27). Thus every adjacent pair contains a phase-safe cell with $|\cos \Theta| \geq r$.

The uniform additive error in Theorem 20.5 is less than $r/2$ for large d . Its positive amplitude is nonzero, so every phase-safe cell has a nonzero coefficient. Pair the chain's first two cells, its next two cells, and so on. This supplies at least $\lfloor n/2 \rfloor$ nonzero cells. No equidistribution or independence assumption is used. $\square$

Define the rectangle centers and half-widths by

$$\kappa_0 = \frac{\kappa_- + \kappa_+}{2}, \quad a_\kappa = \frac{\kappa_+ - \kappa_-}{2}, \quad \bar{\delta} = \frac{\delta_- + \delta_+}{2}, \quad a_\delta = \frac{\delta_+ - \delta_-}{2}.$$

Set

$$\ell = \min\{a_\kappa/2, a_\delta/4\} > 0, \quad v = a_\delta/2 > 0, \quad U_0 = 2\kappa_0 + \bar{\delta}, \quad A_0 = 4 + U_0. \quad (20.28)$$

If $|\kappa - \kappa_0| \leq \ell$ and $|U - U_0| \leq v$, then $\delta = U - 2\kappa$ satisfies $|\delta - \bar{\delta}| \leq v + 2\ell \leq a_\delta$. Thus every such (κ, δ) lies in $\mathcal{R}$.

At fixed derivative order N , the exact coordinate relation is

$$N = 2k + 4d + b + 1, \quad U = \frac{N - 1}{d} - 4 = 2\kappa + \delta. \quad (20.29)$$

Take every integer depth d in the interval

$$\frac{N - 1}{A_0 + v} \leq d \leq \frac{N - 1}{A_0 - v}. \quad (20.30)$$

For each such depth, take integers k satisfying

$$(\kappa_0 - \ell)d \leq k \leq (\kappa_0 + \ell)d, \quad k \equiv d - 1 \pmod{2}, \quad (20.31)$$

and define $b = N - 1 - 2k - 4d$. The preceding inclusion puts $(k/d, b/d)$ in $\mathcal{R}$. In particular $b > 0$ and $k \geq 0$. The parity of b is automatically $N - 1$ modulo 2; no additional filtering is needed. The parity condition in (20.31) gives $\varepsilon = (d - k - 1) \bmod 2 = 0$.

The k values in (20.31) form one chain with step 2, and b changes by -4 at every step. Its length n_d is at least $\ell d - 1$. Lemma 20.7 therefore supplies at least

$$n_d/2 - 1 \geq \ell d/2 - 3/2 \quad (20.32)$$

nonzero cells. All depths in (20.30) grow proportionally to N , so the analytic large-depth threshold is met uniformly for sufficiently large N .

Theorem 20.8 (A strictly larger pointwise quadratic coefficient). *There exist constants $c > 0$ and N_0 such that, for every integer $N \geq N_0$,*

$$a(N) \geq \left(\frac{3}{8} + c\right) N^2. \quad (20.33)$$

For the fixed rectangle chosen above, put

$$C_{\mathcal{R}} = \frac{\ell A_0 v}{(A_0^2 - v^2)^2} > 0. \quad (20.34)$$

More precisely, the proof gives

$$a(N) \geq \beta(N) + C_{\mathcal{R}}(N-1)^2 - O_{\mathcal{R}}(N). \quad (20.35)$$

Any

value of c or N_0 is asserted.

Proof. Sum (20.32) over the integer depths in (20.30). The sum of these depths is one half of the difference of the squared real endpoints, plus $O_{\mathcal{R}}(N)$. Thus the number of selected nonzero cells is at least

$$\begin{aligned} \frac{\ell}{4}(N-1)^2 \left(\frac{1}{(A_0-v)^2} - \frac{1}{(A_0+v)^2} \right) - O_{\mathcal{R}}(N) \\ = C_{\mathcal{R}}(N-1)^2 - O_{\mathcal{R}}(N). \end{aligned}$$

Both the parity restriction and the adjacent-pair restriction have already been included in this count.

The corresponding monomial exponent is

$$j = -q - 1 = -2d - b - 1 = 2k + 2d - N. \quad (20.36)$$

Distinct pairs (d, k) give distinct monomials (j, k) : k is already one monomial coordinate, and then j determines d . There is therefore no double counting. Since $\delta \geq \delta_- > 0$, every selected offset has $b > 0$, so no reflection hole is included. Also $d \geq 1$ and $q = 2d + b \geq 1$, whence $j \leq -2$. The selected terms are disjoint from the source bulk $\beta(N)$, which occupies the nonnegative x -power columns and the positive $j = -1$ boundary.

Adding that bulk gives (20.35). Since $\beta(N) = 3N^2/8 + O(N)$ and $C_{\mathcal{R}} > 0$, every $0 < c < C_{\mathcal{R}}$ gives (20.33) for all sufficiently large N . $\square$

The result concerns every sufficiently large derivative order, not an average or a subsequence. It improves the pointwise quadratic lower-bound coefficient while leaving the full support conjecture and its proposed coefficient $1/2$ unresolved. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.]

20.6 Scope of the quadratic-gain theorem

The theorem adds a positive two-dimensional tail patch to the known bulk at every sufficiently large order. It improves the leading pointwise coefficient above $3/8$, rather than adding only a lower-order correction. The rectangle width, the resulting gain, and the threshold are existential; the proof does not supply numerical values for them.

The asymptotic cosine law is additive. A zero of its leading cosine is not an asserted zero of an integer coefficient. The proof counts cells whose phases are bounded away from such zeros, using adjacent phase separation. It needs no equidistribution or independence hypothesis and makes no claim that every tail cell is nonzero. The full support conjecture and the conjectured leading count $N^2/2$ remain open. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.]

The coefficient model and bulk count are credited to the inspected source. The generating function, harmonic domination, uniform prefactors, global contour comparison, and lattice count are proved explicitly. The exact rational branch certificate is reproducible from the included script. The complete argument received independent mathematical review. It has not been formally verified in Lean or externally refereed, and no exhaustive priority claim is made.

21 Phase curvature and density one in a tail patch (manuscript 45, part 2)

This section proves part 2 of Theorem 11.1. Manuscript 45 takes as input the uniform additive cosine law of manuscript 49, Theorem 20.5. Its citation is “[Theorem 5.1]” of 49, whose article 45’s package included as `inputs/Strict_Quadratic_Gain.pdf` (not shipped). In 45’s notation, $\mathcal{S}_{\text{orig}}(\kappa, \delta) = \Phi(\zeta_+(\kappa, \delta); \kappa, \delta)$ is the stationary upper action of (20.16), and $\mathcal{P}$ is the prefactor (20.20). Then $\Theta_{d,k,b} = d \operatorname{Im} \mathcal{S}_{\text{orig}} + \arg \mathcal{P}$ is the phase (20.23). Manuscript 45’s statement of this input, its Proposition 4.1, with the construction of the action from g and Φ , restates Theorem 20.5 and is not reprinted. Its comment on it is kept: “The companion proves global dominance, both analytic prefactors, and the uniform $O(d^{-1})$ remainder. We use that theorem here rather than assuming that a local saddle alone dominates the original integral.” In this section $w = w(\kappa)$ is the radius (2.8); r and e are independent centered parameters; and ξ and L are local.

21.1 A nonzero fixed-order phase curvature

Introduce independent centered parameters r, e and write $\zeta = x + e$. Define

$$\begin{aligned} f_c(x; r) &= \int_{-r/2}^{r/2} \log(x - t) dt, \\ g_c(z; x, \kappa, r) &= (x - r/2) \log(1 + z) + (-x - r/2) \log(1 - z) - \kappa \log z, \\ \Xi(x; \kappa, r, e) &= f_c(x; r) + g_c(z_s; x, \kappa, r) - \log(x + e). \end{aligned}$$

Let x_s be its upper stationary point and set $\mathcal{T}(\kappa, r, e) = \Xi(x_s; \kappa, r, e)$. These are analytic leading actions near the regular base; no exact finite polynomial for arbitrary nonintegral independent r, e is being asserted. The original action satisfies

$$\mathcal{S}_{\text{orig}}(\kappa, \delta) = \mathcal{T}(\kappa, 4 + \delta, \delta/2). \quad (21.1)$$

At $e = 0$, the saddles are $x_s = iw$ and $z_s = -iv$ with $w, v > 0$. Symmetry of the centered falling interval and the rotated inner exponent give

$$\operatorname{Im} \mathcal{T}(\kappa, r, 0) = \pi(r + \kappa - 1)/2. \quad (21.2)$$

The falling integral contributes $r\pi/2$, the inner action contributes $\kappa\pi/2$, and $-\log(iw)$ contributes $-\pi/2$.

The radius $w = w(\kappa, r)$ solves

$$\mathcal{E}(w, \kappa, r) = 2w \arctan \frac{r}{2w} + w \int_0^\kappa \frac{dt}{\sqrt{w^2 - t(t+r)}} - 1 = 0. \quad (21.3)$$

On the regular branch $\mathcal{E}_w > 0$; the outer angular curvature is $w\mathcal{E}_w$.

The envelope derivative is $\mathcal{T}_e = -1/(x_s + e)$. Hence at $e = 0$,

$$\partial_\kappa \partial_e \operatorname{Im} \mathcal{T} = -w_\kappa/w^2, \quad \partial_r \partial_e \operatorname{Im} \mathcal{T} = -w_r/w^2. \quad (21.4)$$

Also $\partial_e^2 \text{Im } \mathcal{T} = 0$. Indeed Ξ_{xx} at $x = iw$ is real and negative, while $\Xi_{xe} = 1/(iw)^2$ is real. Thus $x_e = -\Xi_{xe}/\Xi_{xx}$ is real, and $\mathcal{T}_{ee} = (x_e + 1)/(iw)^2$ is real.

At fixed N, d , changing the interpolation variable k changes the centered parameters in the direction

$$\mathcal{D} = \partial_\kappa - 2\partial_r - \partial_e.$$

All pure κ, r second derivatives of the imaginary action vanish at $e = 0$ by (21.2). Therefore

$$\mathcal{D}^2 \text{Im } \mathcal{T}(\kappa, 4, 0) = \frac{2w_\kappa - 4w_r}{w^2}. \quad (21.5)$$

This is strictly negative. To verify its sign explicitly, put

$$\xi = \sqrt{w^2 - \kappa(\kappa + r)}, \quad L^2 = w^2 + r^2/4.$$

Differentiation of (21.3) gives

$$\mathcal{E}_\kappa = w/\xi, \quad \mathcal{E}_r = \frac{w^2}{2L^2} + \frac{w(2w^2 - r\kappa)}{4L^2\xi}.$$

Using $w_\kappa = -\mathcal{E}_\kappa/\mathcal{E}_w$ and $w_r = -\mathcal{E}_r/\mathcal{E}_w$, formula (21.5) becomes

$$\mathcal{D}^2 \text{Im } \mathcal{T} = \frac{2w\xi - r\kappa - r^2/2}{\mathcal{E}_w w L^2 \xi} < 0 \quad (r = 4, e = 0). \quad (21.6)$$

The denominator is positive. On the regular base, $w < r/2$ and $\xi \leq w$, so the numerator is strictly negative.

Lemma 21.1 (Uniform row curvature). *There is a nondegenerate compact positive rectangle $\mathcal{R}$ in the region of Theorem 20.5, and constants $0 < \gamma_- < \gamma_+$, such that the following holds. At each fixed d, N , interpolate the phase by*

$$\Theta_{d,N}(X) = d \text{Im } \mathcal{S}_{\text{orig}} \left(\frac{X}{d}, \frac{N-1-4d-2X}{d} \right) + \arg \mathcal{P} \left(\frac{X}{d}, \frac{N-1-4d-2X}{d} \right).$$

On the real interval whose scaled pair lies in $\mathcal{R}$, the function

$$f_{d,N}(X) = \Theta_{d,N}(X)/\pi - 1/2$$

satisfies, for all sufficiently large d ,

$$-\gamma_+/d \leq f_{d,N}''(X) \leq -\gamma_-/d. \quad (21.7)$$

The bounds are uniform in N and d .

Proof. Choose a compact positive regular κ interval. The strict negative curvature in (21.6) has a negative uniform margin there. Analytic continuity preserves that margin for sufficiently small fixed positive δ , giving a rectangle $\mathcal{R}$. The second derivative of the leading phase is $d^{-1}(\partial_\kappa - 2\partial_\delta)^2 \text{Im } \mathcal{S}_{\text{orig}}$, which is exactly the direction already computed. The prefactor argument contributes only $O(d^{-2})$, because it is smooth on the rectangle. Division by π and enlargement of the constants prove (21.7). $\square$

21.2 Integer points near a curved phase

The following elementary estimate converts a uniform cosine error into a bound on actual coefficient cancellations. Its constants do not depend on the position of the interval or on the slope of the curve.

Lemma 21.2 (A shrinking strip around a curved graph). *Fix constants $0 < \gamma_- \leq \gamma_+$, $A > 0$, and $L_0 > 0$. Let $d \geq 1$, and let f be a real C^2 function on an interval J of length at most $L_0 d$. Suppose that throughout J either*

$$\frac{\gamma_-}{d} \leq f'' \leq \frac{\gamma_+}{d} \quad \text{or} \quad \frac{\gamma_-}{d} \leq -f'' \leq \frac{\gamma_+}{d}.$$

Then

$$\#\{x \in J \cap \mathbb{Z} : \text{dist}(f(x), \mathbb{Z}) \leq A/d\} = O_{\gamma_{\pm}, A, L_0}(d^{2/3}). \quad (21.8)$$

Proof. Replacing f by $-f$ if necessary, assume $f'' \geq \gamma_-/d$. For every integer abscissa counted in (21.8), choose an integer ordinate y with $|y - f(x)| \leq A/d$. Partition J into half-open blocks, with the last endpoint included, each of length at most

$$H = cd^{1/3}, \quad c > 0, \quad \gamma_+ c^3/2 < 1/4.$$

Consider three selected points in one block, ordered by $x_1 < x_2 < x_3$. Their determinant

$$\Delta_3 = (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)$$

is an integer. The same expression evaluated on the graph of f is

$$f[x_1, x_2, x_3](x_2 - x_1)(x_3 - x_1)(x_3 - x_2),$$

where the second divided difference has magnitude at most $\gamma_+/(2d)$. The replacement of the three graph ordinates by the selected integers changes the determinant by at most $4AH/d$. Hence

$$|\Delta_3| \leq \frac{\gamma_+ H^3}{2d} + \frac{4AH}{d} < 1$$

for all sufficiently large d , uniformly over all the blocks. It follows that $\Delta_3 = 0$. Thus every triple in a block is collinear.

If a block has at least three selected points, all of them lie on one line. Write their abscissae as $x_1 < \dots < x_n$, let ℓ be the affine function of that line, and put $g = f - \ell$. At every selected abscissa, $|g(x_i)| \leq A/d$. Strong convexity gives, for $1 < i < n$,

$$g(x_i) \leq \frac{x_n - x_i}{x_n - x_1} g(x_1) + \frac{x_i - x_1}{x_n - x_1} g(x_n) - \frac{\gamma_-}{2d} (x_i - x_1)(x_n - x_i).$$

Consequently

$$\frac{\gamma_-}{2d} (x_i - x_1)(x_n - x_i) \leq \frac{2A}{d}.$$

The integer abscissae have spacing at least one. Choosing a middle index therefore yields

$$\left\lfloor \frac{(n-1)^2}{4} \right\rfloor \leq \frac{4A}{\gamma_-}.$$

Each block contains $O_{A, \gamma_-}(1)$ selected points, also when it has fewer than three. There are $O_{\gamma_+, L_0}(1 + d^{2/3})$ blocks. This proves the assertion for large d ; increasing the constant covers the remaining bounded range using $\#(J \cap \mathbb{Z}) \leq L_0 d + 1$. $\square$

21.3 A row bound for actual cancellations

Fix a nondegenerate rectangle

$$\mathcal{R} = [\kappa_-, \kappa_+] \times [\delta_-, \delta_+], \quad 0 < \kappa_- < \kappa_+, \quad 0 < \delta_- < \delta_+,$$

inside the parameter region established above. In particular, the strong cosine law (20.21) and the curvature bounds (21.7) hold uniformly on $\mathcal{R}$. For a fixed order N and depth d , put

$$b = N - 1 - 2k - 4d, \quad J_{d,N} = \left\{ x \in \mathbb{R} : \left(\frac{x}{d}, \frac{N - 1 - 4d - 2x}{d} \right) \in \mathcal{R} \right\}.$$

The set $J_{d,N}$ is an interval, possibly empty, of length at most $(\kappa_+ - \kappa_-)d$.

Proposition 21.3 (Sparse zeros in every row). *For all sufficiently large d , uniformly in the integer N ,*

$$\#\{k \in J_{d,N} \cap \mathbb{Z} : H_{d,k,2d+b} = 0, \ b = N - 1 - 2k - 4d\} = O_{\mathcal{R}}(d^{2/3}). \quad (21.9)$$

Proof. Write the strong cosine law in the form

$$H_{d,k,2d+b} = \mathcal{A}_{d,k,b}(\cos \Theta_{d,k,b} + E_{d,k,b}), \quad \mathcal{A}_{d,k,b} > 0, \quad |E_{d,k,b}| \leq C_{\mathcal{R}}/d.$$

Let $\Theta_{d,N}(x)$ be the smooth row interpolation constructed in the curvature argument, and set

$$f_{d,N}(x) = \frac{\Theta_{d,N}(x)}{\pi} - \frac{1}{2}.$$

For a real angle θ , if $u = \text{dist}(\theta/\pi - 1/2, \mathbb{Z}) \in [0, 1/2]$, then $|\cos \theta| = \sin(\pi u) \geq 2u$. Hence an actual zero of H satisfies

$$\text{dist}(f_{d,N}(k), \mathbb{Z}) \leq \frac{C_{\mathcal{R}}}{2d}.$$

The previously established curvature bounds give $-\gamma_+/d \leq f_{d,N}'' \leq -\gamma_-/d$ on $J_{d,N}$, with fixed positive $\gamma_{\pm}$. Lemma 21.2 proves (21.9). $\square$

This is a necessary condition for an exact cancellation. A zero of the leading cosine is not itself asserted to be a zero of the integer coefficient. The row bound uses the full cosine law and applies to both reflection parities without discarding either class of k .

21.4 Density one in a fixed tail patch

Write the center and half-widths of $\mathcal{R}$ as

$$\kappa_0 = \frac{\kappa_- + \kappa_+}{2}, \quad \bar{\delta} = \frac{\delta_- + \delta_+}{2}, \quad a_{\kappa} = \frac{\kappa_+ - \kappa_-}{2}, \quad a_{\delta} = \frac{\delta_+ - \delta_-}{2}.$$

Define

$$\ell = \min\{a_{\kappa}/2, a_{\delta}/4\}, \quad v = a_{\delta}/2, \quad U_0 = 2\kappa_0 + \bar{\delta}, \quad A_0 = 4 + U_0. \quad (21.10)$$

These constants are positive, and $A_0 > v$. Let $\mathcal{C}_N$ consist of all integer pairs (d, k) with $d \geq 1$ satisfying

$$\frac{N-1}{A_0+v} \leq d \leq \frac{N-1}{A_0-v}, \quad (\kappa_0 - \ell)d \leq k \leq (\kappa_0 + \ell)d. \quad (21.11)$$

For each pair set $b = N - 1 - 2k - 4d$ and $q = 2d + b$.

Theorem 21.4 (Density one and a quantitative deficit). *Put*

$$C_{\text{patch}} = \frac{4\ell A_0 v}{(A_0^2 - v^2)^2} > 0. \quad (21.12)$$

As $N \rightarrow \infty$ through the positive integers,

$$\#\mathcal{C}_N = C_{\text{patch}}(N-1)^2 + O_{\mathcal{R}}(N), \quad (21.13)$$

$$\#\{(d, k) \in \mathcal{C}_N : H_{d,k,q} = 0\} = O_{\mathcal{R}}(N^{5/3}). \quad (21.14)$$

In particular, the proportion of nonzero cells in $\mathcal{C}_N$ tends to one, with deficit $O_{\mathcal{R}}(N^{-1/3})$.

Proof. Put $U = (N-1)/d - 4$. The depth bounds in (21.11) give $|U - U_0| \leq v$. Since $\delta = b/d = U - 2\kappa$ and $|\kappa - \kappa_0| \leq \ell$,

$$|\delta - \bar{\delta}| \leq v + 2\ell \leq a_{\delta}.$$

Also $\ell \leq a_{\kappa}$, so every cell has $(k/d, b/d) \in \mathcal{R}$. In particular $b > 0$ for large N . There is no parity restriction to impose: the defining formula for b automatically gives $b \equiv N-1 \pmod{2}$, and the cosine law applies for every integer k in the indicated interval.

Each integer depth contains $2\ell d + O(1)$ candidate degrees. Summing between the depth bounds gives

$$\begin{aligned} \#\mathcal{C}_N &= 2\ell \sum_{\frac{N-1}{A_0+v} \leq d \leq \frac{N-1}{A_0-v}} d + O_{\mathcal{R}}(N) \\ &= \ell(N-1)^2 \left\{ \frac{1}{(A_0-v)^2} - \frac{1}{(A_0+v)^2} \right\} + O_{\mathcal{R}}(N), \end{aligned}$$

which is (21.13). There are $O_{\mathcal{R}}(N)$ depths and each has $d \asymp_{\mathcal{R}} N$. The uniform row estimate (21.9) therefore sums to (21.14). Dividing by (21.13) proves the density assertion. $\square$

Corollary 21.5 (Pointwise term count; superseded, see Section 25). *Let $\beta(N)$ be the positive bulk count in the coefficient setup. Then*

$$a(N) \geq \beta(N) + C_{\text{patch}}(N-1)^2 - O_{\mathcal{R}}(N^{5/3}). \quad (21.15)$$

Consequently

$$\liminf_{N \rightarrow \infty} \frac{a(N)}{N^2} \geq \frac{3}{8} + C_{\text{patch}}.$$

Proof. The exact coefficient bridge assigns the cell (d, k) the monomial coordinates

$$j = -q - 1 = 2k + 2d - N, \quad k, \quad c_N(j, k) = \binom{N}{k} H_{d,k,q}.$$

The binomial factor is positive. At fixed N , the pair (j, k) determines d , so distinct cells give distinct derivative monomials. Since $d \geq 1$ and $b > 0$ for all sufficiently large N , we have $q \geq 1$ and $j \leq -2$. These cells are disjoint from the source bulk, consisting of the nonnegative x -power columns and the positive $j = -1$ boundary. Thus every nonzero cell counted in Theorem 21.4 adds one term to the bulk count. Equations (21.13)–(21.14) imply (21.15). The final inequality follows from $\beta(N) = 3N^2/8 + O(N)$. $\square$

The rectangle and all implied constants are fixed before N tends to infinity. This density-one conclusion concerns the specified positive-area tail patch. It neither asserts density one throughout the entire conjectured support nor eliminates all exact cancellations in this patch. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.]

[Merge note, batch 72A.] Lemma 21.2 is printed once. Manuscripts 40 (its Lemma 6.1) and 37 (its Lemma 5.1) state and prove the same estimate, as do manuscripts 34, 28, 31 and 26 of the sibling report [2]. Corollary 21.5 strengthens Theorem 20.8: on the same rectangle construction $C_{\text{patch}} = 4C_{\mathcal{R}}$, with an $O(N^{5/3})$ error in place of $O(N)$. It uses Theorem 20.5 as input.

21.5 Scope and proof dependencies

The fixed-offset result leaves only a bounded number of candidate cancellation positions in each specified regular-ratio segment. The two-dimensional result proves density one in a fixed positive-area tail patch. Neither statement proves exact noncancellation everywhere, and neither is a global density theorem for the whole coefficient support.

The source model and derivative bridge are from [1]. The precise regular-saddle and strong cosine inputs are established in the included companions (manuscripts 58 and 49: Part I and Section 20 here; the PDFs that 45 included are not shipped). The sharper rate, phase curvature, near-curve estimate, and density conclusions are proved here. Classical lattice-point results on convex curves provide historical context [9]; the shrinking-strip lemma used here is proved directly and does not invoke a general lattice-point theorem.

The new arguments received independent mathematical review. They have not been formalized in Lean or externally refereed, and no exhaustive priority claim is made. The bounds and thresholds are asymptotic, with no numerical threshold asserted.

22 A direct contour proof for the regular region (manuscript 40)

In this section the branch parameter is written T , because t is the contour variable. Thus $T_0 = t_0$ and $T_c = t_c$, the functions $K(T)$ and $F(T)$ are (4.7) and (4.8), the regular branch is

$$T_0 \leq T < T_c, \quad \kappa = K(T), \quad \kappa_c = K(T_c),$$

and $T = T(\kappa)$ is its inverse. Its needed properties are those of Proposition 4.1. As in Section 2, $R = 2d + q + 1$, $M = k + R$ and $b = q - 2d$. The bridge (2.4) reads $N = 2k + 4d + b + 1$, $j = -q - 1$.

Theorem 22.1 (Direct uniform cosine law). *For every $0 \leq \kappa_1 < \kappa_c$, there is $\delta_0 > 0$ such that the following holds uniformly for integers d, k, b with*

$$d \longrightarrow \infty, \quad 0 \leq k/d \leq \kappa_1, \quad |b|/d \leq \delta_0.$$

Put $\kappa = k/d$, $\delta = b/d$ and

$$L(t) = \log t - \frac{\delta}{2}t + \kappa \log \cosh(t/2) - (\kappa + 4 + \delta) \log \sinh(t/2) - (4 + \delta) \log 2. \quad (22.1)$$

There is an analytic lower saddle $t_s = t_s(\kappa, \delta)$, continuing $-iT(\kappa)$, determined by

$$\kappa = \frac{\sinh t_s}{t_s} - 2 \cosh t_s - 2 - \frac{\delta}{2}(1 + e^{t_s}). \quad (22.2)$$

Set

$$V = t_s^2 L''(t_s), \quad P = \frac{t_s}{4 \sinh^2(t_s/2) \sqrt{V}}, \quad \Theta = d \operatorname{Im} L(t_s) + \arg P, \quad (22.3)$$

$$\mathcal{A} = \frac{2M!}{d! \sqrt{2\pi d}} e^{d \operatorname{Re} L(t_s)} |P| > 0. \quad (22.4)$$

The square root continues the positive root at $\delta = 0$. Then

$$H_{d,k,2d+b} = \mathcal{A}(\cos \Theta + O_{\kappa_1}(d^{-1})). \quad (22.5)$$

The error is additive, including at zeros of the leading cosine. The value of δ_0 is not claimed to be numerical or optimal.

The earlier nested-saddle construction (manuscript 49, Theorem 20.5; not a proof dependency of manuscript 40) treated a smaller specified base interval. The present proof establishes the result near every compact part of the regular branch and supplies a simpler exact prefactor. The full noncancellation conjecture remains open.

22.1 An exact change of contour variable

Cauchy's formula in (2.1), followed by the local analytic substitution $\omega = e^t - 1$, gives

$$H_{d,k,q} = \frac{M!}{d!} 2^{-R-1} \frac{1}{2\pi i} \oint \frac{t^d e^{-bt/2} \cosh^k(t/2)}{\sinh^{M+1}(t/2)} dt. \quad (22.6)$$

The contour initially surrounds only zero. Indeed $d\omega = e^t dt$, and the exponential power is

$$(2d+1) + k/2 - (M+1)/2 = -b/2.$$

The powers $R+1$ and $M+1$ in (22.6) include the Cauchy differential. Since $R = (4+\delta)d+1$, formula (22.6) is exactly

$$H_{d,k,q} = \frac{M!}{d!} \frac{1}{2\pi i} \oint \frac{e^{dL(t)}}{4 \sinh^2(t/2)} dt. \quad (22.7)$$

Branches in L are chosen locally along the two saddle arcs. The original integrand is single-valued for the integer parameters.

Two useful coefficient forms are

$$H_{d,k,q} = \frac{M!}{d!} 2^{d-R} [z^{M-d}] e^{-bz} \cosh^k z \left(\frac{z}{\sinh z} \right)^{M+1}, \quad (22.8)$$

$$= \frac{M!}{d!(M-d)!} \sum_{a=0}^k \binom{k}{a} B_{M-d}^{(M+1)}(2d+1+a). \quad (22.9)$$

The last identity uses the classical higher-order Bernoulli generating function [8]. It identifies an existing polynomial family; it supplies no nonvanishing theorem by itself.

22.2 From local curvature to global angular dominance

Lemma 22.2 (An odd–even product comparison). *For $0 < T < \pi$, set $c = (T/\pi)^2$ and*

$$a(x) = \frac{4cx^2}{(x^2 - c)^2}, \quad A_n = a(2n - 1), \quad B_n = a(2n).$$

For every $z \geq 0$,

$$\frac{\sum_{n \geq 1} \log(1 + A_n z)}{\sum_{n \geq 1} A_n} \leq \frac{\sum_{n \geq 1} \log(1 + B_n z)}{\sum_{n \geq 1} B_n}. \quad (22.10)$$

Proof. The positive sequences are summable and satisfy $A_n > B_n > A_{n+1}$. Moreover

$$(\log a)''(x) = \frac{2(x^4 + 4cx^2 - c^2)}{x^2(x^2 - c)^2} > 0 \quad (x > \sqrt{c}).$$

Thus A_n/B_n decreases with n . Normalize the sequences to probabilities $p_n = A_n/\sum A$ and $q_n = B_n/\sum B$. Their likelihood ratio decreases, so for every increasing bounded sequence g_n , $\mathbb{E}_p g \leq \mathbb{E}_q g$. To see this directly, the covariance under q of the decreasing ratio p_n/q_n and increasing g_n is nonpositive; its double-sum expansion proves the inequality.

The function $f_z(a) = \log(1 + az)/a$ decreases with $a > 0$. Hence $f_z(A_n)$ increases with n , and

$$\mathbb{E}_p f_z(A_n) \leq \mathbb{E}_q f_z(A_n) \leq \mathbb{E}_q f_z(B_n).$$

This is (22.10). All sums converge absolutely, and $z = 0$ follows directly. $\square$

The classical products [6] are

$$\frac{\sinh(t/2)}{t/2} = \prod_{n \geq 1} \left(1 + \frac{t^2}{4\pi^2 n^2}\right), \quad \cosh(t/2) = \prod_{n \geq 1} \left(1 + \frac{t^2}{\pi^2 (2n - 1)^2}\right).$$

For $t = -iT e^{i\theta}$ and $z = \sin^2 \theta$, they give, at $\delta = 0$,

$$\operatorname{Re}\{L(t) - L(-iT)\} = \frac{\kappa}{2} \sum \log(1 + A_n z) - \frac{\kappa + 4}{2} \sum \log(1 + B_n z). \quad (22.11)$$

Put

$$\Delta_{\text{ang}}(T, \kappa) = (\kappa + 4) \sum B_n - \kappa \sum A_n.$$

If $\Delta_{\text{ang}} > 0$, Lemma 22.2 and $\log(1 + x) \geq x/(1 + x)$ imply the global bound

$$\operatorname{Re}\{L(-iT e^{i\theta}) - L(-iT)\} \leq -\frac{\Delta_{\text{ang}}(T, \kappa)}{2(1 + B_1)} \sin^2 \theta. \quad (22.12)$$

Thus the local positive-curvature condition suffices for global dominance of the two imaginary arcs.

That condition holds on the entire regular branch. The branch properties are those of Proposition 4.1; manuscript 40 re-derives them in the variable T .

Logarithmic differentiation of the products yields

$$\sum A_n = \frac{T}{2} \tan \frac{T}{2} + \frac{T^2}{4} \sec^2 \frac{T}{2}, \quad \sum B_n = -\frac{T}{2} \cot \frac{T}{2} + \frac{T^2}{4} \csc^2 \frac{T}{2}.$$

On $\kappa = K(T)$ these simplify to

$$\Delta_{\text{ang}}(T, K(T)) = \frac{F(T)}{\sin T} > 0. \quad (22.13)$$

This is also the angular saddle curvature $t_s^2 L''(t_s)$ at $t_s = -iT$. Indeed

$$L'(t) = \frac{\sinh(t)/t - 2 \cosh t - 2 - \kappa}{\sinh t} \quad (\delta = 0).$$

The global margin (22.12) is therefore uniform for $0 \leq \kappa \leq \kappa_1 < \kappa_c$.

22.3 The perturbed circle and the exact prefactor

The nonzero Hessian in (22.13) and the implicit function theorem give a lower saddle $t_s(\kappa, \delta)$, analytic near every regular base point. Compactness gives one sufficiently small δ_0 for $0 \leq \kappa \leq \kappa_1$; take $\delta_0 < 1$. The conjugate upper saddle has the same radius. Both lie on the circle $|t| = |t_s|$, whose radius remains strictly between zero and π .

This contour enlargement is legitimate for the exact integrand: its other poles occur at $2\pi i\mathbb{Z} \setminus \{0\}$, so no pole is crossed. We next check global dominance, rather than infer it solely from a local stationary point.

On the perturbed radius $T' = |t_s|$, the unperturbed deficit $\Delta_{\text{ang}}(T', \kappa)$ remains uniformly positive by continuity. Formula (22.12) therefore gives a fixed angular gap outside neighborhoods of the two imaginary points. The change in $\text{Re } L$ caused by δ is at most $C|\delta|$ on the full circle, since t , $\sinh(t/2)$, and $\cosh(t/2)$ stay separated from zero there. The actual saddle angle differs from the imaginary angle by $O(\delta)$, and the reference modulus at this displaced angle differs by $O(\delta^2)$ in its logarithm. Choose δ_0 so these changes are smaller than the fixed far-arc gap. The other arcs are then exponentially smaller than the saddle arcs.

Near each actual saddle, the angular first derivative vanishes and the real part of the angular Hessian remains positive. Uniform Taylor bounds give local Gaussian decay. The analytic prefactor $1/(4\sinh^2(t/2))$ is bounded above and away from zero on these circles. These facts justify a uniform saddle expansion throughout the stated parameter rectangle.

Parametrize the lower arc by $t = t_s e^{i\theta}$. The differential supplies the factor t_s , so its contribution to (22.7) is

$$\frac{M!}{d!} \frac{e^{dL(t_s)}}{\sqrt{2\pi d}} \left\{ \frac{t_s}{4\sinh^2(t_s/2)\sqrt{V}} + O(d^{-1}) \right\}.$$

The error is relative to the nonzero analytic prefactor. Expand the exponent through the cubic term and the prefactor through the linear term on a symmetric local interval. The odd Gaussian terms vanish. The remaining normalized moments of θ^2 , $d\theta^4$, and $d^2\theta^6$ are all $O(d^{-1})$. The far arcs are exponentially small. This proves the indicated rate uniformly, and the conjugate upper contribution completes Theorem 22.1.

There is a useful exact parity check. At $\delta = 0$ on the lower branch, $\arg P = \pi/2$ and

$$d \text{Im } L(t_s) = (k + 3d)\pi/2.$$

Hence $\Theta = (k + 3d + 1)\pi/2$, which is congruent to $(k - d + 1)\pi/2$ modulo 2π . The leading cosine has exactly the proved reflection parity $k + d$ even. Those holes are exact by the source symmetry; the approximation alone is not used to prove their exact vanishing.

22.4 A short formula for the phase curvature

The direct action also simplifies the density argument. Introduce independent centered parameters r, e and set

$$L_c(t; \kappa, r, e) = \log t - et + \kappa \log \cosh(t/2) - (\kappa + r) \log \sinh(t/2) - r \log 2.$$

Let $S(\kappa, r, e)$ be its value at the continued lower saddle. At $e = 0$ the saddle is $-iT$, where

$$\kappa = K_r(T) := \frac{\sin T}{T} - \frac{r}{2}(1 + \cos T), \quad \text{Im } S(\kappa, r, 0) = \frac{\pi}{2}(\kappa + r - 1). \quad (22.14)$$

The envelope derivative gives $S_e = -t_s$, hence

$$\partial_e \text{Im } S = T, \quad \partial_\kappa \partial_e \text{Im } S = T_\kappa, \quad \partial_r \partial_e \text{Im } S = T_r.$$

Also $\partial_e^2 \text{Im } S = 0$: the equation $L'_c(t_s) = 0$ gives $\partial_e t_s = 1/L'_c(t_s)$, which is real at the base. All pure κ, r second derivatives of the imaginary action vanish by (22.14).

At a fixed order N and depth d , changing k follows the centered direction

$$\mathcal{D} = \partial_\kappa - 2\partial_r - \partial_e, \quad r = 4 + \delta, \quad e = \delta/2.$$

Implicit differentiation of (22.14) yields

$$T_\kappa = \frac{1}{K'_r(T)}, \quad T_r = \frac{1 + \cos T}{2K'_r(T)}.$$

Therefore the curvature is simply

$$\mathcal{D}^2 \text{Im } S(\kappa, 4, 0) = -2T_\kappa + 4T_r = \frac{2 \cos T}{K'(T)} = \frac{2T^2 \cos T}{F(T)} < 0. \quad (22.15)$$

The strict sign is uniform on every compact regular interval. It persists on a sufficiently small two-dimensional neighborhood by continuity. The argument of the analytic prefactor contributes only $O(d^{-2})$ to the second derivative of the interpolated row phase. Thus, after shrinking the same δ_0 in Theorem 22.1, the negative curvature is uniform on the entire closed strip $[0, \kappa_1] \times [-\delta_0, \delta_0]$, including $\kappa = 0$.

22.5 Density one on compact analytic patches

We give a general consequence that allows the curvature to vanish on an exceptional analytic set. It applies only where the strong uniform cosine law has already been established.

The integer-point estimate used below is Lemma 21.2. Manuscript 40 states and proves the same lemma, with the same proof, and it is printed once.

Corollary 22.3 (Quantitative density on the full regular strip; the count bound is superseded, see Section 25). *Fix $0 < \kappa_1 < \kappa_c$ and choose δ_0 as above. Let $\Omega_0 = [0, \kappa_1] \times [-\delta_0, \delta_0]$. At order N , the integer cells whose ratios lie in Ω_0 number*

$$C_0(N-1)^2 + O(N), \quad C_0 = \frac{\delta_0}{2} \left\{ \frac{1}{16 - \delta_0^2} - \frac{1}{(4 + 2\kappa_1)^2 - \delta_0^2} \right\} > 0.$$

Only $O(N^{5/3})$ of those exact coefficients vanish, including any reflection holes. Their nonzero proportion tends to one with deficit $O(N^{-1/3})$. Consequently the derivative term count satisfies

$$a(N) \geq (3/8 + C_0)N^2 - O(N^{5/3}).$$

Proof. The uniform curvature just proved gives second-derivative bounds of order $1/d$ for the normalized row phase. An exact zero in the additive cosine law forces that phase within $O(1/d)$ of an integer. Lemma 21.2 bounds the zeros by $O(d^{2/3})$ in each row. The parameter map and its Jacobian, computed in the proof below, give the candidate constant $\int_{\Omega_0} (4 + 2\kappa + \delta)^{-3} d\kappa d\delta$; elementary integration gives the displayed C_0 . There are $O(N)$ rows with $d \asymp N$, so the zero bound sums to $O(N^{5/3})$. This argument includes the endpoint $\kappa = 0$; no singular nested parametrization is used. Since $\delta_0 < 1$, all these cells have $q \geq 1$ for large N and add distinct terms outside the positive bulk. Combining their nonzero count with $\beta(N) = 3N^2/8 + O(N)$ proves the last inequality. $\square$

Theorem 22.4 (Analytic exceptional sets). *Let U be a connected open region in the positive (κ, δ) plane with a uniform additive cosine law of the form (22.5) on every compact subset, with*

phase $\Theta = d \operatorname{Im} S_{\text{orig}} + \arg P$, where P is analytic and nonzero. Suppose its leading stationary action is real analytic and its directional curvature

$$C = (\partial_\kappa - 2\partial_\delta)^2 \operatorname{Im} S_{\text{orig}}$$

is not identically zero. Let Ω be a finite union of closed nondegenerate axis-parallel rectangles, compactly contained in U . At order N , count all integer (d, k) whose ratios

$$\left(\frac{k}{d}, \frac{N-1-4d-2k}{d} \right) \in \Omega.$$

Their number is

$$(N-1)^2 \int_{\Omega} \frac{d\kappa d\delta}{(4+2\kappa+\delta)^3} + O_{\Omega}(N), \quad (22.16)$$

and only $o_{\Omega}(N^2)$ of their exact coefficients vanish. Thus the nonzero proportion tends to one. No power-saving rate is asserted for this general exceptional-set argument.

Proof. A nonzero real-analytic function on a connected open set has a zero set of area zero. For completeness, at each zero some directional derivative of finite order is nonzero; otherwise its Taylor series and then analytic continuation would make the function identically zero. After rotating coordinates, that derivative is an x derivative and stays nonzero in a neighborhood. Rolle's theorem bounds the number of zeros on every horizontal segment by the derivative order, and Fubini's theorem gives area zero. A countable cover completes this local argument.

Set $x = d/(N-1)$ and $y = k/(N-1)$. The inverse ratio map is

$$x = (4+2\kappa+\delta)^{-1}, \quad y = \kappa(4+2\kappa+\delta)^{-1},$$

with Jacobian $(4+2\kappa+\delta)^{-3}$. The image K of Ω has piecewise smooth boundary, so its grid-point count proves (22.16) with the $O(N)$ boundary error.

Fix $\epsilon > 0$. Cover the compact image in K of the zero set of C by the interiors of finitely many axis-parallel boxes of total area less than ϵ . Their grid-point count is at most

$$\epsilon(N-1)^2 + O_{\epsilon}(N).$$

This bounds every candidate in those boxes without a phase assumption. The remaining compact set is separated from the curvature-zero set and has a finite cover by parameter rectangles on each of which C has constant sign and is bounded away from zero.

At fixed N, d , interpolate

$$f_{d,N}(X) = \frac{1}{\pi} \left[d \operatorname{Im} S_{\text{orig}} \left(\frac{X}{d}, \frac{N-1-4d-2X}{d} \right) + \arg P \right] - \frac{1}{2}.$$

Its second derivative has constant sign and magnitude between fixed positive multiples of d^{-1} on each rectangle. If $H = 0$, the additive cosine error gives $\operatorname{dist}(f_{d,N}(k), \mathbb{Z}) = O(d^{-1})$. Lemma 21.2 therefore bounds such zeros by $O_{\epsilon}(d^{2/3})$ per row. There are $O(N)$ relevant rows, all with $d \asymp N$, giving $O_{\epsilon}(N^{5/3})$ zeros outside the boxes.

After division by N^2 , the limsup of the total zero count is at most ϵ . Send N to infinity first and then ϵ to zero. The finite covers and constants stay fixed during the first limit, proving the claim. $\square$

Theorem 22.1 supplies such connected regions near every compact regular base interval, and (22.15) verifies the nontrivial curvature. On a compact rectangle where its magnitude stays bounded away from zero, the stronger $O(N^{5/3})$ zero bound follows directly. The theorem does not continue the cosine law across an unproved dominance boundary.

All these positive-offset cells have $q \geq 1$ and $j \leq -2$ for large N . The bridge (2.4) is injective in (d, k) at fixed N , so they contribute new terms beyond the source’s proved positive bulk $\beta(N) = 3N^2/8 + O(N)$. In particular, for any such patch,

$$\liminf_{N \rightarrow \infty} \frac{a(N)}{N^2} \geq \frac{3}{8} + \int_{\Omega} \frac{d\kappa d\delta}{(4 + 2\kappa + \delta)^3}.$$

This is a local support-density conclusion and a global lower bound, not a proof of the conjectured leading constant $1/2$. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.]

22.6 Normalization checks and limitations

For comparison with the nested formula, write $r = 4 + \delta$, $e = \delta/2$, $x = 1/t_s - e$, and

$$z = \frac{\kappa}{\kappa + r} \tanh(t_s/2).$$

Eliminating the inner saddle gives exactly (22.2). Its stationary action differs from $L(t_s)$ by the real quantity

$$1 - r + (\kappa + r) \log(\kappa + r) - \kappa \log \kappa.$$

For $\kappa > 0$, the exact analytic prefactors satisfy

$$P_{\text{nested}}^2/P^2 = (\kappa + r)^3/\kappa, \quad P_{\text{nested}} = (\kappa + r)^{3/2}P/\sqrt{\kappa}.$$

The second identity follows from the matching positive-imaginary base values and analytic continuation. Stirling’s formula then matches the two full amplitudes with relative error $O(d^{-1})$ on compact positive κ intervals. The new contour formula itself has no singularity at $\kappa = 0$.

The accompanying standard-library verifier checks (22.8) against the independent falling-factorial/three-term-recurrence model in 112 exact cases. A separate numerical diagnostic compares exact integer coefficients at $d = 200, 400, 800$ with the asymptotic formula and checks both nested actions and prefactors. These finite checks detect normalization mistakes; the theorems are proved analytically.

The new arguments received independent mathematical review. They are not Lean-formalized or externally refereed, and no exhaustive priority claim is made. The offset width, error constants, and eventual thresholds remain nonnumerical. The global support conjecture and behavior outside established dominance regions remain open.

23 A global 0.49 lower bound (manuscript 37)

Manuscript 37 combines two things. One is a horizontal-contour cosine law with density one in a wide cone. The other is a real-saddle positivity theorem on a triangle near $q = 0$. In this section the branch parameter is written T , as in Section 22. Euler’s number is written e , and e is the tilt $|\delta|/2$.

Theorem 23.1 (A global pointwise lower bound; part (3) is superseded, see Section 25). *The following conclusions hold.*

1. *At derivative order N , the tail cone $10q > 3d$ has $\frac{5}{46}(N - 1)^2 + O(N)$ candidate cells and only $o(N^2)$ exact cancellations.*

2. There is d_0 such that $H_{d,k,q} > 0$ whenever $d \geq d_0$ and

$$k, q \geq 0, \quad 40k + 420q \leq 91d.$$

At order N , this triangle contributes $\frac{169}{19912}(N-1)^2 + O(N)$ cells outside the bulk, disjoint from the preceding cone.

3. Consequently

$$\liminf_{N \rightarrow \infty} \frac{a(N)}{N^2} \geq \frac{3}{8} + \frac{5}{46} + \frac{169}{19912} = \frac{28176}{57247} > \frac{49}{100}. \quad (23.1)$$

In particular, $a(N) \geq 49N^2/100$ for all sufficiently large N .

No numerical eventual threshold is asserted.

The full outside-bulk candidate set has $N^2/8 + O(N)$ cells. Together with (2.5), this yields the possible leading count $N^2/2$. Theorem 23.1 supplies a pointwise lower constant; it does not settle that leading asymptotic. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.]

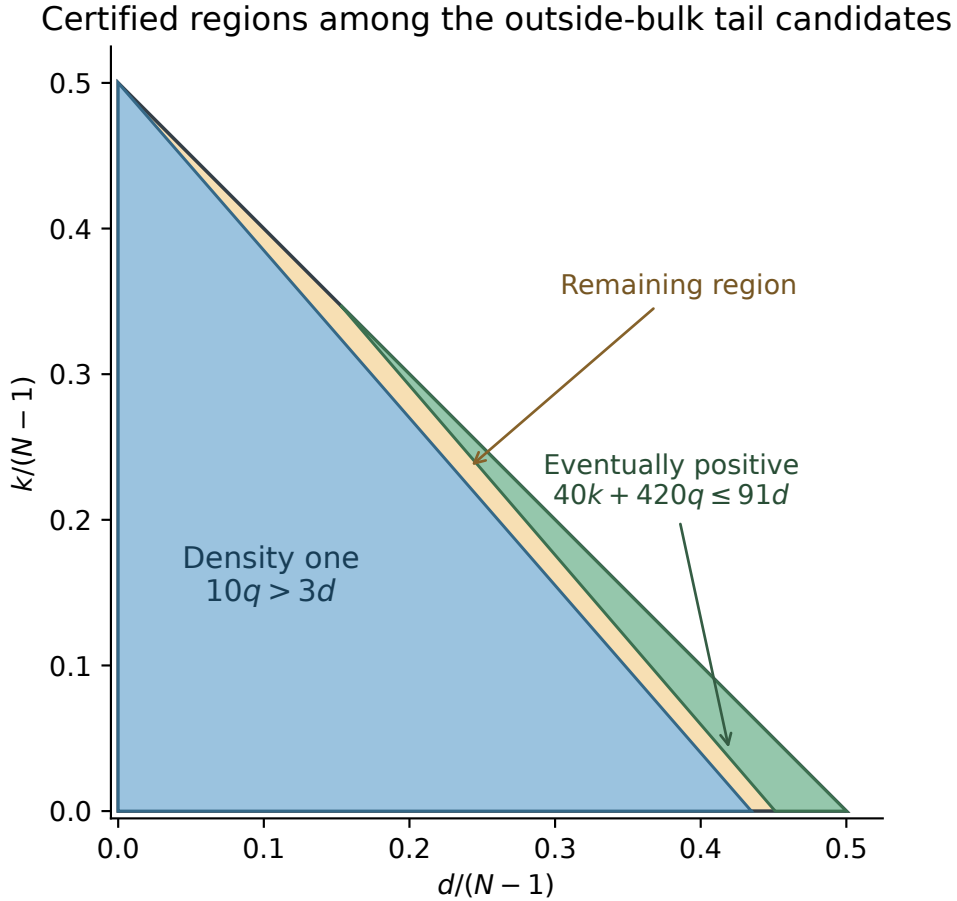


Figure 1: The tail candidate triangle in coordinates $(d/(N-1), k/(N-1))$. Its area is $1/8$. The two proved regions have areas $5/46$ and $169/19912$; the positive bulk of leading size $3N^2/8$ is not shown. The remaining region is unresolved and is not asserted to contain zeros.

23.1 An exact contour and the favored critical curve

Put

$$b = q - 2d, \quad \kappa = k/d, \quad \delta = b/d, \quad r = 4 + \delta.$$

The coefficient has the exact single-contour representation (22.7) with the action (22.1), whose $4 + \delta$ is r . Manuscript 37 derives it by the same substitution $\omega = e^t - 1$, with $R = rd + 1$. The original integer powers make the integrand single-valued despite the local logarithms, and its other poles are at $2\pi i\mathbb{Z} \setminus \{0\}$.

For the horizontal construction assume $\delta > -17/10$, $\delta \neq 0$; the real-saddle region will be treated separately in Theorem 23.6. Write e for Euler's number. For modulus estimates let

$$e = |\delta|/2, \quad h = r/2 - e > 0.$$

The left half-plane is favored when $\delta > 0$, and the right half-plane when $\delta < 0$. Reflecting the real coordinate reduces both modulus problems to

$$L_c(t) = \log t - et + \kappa \log \cosh(t/2) - (\kappa + r) \log \sinh(t/2) - r \log 2. \quad (23.2)$$

For positive offsets, $h = 2$ and $e = \delta/2$. For negative offsets,

$$h = 2 + \delta \in (0, 2), \quad r = h + 2, \quad e = (2 - h)/2. \quad (23.3)$$

Reflection sends a lower-left critical point of (23.2) to the original lower-right critical point and preserves real action values.

Write $t = -s - iy$ with $s > 0$, $0 < y < \pi$. Define

$$\mathcal{U}(s, y) = \frac{s \coth s - y \cot y}{s^2 + y^2} = 2 \sum_{n \geq 1} \frac{(n\pi)^2}{(s^2 + (n\pi)^2)((n\pi)^2 - y^2)}. \quad (23.4)$$

The identity follows from the classical partial fractions [6, 7]. The series and its derivatives converge locally uniformly. In particular

$$\mathcal{U}_s < 0, \quad \mathcal{U}_y > 0,$$

while $\mathcal{U}(s, 0)$ decreases from $1/3$ to zero and $\mathcal{U}(s, y) \rightarrow \infty$ as $y \uparrow \pi$.

The critical equation is $F_{r,e}(t) = \kappa$, where

$$F_{r,e}(t) = \frac{\sinh t}{t} - \frac{r}{2}(1 + \cosh t) - e \sinh t.$$

Its imaginary part factors as

$$\operatorname{Im} F_{r,e}(-s - iy) = \sinh s \sin y \left\{ \mathcal{U}(s, y) - h + \frac{2e}{e^{2s} - 1} \right\}. \quad (23.5)$$

Assume $e > 0$, as will hold off the line $\delta = 0$. The right side of

$$\mathcal{U}(s, y) = h - \frac{2e}{e^{2s} - 1} \quad (23.6)$$

increases from $-\infty$ to $h > 0$. It first meets $\mathcal{U}(s, 0)$ at a unique $s_0 > 0$. For $s > s_0$ there is one smooth curve $y(s) \in (0, \pi)$, strictly increasing from zero to π .

Along this curve,

$$\operatorname{Re} F'_{r,e}(-s - iy) = -\sinh s \sin y \mathcal{U}_y < 0.$$

The Cauchy–Riemann equations therefore give

$$\kappa'(s) = -\operatorname{Re} F'_{r,e}(-s - iy)(1 + y'(s)^2) > 0, \quad \kappa(s) = \operatorname{Re} F_{r,e}(-s - iy(s)). \quad (23.7)$$

Also $\kappa(s) \rightarrow \infty$: as $y \rightarrow \pi$, the positive real term grows with leading size $he^s/2$, whereas the term $\sinh t/t$ is smaller by a factor $O(1/s)$. A point on the curve with negative κ therefore ensures one simple favored critical point for each $\kappa \geq 0$ at all later heights.

23.2 A rational lower bound on the saddle height

The next step supplies a uniform geometric condition for global contour selection.

Lemma 23.2 (The saddle stays above height $\pi/2$). *For every $\kappa \geq 0$ and $\delta > -17/10$, $\delta \neq 0$, there is a unique simple favored critical point in the lower strip $-\pi < \text{Im } t < 0$. Its height is strictly greater than $\pi/2$.*

Proof. Set $C = \cosh s$, $S = \sinh s$, $c = \cos y$, and $\Sigma = s^2 + y^2$. Eliminating e from the imaginary critical equation gives the exact identity

$$\frac{\kappa}{C + c} = e^s \left\{ \frac{s + y(e^{-s} - c)/\sin y}{\Sigma} - h \right\}. \quad (23.8)$$

For positive offsets $h = 2$. The numerator inside braces, after subtracting $h\Sigma$, has derivative

$$1 - \frac{y}{\sin y} e^{-s} - 2hs < 1 - e^{-s} - s < 0.$$

At $y = \pi/2$ its value at $s = 0$ is $y - 2y^2 < 0$. Thus the point of the curve at this height has negative κ .

For negative offsets, use (23.3) and put $a = \pi/2$. At height a , equation (23.6) becomes

$$h = B_h(s) := 2e^{-2s} + (1 - e^{-2s}) \frac{s \coth s}{s^2 + a^2}. \quad (23.9)$$

This decreases strictly from 2 to zero: $\mathcal{U}_s < 0$ and $\mathcal{U}(s, a) \leq \mathcal{U}(0, a) = 1/a^2 < 2$ make its derivative negative. At this height, the sign of κ is the sign of

$$J(s) = a - e^{-s}(s + 2s^2 + 2a^2).$$

Moreover

$$J'(s) = e^{-s}(2s^2 - 3s + 2a^2 - 1) > 0,$$

since the quadratic has discriminant $17 - 4\pi^2 < 0$.

Take $s_* = 51/20$. The elementary bounds

$$\pi > 157/50, \quad e^{51/10} > 160, \quad e^{51/20} < 13$$

give

$$B_h(s_*) < \frac{1}{80} + \frac{(51/20)(161/160)}{(51/20)^2 + (157/100)^2} = \frac{267803}{896740} < \frac{3}{10}. \quad (23.10)$$

Also

$$e^{s_*} < 13 < \pi + \frac{3111}{100\pi},$$

so $J(s_*) < 0$. For the last comparison, $x + 3111/(100x)$ decreases on $(3, 22/7)$ and its value at $22/7$ exceeds 13 by $639/15400$. The exponential inequalities follow respectively from the positive Taylor sum through order 10 and the sum through order 4 plus a geometric tail. The accompanying verifier replays these rational certificates exactly.

If $h \geq 3/10$, the solution $B_h(s) = h$ has $s < s_*$ and therefore $J(s) < 0$. This proves that the height- $\pi/2$ crossing has negative κ in both cases. Monotonicity (23.7) and the increasing height curve now give a unique critical point for every $\kappa \geq 0$, with height above $\pi/2$. Simplicity follows from the strictly negative real part of $F'_{r,e}$ in (23.7). $\square$

23.3 Global selection on horizontal contours

Lemma 23.3 (A unique maximum on each horizontal line). *At a favored critical point of height $y \in (\pi/2, \pi)$, the lower horizontal line has exactly one global modulus maximum, at that point. The upper line has its conjugate maximum. Both maxima have nondegenerate horizontal curvature.*

Proof. In the reflected favored model, put $J_y(s) = \operatorname{Re} L_c(-s - iy)$. Differentiation gives

$$J'_y(s) \frac{C - c}{S} = A_h(s) + eB_0(s) - \frac{r}{2} - \frac{\kappa c}{C + c}, \quad (23.11)$$

where

$$A_h(s) = \frac{s(C - c)}{(s^2 + y^2)S}, \quad B_0(s) = \frac{C - c}{S}.$$

Every variable term on the right decreases when $c < 0$. For A_h , the partial fraction identity gives

$$\frac{1}{A_h(s)} = 2 \sum_{n \in \mathbb{Z}} \frac{s^2 + y^2}{s^2 + (y + 2\pi n)^2}. \quad (23.12)$$

The $n = 0$ term is constant. Every other term increases strictly with s , since $|y + 2\pi n| > y$ for $0 < y < \pi$. Hence A_h decreases strictly. Also

$$B'_0(s) = \frac{C \cos y - 1}{S^2} < 0,$$

and the last term in (23.11) is decreasing because $c < 0$.

The right side of (23.11) tends to $+\infty$ at zero, due to $e > 0$, and to $e - r/2 = -h < 0$ at infinity. It has one zero, and its derivative there is negative. The classified saddle is therefore the unique favored-side maximum, with $J''_y < 0$. At the opposite point of real coordinate s , the action's real part is smaller by $2es$. Thus the maximum is unique on the whole horizontal line. Reflection returns the same assertion in the original coordinates, where

$$\operatorname{Re}(-L''(t_s)) > 0.$$

□

The small contour in (22.7) can be enlarged to the boundary of $|\operatorname{Im} t| \leq y$, with vertical sides tending to infinity. Since $y < \pi$, no other pole is enclosed. On the ends, uniformly on compact parameter sets,

$$\operatorname{Re} L(t) \leq \log(1 + |\operatorname{Re} t|) - h|\operatorname{Re} t| + C.$$

Thus the vertical integrals vanish. Uniqueness and compactness give a uniform gap away from the two saddle neighborhoods; the negative horizontal Hessian supplies uniform Gaussian decay there. This is a global selection argument, not a conclusion from the critical equation alone.

Theorem 23.4 (Global compact-uniform cosine law). *For the original lower favored saddle t_s , put*

$$W = -L''(t_s), \quad P_h = \frac{-i}{4 \sinh^2(t_s/2) \sqrt{W}}, \quad \Theta = d \operatorname{Im} L(t_s) + \arg P_h, \quad (23.13)$$

$$\mathcal{A}_h = \frac{2M!}{d! \sqrt{2\pi d}} e^{d \operatorname{Re} L(t_s)} |P_h| > 0. \quad (23.14)$$

The square root has positive real part. On every compact subset of $\kappa \geq 0$, $\delta > -17/10$, $\delta \neq 0$,

$$H_{d,k,2d+b} = \mathcal{A}_h (\cos \Theta + O(d^{-1})) \quad (23.15)$$

uniformly as $d \rightarrow \infty$. The error is additive, including at leading cosine zeros.

Proof. On the lower horizontal line, $dt = dx$. Its local Gaussian contribution, before multiplication by $M!/d!$, is

$$e^{dL(t_s)} \frac{1}{4 \sinh^2(t_s/2)} \sqrt{\frac{2\pi}{dW}} (1 + O(d^{-1})).$$

The counterclockwise rectangle traverses the lower line from left to right and the upper line from right to left. The upper integral is the conjugate of the lower one. Division by $2\pi i$ therefore extracts the lower imaginary part, giving precisely the factor $-i$ in (23.13).

The rate follows by expanding the local analytic exponent through its cubic term and the prefactor through its linear term on a symmetric interval. Odd Gaussian terms vanish. The remaining normalized moments of x^2 , dx^4 , and d^2x^6 are $O(d^{-1})$. The other arcs are exponentially small by Lemma 23.3. Summing the conjugate contributions yields (23.15); no phase is divided out. $\square$

At the regular boundary $\delta = 0$, $t_s = -iT$ and $\sqrt{t_s^2 L''} = it_s \sqrt{-L''}$. Thus (23.13) agrees with the circular prefactor $t_s/(4 \sinh^2(t_s/2) \sqrt{t_s^2 L''})$, including its positive-imaginary base value. This checks the differential and orientation factors.

23.4 From the phase law to density in the wide cone

The action and nonzero prefactor are analytic on each of the two connected regions

$$\mathcal{W}_+ = \{\kappa > 0, \delta > 0\}, \quad \mathcal{W}_- = \{\kappa > 0, -17/10 < \delta < 0\}.$$

The critical curve is unique and simple, and the logarithms can be chosen continuously in the lower strip. Consider the directional phase curvature

$$C(\kappa, \delta) = (\partial_\kappa - 2\partial_\delta)^2 \operatorname{Im} L(t_s). \quad (23.16)$$

It is not identically zero in either region. To check this directly, let $K(T)$ and $F(T)$ be (4.7) and (4.8).

At $T = 14/5$, alternating Taylor bounds give $K(T) > 0$ and $F(T) > 0$, so $t_s = -iT$ is a nondegenerate base saddle with $L''(t_s) < 0$. Its continuation enters the favored half-plane on each side of $\delta = 0$, since $\operatorname{Re}(\partial_\delta t_s) = 1/(2L''(t_s)) < 0$. Since the action is affine in κ, δ and

$$L_{t\kappa} - 2L_{t\delta} = 1 + \coth t,$$

stationary differentiation gives

$$(\partial_\kappa - 2\partial_\delta)^2 L(t_s) = -\frac{(1 + \coth t_s)^2}{L''(t_s)}.$$

At $t_s = -iT$ its imaginary part is $2T^2 \cos T/F(T) < 0$. The analytic continuations from positive and negative δ have this same nonzero limit.

[Merge note, batch 72A.] Manuscript 40's formula (22.15) gives this value on the whole regular branch, and it is negative there. The integer-point estimate used next is Lemma 21.2. Manuscript 37 states and proves the same lemma, and it is printed once.

On a compact parameter rectangle where $|C|$ is bounded away from zero, interpolate at fixed N, d

$$f_{d,N}(X) = \frac{1}{\pi} \left[d \operatorname{Im} L \left(t_s \left(\frac{X}{d}, \frac{N-1-4d-2X}{d} \right) \right) + \arg P_h \right] - \frac{1}{2}.$$

Its second derivative is bounded between fixed nonzero multiples of $1/d$ in one sign; the prefactor argument contributes only $O(d^{-2})$. If $H = 0$, the additive cosine law forces $\text{dist}(f_{d,N}(k), \mathbb{Z}) = O(d^{-1})$. Lemma 21.2 then gives $O(d^{2/3})$ zeros per row, hence $O(N^{5/3})$ over any such fixed patch.

To remove the curvature restriction on a compact patch, use real analyticity. The zero set of the nonzero analytic function C has area zero, by the elementary argument in the proof of Theorem 22.4, which manuscript 37 repeats.

The exact ratio map is

$$x = \frac{d}{N-1} = \frac{1}{4+2\kappa+\delta}, \quad y = \frac{k}{N-1} = \frac{\kappa}{4+2\kappa+\delta}, \quad \left| \frac{\partial(x,y)}{\partial(\kappa,\delta)} \right| = (4+2\kappa+\delta)^{-3}. \quad (23.17)$$

For a fixed finite union Ω of compact rectangles inside $\mathcal{W}_+ \cup \mathcal{W}_-$, the image has piecewise smooth boundary. Its candidate count is

$$(N-1)^2 I_\Omega + O_\Omega(N), \quad I_\Omega = \int_\Omega (4+2\kappa+\delta)^{-3} d\kappa d\delta. \quad (23.18)$$

Cover the compact image of $\{C = 0\}$ by finitely many open boxes of total area less than ϵ . They contain at most $\epsilon(N-1)^2 + O_\epsilon(N)$ lattice candidates. The remaining compact set has a finite cover by curvature rectangles, on which the zero count is $O_\epsilon(N^{5/3})$. Taking $N \rightarrow \infty$ first and then $\epsilon \downarrow 0$ proves $o(N^2)$ exact cancellations in Ω .

Finally exhaust $\mathcal{W}_+ \cup \mathcal{W}_-$ by finite unions of compact rectangles, for example

$$\kappa \in [1/m, m], \quad \delta \in [-17/10 + 1/m, -1/m] \cup [1/m, m].$$

Their integrals increase to

$$\int_0^\infty \int_{-17/10}^\infty (4+2\kappa+\delta)^{-3} d\delta d\kappa = \frac{5}{46}. \quad (23.19)$$

There is no uniform analytic error claim over this noncompact set. For each fixed exhaustion set first take the limit in N , and only then enlarge the set.

For completeness this also gives the global $o(N^2)$ cancellation count in the stated cone. The full cone $10q > 3d$ is, in (x, y) coordinates, the triangle

$$x \geq 0, \quad y \geq 0, \quad (23/10)x + 2y < 1,$$

with $5/46$ area and $O(N)$ lattice boundary error. Outside any fixed exhaustion set, the candidate count divided by N^2 has limsup at most $5/46 - I_\Omega$. Letting Ω exhaust proves that the whole cone has only $o(N^2)$ exact zeros. This is a qualitative global density statement; the $O(N^{5/3})$ estimate was used only on fixed curvature patches. The line $\delta = 0$ and coordinate boundaries have zero area and are harmless in this argument.

23.5 Angular selection below radius three

We need a second contour argument for a disjoint region near $q = 0$. It gives a positive real saddle rather than an oscillating conjugate pair. Throughout this section, L is the leading action in (22.1).

Lemma 23.5 (A positive real saddle is globally selected). *Suppose $\kappa \geq 0$, $-2 \leq \delta < 0$, and $T \in (0, 3]$ satisfies*

$$L'(T) = 0, \quad V = T^2 L''(T) > 0.$$

Then $t = T$ is the unique global maximum of $\operatorname{Re} L(t)$ on $|t| = T$. On a compact family of such saddles, the resulting one-arc asymptotic is uniform:

$$H_{d,k,q} = \frac{M!}{d!} e^{dL(T)} \frac{T}{4 \sinh^2(T/2) \sqrt{2\pi dV}} (1 + O(d^{-1})). \quad (23.20)$$

In particular the coefficient is strictly positive for sufficiently large d , uniformly on the compact family.

Proof. Put $c = (T/\pi)^2$. Since $T \leq 3$ and $\pi > 157/50$, we have $0 < c < 23/25$. The hyperbolic products give the factors

$$a(x) = \frac{4cx^2}{(x^2 - c)^2}, \quad A_n = a(2n - 1), \quad B_n = a(2n).$$

For $0 \leq z \leq 1$ define their transformed version

$$a_z(x) = \frac{a(x)}{1 + za(x)} = \frac{4cx^2}{x^4 + (4z - 2)cx^2 + c^2}.$$

It is strictly decreasing for $x \geq 1$: its logarithmic first derivative has numerator $2(c^2 - x^4)$. We claim that the ratios $a_z(2n - 1)/a_z(2n)$ are nonincreasing in n .

First, writing $D_z(x) = x^4 + (4z - 2)cx^2 + c^2$, differentiation gives

$$(\log a_z)''(x) = \frac{2}{x^2 D_z(x)^2} \left\{ x^8 + (2 - 4z)cx^6 - 8c^2x^4 + (6 - 12z)c^3x^2 - c^4 \right\}.$$

For $x \geq 3$, the numerator in braces is bounded below by

$$x^8 - 2x^6 - 8x^4 - 6x^2 - 1 \geq x^8 \left(1 - \frac{2}{9} - \frac{8}{81} - \frac{6}{729} - \frac{1}{6561} \right) > 0.$$

Thus logarithmic convexity proves the claim for the successive ratios starting with $n = 2$. The remaining first comparison is exact algebra:

$$\begin{aligned} a_z(1)a_z(4) - a_z(2)a_z(3) &= \frac{-64c^2Q(c, z)}{D_z(1)D_z(2)D_z(3)D_z(4)}, \\ Q(c, z) &= 5c^4 + (404z - 202)c^3 + 1925c^2 + (2304z - 1152)c - 2880. \end{aligned}$$

The polynomial Q increases with z . Moreover

$$Q(c, 1) = 5c^4 + 202c^3 + 1925c^2 + 1152c - 2880$$

is increasing for $c > 0$, and its value at $23/25$ is $-2340864/78125 < 0$. All denominators are positive. This proves the claimed transformed ratio ordering.

Set

$$S_A(z) = \sum_{n \geq 1} \frac{A_n}{1 + A_n z}, \quad S_B(z) = \sum_{n \geq 1} \frac{B_n}{1 + B_n z}, \quad \varrho(z) = \frac{S_A(z)}{S_B(z)}.$$

Normalize the two summand sequences to probabilities p_n, q_n . Their likelihood ratio is nonincreasing in n . Since $a_z(2n - 1)$ decreases, the elementary covariance comparison yields

$$\mathbb{E}_p a_z(2n - 1) \geq \mathbb{E}_q a_z(2n - 1) > \mathbb{E}_q a_z(2n).$$

Termwise differentiation is justified by absolute convergence, and therefore

$$\frac{\varrho'(z)}{\varrho(z)} = -\mathbb{E}_p a_z(2n - 1) + \mathbb{E}_q a_z(2n) < 0. \quad (23.21)$$

A further elementary bound will control the tilt. The function $h_B(z) = \sqrt{z} S_B(z)$ is strictly increasing for $0 < z \leq 1$, because

$$2\sqrt{z} h'_B(z) = \sum_{n \geq 1} \frac{B_n(1 - B_n z)}{(1 + B_n z)^2} > 0.$$

Here is an explicit verification of the strict sign. For $n = 1$, $B_1/c \leq 16/9$ and $B_1 z \leq 16/9$, so its summand is at least $-112c/625$. For $n \geq 2$,

$$\frac{c}{n^2} \leq B_n \leq \frac{64}{225}, \quad \frac{1 - B_n z}{(1 + B_n z)^2} \geq \frac{36225}{83521}.$$

Using $\sum_{n \geq 2} n^{-2} > 1/2$, already true of the terms $2 \leq n \leq 7$, we obtain

$$2\sqrt{z} h'_B(z) \geq c \left(\frac{36225}{167042} - \frac{112}{625} \right) > 0. \quad (23.22)$$

Write $\nu = -\delta > 0$ and $r = 4 + \delta > 0$. On a quadrant in the positive half-circle, use $z = (\operatorname{Re} t/T)^2$. Up to a constant, the real action is

$$G(z) = \frac{\kappa}{2} \sum \log(1 + A_n z) - \frac{\kappa + r}{2} \sum \log(1 + B_n z) + \frac{\nu T}{2} \sqrt{z}.$$

Its derivative has the sign of

$$\kappa \varrho(z) - (\kappa + r) + \frac{\nu T}{2h_B(z)}, \quad (23.23)$$

which is strictly decreasing by (23.21)–(23.22). At the real saddle, the expansion $z = \cos^2 \theta = 1 - \theta^2 + O(\theta^4)$ gives $G'(1) = V/2 > 0$. Thus (23.23) is positive on $(0, 1]$, and G is strictly increasing. Every point in the negative half-circle has smaller real action than its reflected positive-half point, by $\nu |\operatorname{Re} t|$. The imaginary points are also smaller than T . Hence T is the unique global maximum.

The circle has radius less than π , so the exact contour can be enlarged to it without crossing another pole. On compact saddle families, strict global dominance and $V > 0$ give uniform far-arc suppression and local Gaussian decay. In $t = T e^{i\theta}$, the Cauchy differential supplies T , and the analytic factor is $1/(4 \sinh^2(T/2))$. The single Gaussian integral gives (23.20); there is no conjugate-pair factor of two. Odd local terms integrate to zero and the next normalized moments are $O(d^{-1})$. Every leading factor in (23.20) is positive. $\square$

23.6 An explicit positive coefficient cone

Theorem 23.6 (Uniform eventual positivity). *For every sufficiently large integer d , uniformly over all integers $k, q \geq 0$ satisfying*

$$40k + 420q \leq 91d, \quad (23.24)$$

one has $H_{d,k,q} > 0$. The threshold is uniform but is not claimed to be numerical.

Proof. Put $\eta = q/d = \delta + 2$. The scaled parameter triangle is

$$\kappa \geq 0, \quad \eta \geq 0, \quad 40\kappa + 420\eta \leq 91.$$

In particular $0 \leq \eta \leq 13/60$, so $\delta < 0$. On the positive real axis, the critical equation is $F_\delta(T) = \kappa$, with

$$F_\delta(T) = \frac{\sinh T}{T} - e^{-T} - 1 - \frac{\eta}{2}(1 + e^T) = \frac{1 + e^T}{2}(g(T) - \eta),$$

$$g(T) = \frac{1 - e^{-T}}{T} - 2e^{-T}.$$

For $0 < T \leq 3$,

$$g'(T) = \frac{e^{-T}(2T^2 + T + 1) - 1}{T^2} > 0.$$

Indeed, the positive Taylor series makes $(e^T - 1 - T)/T^2$ increasing, and at $T = 3$ its value is $(e^3 - 4)/9 < 17/9 < 2$, using $e^3 < 21$.

At $T = 1$,

$$F_\delta(1) \leq \frac{(e - 3)(e + 1)}{2e} < 0.$$

At $T = 3$, put $E_3 = e^3 > 20$. Since $\eta \leq 13/60 < 1/3$,

$$\begin{aligned} F_\delta(3) &= E_3 \left(\frac{1}{6} - \frac{\eta}{2} \right) - \frac{7}{6E_3} - 1 - \frac{\eta}{2} \\ &> \frac{91}{40} - \frac{21}{2}\eta \geq \kappa. \end{aligned}$$

Once $F_\delta \geq 0$, its derivative is strictly positive by the factored form and $g' > 0$. There is consequently exactly one saddle $T \in (1, 3)$ at every parameter point of the triangle. At this saddle,

$$V = T^2 L''(T) = \frac{T^2 F'_\delta(T)}{\sinh T} > 0.$$

The strict endpoint inequalities, compactness of the triangle, and the implicit function theorem make the radius bounds and positive curvature uniform, including $\kappa = 0$ and $\eta = 0$. Apply Lemma 23.5 to obtain the uniform positivity. $\square$

Corollary 23.7 (Additional positive terms). *At derivative order N , the cone (23.24) supplies*

$$\frac{169}{19912}(N - 1)^2 + O(N) \tag{23.25}$$

nonzero monomials outside the proved positive bulk. These cells are also disjoint from the oscillatory region $q/d > 3/10$.

Proof. Set $x = d/(N - 1)$ and $y = k/(N - 1)$. The exact bridge $N - 1 = 2k + 2d + q$ turns the conditions into a fixed triangle with vertices

$$\left(\frac{1}{2}, 0 \right), \quad \left(\frac{420}{931}, 0 \right), \quad \left(\frac{20}{131}, \frac{91}{262} \right).$$

Its area is

$$\frac{1}{2} \left(\frac{1}{2} - \frac{420}{931} \right) \frac{91}{262} = \frac{169}{19912}.$$

The grid-point count has this area times $(N - 1)^2$, with boundary error $O(N)$. Every depth is comparable to N , so Theorem 23.6 applies to all these cells once N is large. Excluding the edge $q = 0$ removes only $O(N)$ already-bulk cells. For the others, $j = -q - 1 \leq -2$, and the factor $\binom{N}{k}$ in the coefficient bridge is positive. At fixed N , the coordinates (j, k) determine d , so distinct cells give distinct monomials. Finally $q/d \leq 13/60 < 3/10$, proving disjointness from the oscillatory region. $\square$

23.7 Combining the regions and the remaining question

The positive-real triangle has $\delta \leq -2 + 91/420 < -17/10$, so it is disjoint from the wide oscillatory cone. Its $q = 0$ boundary belongs to the earlier bulk and contributes only $O(N)$ cells. All other counted cells have $q \geq 1$, hence $j \leq -2$; injectivity of (2.4) prevents double counting.

The wide cone contributes $(5/46)N^2 + o(N^2)$ nonzero terms, and the positive-real triangle contributes $(169/19912)N^2 + O(N)$ for all sufficiently large N . Adding (2.5) proves (23.1). The exact excess over $49/100$ is

$$\frac{28176}{57247} - \frac{49}{100} = \frac{12497}{5724700} > 0.$$

This strict margin gives the eventual pointwise inequality in Theorem 23.1.

The proof does not establish exact noncancellation in the oscillatory cone, determine every coefficient in the intervening region, or prove $a(N) \sim N^2/2$. [Added 1 October 2026, batch 72A: the leading asymptotic $a(N) = N^2/2 + o(N^2)$ was proved the same day by manuscript 34, written after this manuscript, and is printed in the sibling report *power-tower-exponent-supports* [2]. That theorem is ineffective, so the explicit bounds of the source and of its batch-72A count-bounds addition remain the best effective ones (Section 25). The full support conjecture remains open.] The full tail candidate area is $1/8$; the two regions proved here have total area $5/46 + 169/19912$. Figure 1 displays the remaining gap.

The result and its complete proofs received independent mathematical review. The accompanying rational verifier checks all elementary Taylor, angular-selection, and area certificates; finite numerical experiments are not used to prove the saddle or density assertions. The arguments have not been formalized in Lean or externally refereed, and no exhaustive priority claim is made.

24 A pointwise three-halves correction (manuscript 53)

This section derives manuscript 53's count from its growing-offset phase law, Theorem 12.2 and Corollary 12.3 (Section 12).

Theorem 24.1 (Pointwise three-halves correction; superseded, see Section 25). *There are constants $C > 0$ and N_0 such that, for every integer $N \geq N_0$,*

$$a(N) \geq \beta(N) + CN^{3/2} \geq \frac{3}{8}N^2 + CN^{3/2}. \quad (24.1)$$

This is a pointwise bound, not a summatory or average statement. Neither a numerical value of C nor a numerical threshold N_0 is claimed. The leading quadratic coefficient remains $3/8$; no conjectural $N^2/2$ asymptotic is inferred. [Added 1 October 2026, batch 72A: the leading coefficient of the pointwise lower bound was already above $3/8$ by manuscript 49 (Theorem 20.8), and $a(N) = N^2/2 + o(N^2)$ is manuscript 34's theorem [2]; see Section 25.]

24.1 Counting a positive fraction of phase-safe depths

We now use the proved growing-offset phase estimate to count terms at each derivative order. Fix

$$I = [\kappa_-, \kappa_+] = [1/100, 1/50] \subset (0, \kappa_c).$$

The phase-safe consequence 12.3 supplies constants $\eta_* > 0$ and d_* such that, for $d \geq d_*$, $k/d \in I$, $1 \leq b \leq \eta_*\sqrt{d}$, and $\varepsilon = 0$,

$$|\cos(b\psi(k/d))| \geq \frac{1}{2} \implies H_{d,k,2d+b} \neq 0. \quad (24.2)$$

Indeed choose η_* so that the normalized error is at most $1/4$ for large d . Shrink η_* further if necessary so that $\eta_* \leq 1$.

Define the positive constants

$$\Delta\psi = \psi(\kappa_-) - \psi(\kappa_+) > 0, \quad G = \max_{\kappa \in I} \frac{(\kappa + 2)^2}{2K'(t(\kappa))} < \infty. \quad (24.3)$$

Their positivity and finiteness follow from the strictly positive derivative of the regular branch on I .

Lemma 24.2 (Phase-safe depths with the required parity). *Fix a derivative order N and a positive integer b with $b \equiv N - 1 \pmod{2}$. Put $n_b = (N - 1 - b)/2$. If*

$$b \geq B_0 := \left\lceil \frac{6\pi}{\Delta\psi} \right\rceil,$$

then the number of integer depths d satisfying

$$\frac{k}{d} \in I, \quad k = n_b - 2d, \quad \varepsilon = 0, \quad |\cos(b\psi(k/d))| \geq \frac{1}{2}$$

is at least

$$\frac{n_b \Delta\psi}{6G} - 2 - \frac{2b \Delta\psi}{\pi}, \quad (24.4)$$

provided $n_b > 0$.

Proof. The real depths with $\kappa(d) = k/d = n_b/d - 2 \in I$ form the interval

$$J_{N,b} = \left[\frac{n_b}{\kappa_+ + 2}, \frac{n_b}{\kappa_- + 2} \right].$$

On it, $\theta_b(d) = b\psi(\kappa(d))$ increases strictly through an angle interval of length $b\Delta\psi$. Differentiating $\psi(\kappa) = (\pi - t(\kappa))/2$ gives

$$\theta'_b(d) = \frac{b(\kappa(d) + 2)^2}{2n_b K'(t(\kappa(d)))} \leq \frac{bG}{n_b}.$$

In an arbitrary real interval of length L , the set where $|\cos \theta| \geq 1/2$ has measure at least $(2/3)L - 2\pi$. For $b \geq B_0$, this is at least $b\Delta\psi/3$. Its preimage in $J_{N,b}$ consequently has total length at least $n_b \Delta\psi / (3G)$.

The safe phase set has at most $2 + 2b\Delta\psi/\pi$ connected components; monotonicity gives the same bound for its preimage. Meanwhile

$$d - k - 1 = 3d - n_b - 1,$$

so the exact parity requirement $\varepsilon = 0$ is

$$d \equiv n_b + 1 \pmod{2}. \quad (24.5)$$

Each real interval of length ℓ contains at least $\ell/2 - 1$ integers in this prescribed parity class. Summing over the components proves (24.4). This is interval counting and requires no equidistribution assumption. $\square$

Theorem 24.3 (A pointwise correction of order $N^{3/2}$). *There exist constants $C > 0$ and N_0 such that, for every integer $N \geq N_0$,*

$$a(N) \geq \beta(N) + CN^{3/2} \geq \frac{3}{8}N^2 + CN^{3/2}, \quad (24.6)$$

where

$$\beta(N) = \begin{cases} (3N^2 + 10N + 8)/8, & N \text{ even}, \\ (3N^2 + 12N + 1)/8, & N \text{ odd} \end{cases}$$

is the established positive-bulk count. In terms of (24.3) and the analytic choice of η_* , any positive C smaller than

$$\frac{\eta_* \Delta \psi}{144\sqrt{6}G} \quad (24.7)$$

may be used. Neither a numerical value of η_* nor a numerical threshold N_0 is asserted.

Proof. Put $R_N = \eta_* \sqrt{N/6}$ and $B_N = \lfloor R_N \rfloor$. Keep the offsets

$$B_0 \leq b \leq B_N, \quad b \equiv N - 1 \pmod{2}. \quad (24.8)$$

For all sufficiently large N , one has $b \leq \sqrt{N}$, $N - 1 - b \geq 3N/4$, and hence $n_b \geq 3N/8 \geq N/3$. Every real depth in $J_{N,b}$ then satisfies

$$d \geq \frac{3N/8}{\kappa_+ + 2} > \frac{N}{6},$$

because $\kappa_+ + 2 < 9/4$. In particular

$$b \leq \eta_* \sqrt{N/6} \leq \eta_* \sqrt{d},$$

and $d \geq d_*$ once N is large enough. Thus every phase-safe integer depth supplied by Lemma 24.2 meets the analytic hypotheses of (24.2).

The error term in (24.4) is $O(b + 1) = O(\sqrt{N})$, uniformly in the selected offsets. It follows that, for all sufficiently large N , each such b supplies at least

$$\frac{N \Delta \psi}{36G}$$

nonzero cells. Once $R_N \geq 2B_0 + 6$, the number of offsets in (24.8) is at least $R_N/4$. Their product gives the following lower bound on nonzero cells:

$$\frac{\eta_* \Delta \psi}{144\sqrt{6}G} N^{3/2}.$$

It remains to check that these are distinct new derivative terms. Their coordinates are

$$q = 2d + b \geq 1, \quad k = \frac{N - 1 - b - 4d}{2} \geq 0, \quad j = -2d - b - 1, \quad N = 2k + 4d + b + 1. \quad (24.9)$$

There are no reflection holes because $b > 0$. If two pairs (b, d) gave the same (j, k) , equality of j would give $\Delta b + 2\Delta d = 0$, whereas equality of k would give $\Delta b + 4\Delta d = 0$. Both differences must vanish. The resulting monomials are therefore distinct.

The source bulk consists of the nonnegative x -power columns and the positive $j = -1$ boundary. Our cells have $j \leq -2$ and $d \geq 1$, so they are outside that bulk. Equivalently, they occupy the disjoint tail region $d \geq 1$, $q \geq 1$. Adding the source bulk count proves (24.6). $\square$

The new cells have depths $d \geq N/6$. Consequently they are also disjoint, for large N , from any separately certified shallow-depth region with depth $O(\log N)$. The proof therefore does not replace or double-count such earlier contributions. The leading lower-bound coefficient remains $3/8$ [Added 1 October 2026, batch 72A: stale against manuscripts 49 and 34, see the previous note; the shallow-depth region with depth $O(\log N)$ alluded to is that of the source's Section 3.23 and of its batch-72A count-bounds addition (manuscript 68), which 53 does not cite]; the new unconditional pointwise improvement is the positive correction of order $N^{3/2}$.

24.2 Scope, verification, and the remaining problem

The new contribution is a pointwise lower bound at every sufficiently large derivative order. It is stronger than a pointwise correction of order $N \log N$, while remaining compatible with separate stronger summatory lower bounds. [Added 1 October 2026, batch 72A: the order- $N \log N$ correction and the summatory bounds are manuscript 68's, now an addition to the source report; manuscript 53 does not cite it.] The constant C is positive by the proof, but the saddle threshold and an admissible numerical η are not evaluated here.

The argument proves only selected coefficients nonzero. It does not prove the full support conjecture, does not identify the conjectured $N^2/2$ leading asymptotic, and does not claim exact zeros from phase crossings. The growing-offset estimate intentionally uses a uniform error of order $|b|/\sqrt{d} + b^2/d$; no fixed- b $O(1/d)$ remainder is asserted in this report. [Added 1 October 2026, batch 72A: manuscript 45's Theorem 11.3 supplies that remainder at fixed b (Section 11).]

The coefficient model and bulk region are credited to the inspected source. The regular saddle geometry is verified through the displayed identities, with its rational subinterval checked by exact fractions. All other arguments are analytic or elementary interval counting. The proof received independent mathematical review, including the full offset dependence and parity count. It has not been formalized in Lean or externally refereed, and no exhaustive priority claim is made.

25 The headline lower bounds and what supersedes them

Each of the five manuscripts of this part ends in a pointwise lower bound for $a(N)$. Each bound holds for every $N \geq N_0$ with an N_0 that is not numerical, and each manuscript says so. Table 2 lists them.

Table 2: Headline lower bounds of Part III, all ineffective and all implied by $a(N) = N^2/2 + o(N^2)$ (manuscript 34, [2]). The right column lists what each manuscript proves that a count asymptotic does not imply.

MS	Headline bound	Kept here, not implied by the count asymptotic
37	Theorem 23.1(3): $\liminf a(N)/N^2 \geq 28176/57247 = 0.49218\dots$, so $a(N) \geq 0.49N^2$ eventually	the horizontal-contour cosine law on the wide cone (Theorem 23.4) with its rational height certificate; uniform eventual positivity of <i>every</i> coefficient on the triangle $40k + 420q \leq 91d$ (Theorem 23.6), by real-saddle angular selection
49	Theorems 20.1 and 20.8: $a(N) \geq (3/8 + c)N^2$; in fact $a(N) \geq \beta(N) + C_{\mathcal{R}}(N-1)^2 - O(N)$	the global inner-saddle bound, the two-saddle additive cosine law with explicit amplitude and phase (Theorem 20.5), and the separated adjacent phases (Theorem 20.6)
45	Corollary 21.5: $a(N) \geq \beta(N) + C_{\text{patch}}(N-1)^2 - O(N^{5/3})$, $C_{\text{patch}} = 4C_{\mathcal{R}}$	the fixed-offset $O(1/d)$ rate and bounded strips (Section 11); the explicit negative row curvature (21.6); $O(d^{2/3})$ zeros per row and $O(N^{5/3})$ per patch
40	Corollary 22.3: $a(N) \geq (3/8 + C_0)N^2 - O(N^{5/3})$; and $\liminf a(N)/N^2 \geq 3/8 + \int_{\Omega} (4+2\kappa+\delta)^{-3}$ for every admissible patch	the cosine law through $\delta = 0$ and $\kappa = 0$ (Theorem 22.1); the curvature $2T^2 \cos T/F(T) < 0$ on the whole branch (22.15); the odd-even product comparison (Lemma 22.2); the density theorem for analytic exceptional sets (Theorem 22.4)
53	Theorem 24.1: $a(N) \geq \beta(N) + CN^{3/2}$	the phase law uniform in $ b \leq \sqrt{d}$ with explicit error (Theorem 12.2)

The superseding relation. Manuscript 34 proves $a(N)/N^2 \rightarrow 1/2$; more precisely, the outside-bulk tail has only $o(N^2)$ exact zeros. Its proof is the $a = 2$ case of manuscript 28’s theorem for x^{x^a} , and both are printed in the sibling report *power-tower-exponent-supports* [2]. Every bound in Table 2 has leading constant at most $28176/57247 < 1/2$, so each holds for all sufficiently large N as a consequence. Manuscript 34 asserts no numerical error rate or order threshold, so no effectivity is lost; the headline bounds were never effective either. Manuscript 34 does not imply the statements in the right column. Those are statements about individual coefficients: cosine laws, separated phases, positivity, bounded strips and curvature formulas. The density-one statements on fixed patches (45, 40, 37) overlap manuscript 34’s own compact-patch argument, which uses the same integer-triangle lemma; manuscript 34 proves those patch statements away from the line $\delta = 0$ only, which its argument discards as a set of zero area, while 40’s strip contains that line. All five manuscripts were written before 34, and none cites it.

Effective bounds. The best effective lower bounds are not in this report. The source proves, for every $N \geq 0$, the bulk count plus the classified cells of depths one to four (its Corollary `xxb:dc:cor:lower`). The batch-72A count-bounds addition to the source (manuscript 68, code shipped there as `code/07-count-bounds-verify_counts.py`) proves

$$a(N) \geq \frac{3}{8}N^2 + \frac{N}{2} \log N - \frac{N}{2} \log \log N - \frac{5}{4}N \quad (N \geq e^{10}),$$

and an exact summatory bound with cubic constant $11/72$. Manuscript 34’s asymptotic implies neither.

Part IV

Arithmetic at unbounded deficit

Part IV is manuscript 64, its Sections 1–4, 7 and 8; its Sections 5 and 6 are the case $\kappa = 0$ in Part I. Unlike Parts I–III, its congruence families and central strips hold at every depth in the stated families, not only eventually.

In this part $r = 2d + q + 1$ and $M = k + r$, as in Section 2. The polynomials $G_j(c, r)$ are (2.2); manuscript 64 wrote their variable as a , renamed c here. Throughout, p is an odd prime and v_p the p -adic valuation. In Theorem 27.2, $r = Ap + s$ and $k = Bp + t$ with residues s, t , so s is not $|b|$ there, and $C = A + B$.

Status and attribution. The results have ordinary mathematical proofs and reproducible exact checks, with independent mathematical review. They are not formally verified in Lean or externally refereed. Consecutive-factor congruences, the classical Meixner–Pollaczek framework, and Bernoulli local limit theory are credited inputs. The contribution is their stated application to these coefficient families, the exact valuations, and the central-strip and phase conclusions. Worldwide priority is not asserted.

26 The scope of the arithmetic results (manuscript 64)

The desired full conjecture says that this integer vanishes precisely when $q = 2d$ and $k + d$ is even. Those symmetry zeros are already proved; the converse is not assumed here. The source report completes depths one through four; an earlier companion completes depths five and six

and supplies exponential fixed-depth cutoffs [Added 1 October 2026, batch 72A: the companion is manuscript 70, now the batch-72A addition on depths five and six to the source report; manuscript 64 does not cite it]. The present families allow the depth itself to grow without bound. They do not classify every cell at any further complete depth.

27 A block congruence for the centered recurrence

The calculations in this section use classical finite-field algebra. The underlying consecutive-factor congruence is the standard mechanism behind Stirling congruences; compare [10]. We give all needed proofs.

Lemma 27.1 (A prime-length recurrence block). *For every odd prime p ,*

$$G_p(c, r) \equiv c^p - c \pmod{p} \quad (27.1)$$

as a polynomial in c, r . Consequently, for $B \geq 0$ and $0 \leq t < p$,

$$G_{Bp+t}(c, r) \equiv (c^p - c)^B G_t(c, r) \pmod{p}. \quad (27.2)$$

Proof. First fix $c, r \in \mathbb{F}_p$. Choose representatives $A_0, B_0 \in \{0, \dots, p-1\}$ satisfying $A_0 - B_0 = c$ and $-(A_0 + B_0) = r + 1$ in the field. This is possible because two is invertible. The polynomial

$$Y_p(z) = (1+z)^{A_0}(1-z)^{B_0}$$

satisfies $(1-z^2)Y'_p = (c+(r+1)z)Y_p$. Its derivatives at zero satisfy (2.2), so $G_p(c, r) = Y_p^{(p)}(0) = 0$ in characteristic p . For fixed r , monicity and degree p in c imply (27.1). Every coefficient has degree at most $(p-1)/2$ in r , by the recurrence; agreement at all $r \in \mathbb{F}_p$ therefore proves the polynomial identity.

At an index Bp , the recurrence multiplier $j(j+r)$ vanishes modulo p . The following block restarts from a scalar multiple of G_0, G_1 , and its coefficients are periodic modulo p . Induction on the blocks gives (27.2). $\square$

Theorem 27.2 (A residual polynomial of bounded degree). *Fix an odd prime p , and write*

$$r = Ap + s, \quad k = Bp + t, \quad 0 \leq s, t < p, \quad C = A + B.$$

Over $\mathbb{F}_p$, put

$$W(u) = \prod_{j=0}^{s-1} (2d - j + u) G_t(2u + 2d - q, s), \quad \deg W \leq 2p - 2.$$

Then

$$H_{d,k,q} \equiv 2^B (-1)^C \sum_{\ell=0}^C (-1)^\ell \binom{C}{\ell} [u^{d-C-\ell(p-1)}] W(u) \pmod{p}. \quad (27.3)$$

Out-of-range coefficients are zero. At most two terms on the right can be nonzero.

Proof. Each complete group of p consecutive factors in (2.3) is $u^p - u$ modulo p . Lemma 27.1, with $c = 2u + 2d - q$, gives $c^p - c = 2(u^p - u)$. Hence the entire polynomial in (2.3) is congruent to

$$2^B (u^p - u)^C W(u) = 2^B (-1)^C u^C (1 - u^{p-1})^C W(u).$$

Extracting u^d proves (27.3).

The sampled degrees of W differ by $p - 1$. Its degree bound permits three sampled degrees only when $s = t = p - 1$, in which case they must be $0, p - 1, 2p - 2$. We show $W(0) = 0$ in that case. If the least residue of $2d$ is at most $p - 2$, the remaining consecutive product contains a zero constant factor. Otherwise $2d \equiv -1$, and $2d - q = 4d + 1 - r \equiv 0$, while $r \equiv -1$. The recurrence gives $G_j(0, -1) = 0$ for every $j \geq 1$. Thus again $W(0) = 0$, removing the third possible term. $\square$

A nonzero residue in (27.3) proves exact nonvanishing. A zero residue proves only divisibility by p and must not be treated as an exact cancellation. In particular, when $C > d$ every coefficient in (27.3) is zero; such a prime supplies no nonvanishing certificate.

28 Prime units and exact valuations at unbounded depth

Theorem 28.1 (A two-parameter prime-unit family). *Let p be an odd prime, $A \geq 1$, $B \geq 0$, and suppose*

$$d = A + B, \quad k = (B + 1)p - 1, \quad q = (p - 2)A - 2B - 1 \geq 0.$$

Let s_0 be the least nonnegative residue of $2d$ modulo p . Then

$$H_{d,k,q} \equiv (-1)^{d+s_0+1} 2^B \not\equiv 0 \pmod{p}. \quad (28.1)$$

For $N = p(A + 2B + 2) - 2$ and $j = 2B - (p - 2)A$, the original derivative coefficient satisfies

$$v_p(c_N(j, k)) = 1 + v_p(d + 1) + v_p \binom{d + B + 1}{B}. \quad (28.2)$$

Proof. Here $r = 2d + q + 1 = Ap$, and the two complete block counts add to $A + B = d$. Theorem 27.2 gives

$$H_{d,k,q} \equiv 2^B (-1)^d G_{p-1}(4d + 1, 0) \pmod{p}.$$

For an explicit last factor, take $A_0 = s_0$ and $B_0 = p - 1 - s_0$. The polynomial $(1 + z)^{A_0}(1 - z)^{B_0}$ has degree $p - 1$, and its derivatives satisfy the recurrence with $c = 4d + 1, r = 0$ modulo p . Its top coefficient and Wilson's theorem give

$$G_{p-1}(4d + 1, 0) \equiv (p - 1)! (-1)^{B_0} = (-1)^{s_0+1} \pmod{p}.$$

This proves (28.1).

The factor H is a p -unit, so (2.4) reduces the valuation to that of $\binom{N}{k}$. Set $L = A + 2B + 2$, $M = B + 1$, $T = d + 1$, so $L = M + T$, $N = pL - 2$, $k = pM - 1$, and $N - k = pT - 1$. The factorial valuation identity $v_p(n!) = \lfloor n/p \rfloor + v_p(\lfloor n/p \rfloor!)$ yields

$$v_p \binom{N}{k} = 1 + v_p \frac{(L - 1)!}{(M - 1)!(T - 1)!} = 1 + v_p \left((d + 1) \binom{d + B + 1}{B} \right),$$

which is (28.2). $\square$

For example, take $p = 3$ and $A = 2B + 3$. Then $q = 2$, and for every $B \geq 0$,

$$c_{12B+13}(-3, 3B + 2) \not\equiv 0, \quad v_3(c_{12B+13}(-3, 3B + 2)) = 1 + v_3 \binom{4B + 4}{B}.$$

A complementary central family is $p = 5$, $A = 4B + 1$, for which

$$H_{5B+1, 5B+4, 10B+2} \equiv (-2)^B \not\equiv 0 \pmod{5}.$$

Here both depth and logarithmic degree grow while $q = 2d$.

At $B = 333$ in the preceding $p = 3$ family, this certifies $c_{4009}(-3, 1001)$ at depth 1002, with exact valuation one. This is a consequence of the all-parameter proof, not of a finite scan through derivative order 4009.

Corollary 28.2 (A logarithm-free prime-block family). *For an odd prime p , $d \geq 1$, and $0 \leq s \leq (2d \bmod p)$, set $q = (p - 2)d + s - 1$. Then*

$$H_{d,0,q} \equiv (-1)^d \prod_{j=0}^{s-1} (2d - j) \not\equiv 0 \pmod{p}.$$

In particular $H_{d,0,d-1}$ is nonzero modulo three for every $d \geq 1$.

Proof. The consecutive product has length $pd + s$, and its complete blocks contribute $(u^p - u)^d$. Extract its lowest degree d and then the constant term of the remaining product. The bound on s ensures that every remaining constant is nonzero modulo p . $\square$

29 Uniform logarithm-free strips near the symmetry line

[Merge note, batch 72A.] In this section manuscript 64 wrote $b = |2d - q|$ and used s as a summation index. Here the unsigned offset is $s = |q - 2d|$, consistent with $s = |b|$ elsewhere, and the index is ℓ .

Put

$$H_{d,0,q} = [u^d]u \prod_{j=1}^{2d} (u + j) \prod_{j=1}^q (u - j), \quad d \geq 1, \quad q \geq 0.$$

These are the $k = 0$ tail coefficients in the established depth coordinates; no depth-zero or structural low-degree cases are included.

Proposition 29.1 (An elementary-symmetric sufficient condition). *Write*

$$h = \left\lfloor \frac{d-1}{2} \right\rfloor, \quad \varepsilon = d - 1 - 2h \in \{0, 1\}, \quad m = \min(2d, q), \quad s = |2d - q|.$$

Suppose $m \geq h$, $s \geq \varepsilon$, and

$$h^2(s - \varepsilon)(s - \varepsilon - 1) \leq (m - h + 1)(m + 1)(\varepsilon + 1)(\varepsilon + 2). \quad (29.1)$$

Then $H_{d,0,q} \neq 0$. More precisely, let $\sigma = 1$ when $q \leq 2d$ and $\sigma = -1$ otherwise. Its sign is $(-1)^{q+h}\sigma^\varepsilon$.

Proof. Let $E_r = e_r(1, 1/2^2, \dots, 1/m^2)$ and $B_\ell = e_\ell(1/(m+1), \dots, 1/(m+s))$, with $e_0 = 1$ and out-of-range elementary symmetric functions zero. Pairing opposite factors gives

$$H_{d,0,q} = (-1)^q (2d)!q! [u^{d-1}] \prod_{j=1}^m (1 - u^2/j^2) \prod_{j=m+1}^{m+s} (1 + \sigma u/j).$$

Thus, after removing the nonzero factor $(-1)^{q+h}\sigma^\varepsilon(2d)!q!$, the coefficient is

$$\sum_{j=0}^{\min(h, \lfloor (s-\varepsilon)/2 \rfloor)} (-1)^j T_j, \quad T_j = E_{h-j} B_{\varepsilon+2j} > 0.$$

If this sum has one term, the claim is immediate. Otherwise we prove that its terms strictly decrease.

For $1 \leq r \leq m$, the identity

$$rE_r = \sum_{|S|=r-1} \left(\prod_{i \in S} i^{-2} \right) \sum_{j \notin S} j^{-2}$$

and the fact that the smallest possible remaining sum is at least $\sum_{j=r}^m j^{-2} > 1/r - 1/(m+1)$ imply

$$\frac{E_{r-1}}{E_r} < \frac{r^2(m+1)}{m-r+1}.$$

Applying the corresponding identity twice to B_ℓ , with every variable at most $1/(m+1)$, gives

$$\frac{B_{\ell+2}}{B_\ell} \leq \frac{(s-\ell)(s-\ell-1)}{(\ell+1)(\ell+2)(m+1)^2} \quad (0 \leq \ell \leq s-2).$$

For adjacent terms, put $r = h - j$ and $\ell = \varepsilon + 2j$. Since $r \leq h$ and $\ell \geq \varepsilon$, these inequalities yield

$$\frac{T_{j+1}}{T_j} < \frac{h^2(s-\varepsilon)(s-\varepsilon-1)}{(m-h+1)(m+1)(\varepsilon+1)(\varepsilon+2)} \leq 1.$$

The alternating sum is strictly positive, by grouping consecutive pairs (and retaining a positive last term if necessary). This proves the nonvanishing and sign claims. $\square$

Theorem 29.2 (Two all-depth central strips). *For every $d \geq 1$ and $q \geq 0$:*

1. *if d is odd and $|q - 2d| \leq 5$, then $H_{d,0,q} \neq 0$;*
2. *if d is even and $1 \leq |q - 2d| \leq 9$, then $H_{d,0,q} \neq 0$.*

At $q = 2d$, the coefficient is zero exactly when d is even.

Proof. For odd $d = 2h + 1 \geq 15$, one has $h \geq 7$, $\varepsilon = 0$, $s \leq 5$, and $m \geq 4h - 3 \geq h$. The worst instance of (29.1) is implied by

$$20h^2 \leq 2(3h-2)(4h-2),$$

whose right side minus left side is $4h^2 - 28h + 8 > 0$ for $h \geq 7$.

For even $d = 2h + 2 \geq 22$, one has $h \geq 10$, $\varepsilon = 1$, $1 \leq s \leq 9$, and $m \geq 4h - 5 \geq h$. It suffices that

$$56h^2 \leq 6(3h-4)(4h-4).$$

The difference is $16h^2 - 168h + 96 > 0$ for $h \geq 10$. Proposition 29.1 proves both ranges.

The remaining odd depths 1, 3, 5, 7, 9, 11, 13 and even depths 2, 4, ..., 20 form 258 cases, including the ten even-depth symmetry cells. Two independent exact-integer evaluations, one by multiplication of shifted factors and one by the signed-Stirling expansion, verify precisely the stated zeros. The largest polynomial degree needed is 90. The supplied script prints the count and rejects any disagreement; these cases also lie within the source's existing order-3000 certificate.

Finally, at $q = 2d$ the polynomial is $u \prod_{j=1}^{2d} (u^2 - j^2)$. Its even-degree coefficients vanish, and all its odd-degree coefficients in the relevant range are nonzero. $\square$

30 Why root geometry is not enough

The source's factorization separates consecutive real roots from purely imaginary centered roots. These geometric facts alone cannot imply the desired coefficient nonvanishing. For an exact example put

$$F_{\sharp}(u) = u \prod_{j=1}^6 (u^2 - j^2), \quad Q_{\sharp}(u) = 5369u^2 + 3600.$$

The first factor has consecutive simple real roots; the second has positive integer coefficients, even parity, and two purely imaginary simple roots. Nevertheless,

$$[u]F_{\sharp} = 518400, \quad [u^3]F_{\sharp} = -773136, \quad [u^3](F_{\sharp}Q_{\sharp}) = 0.$$

This matches the consecutive-factor part of the depth-three central case, but $Q_{\sharp}$ is not its prescribed recurrence polynomial. The actual factor is $G_2(2u, 13) = 4u^2 + 14$, for which the same coefficient is $-8750304 \neq 0$. Thus this example refutes only a proposed argument based on root geometry alone; it is not a counterexample to A290268.

A stronger structural assertion was also tested: replace the evaluation $2d$ by any integer $h' \geq 2d$ in

$$[u^d]((h' + u))_{h'+q+1} G_k(2u + h' - q, h' + q + 1).$$

The candidate statement is that only $q = h'$, $k + d$ even can make this zero. It would imply the A290268 conjecture. An exact diagnostic scan over $1 \leq d \leq 12$, $2d \leq h' \leq 4d$, $1 \leq k \leq 40$, and $0 \leq q \leq 120$ checked 813120 cells and found no other zero. This is an unproved candidate lemma, not a theorem or an assumption used in any result above.

31 Evidence, literature, and remaining questions (manuscript 64)

The proof mechanisms have established antecedents. Poon [11] develops harmonic elementary-symmetric descriptions of falling-factorial derivatives; these identities do not by themselves give the present nonvanishing statements. Consecutive residue blocks and related Stirling congruences are classical; Guillot and Ségalat [10] explicitly acknowledge the earlier work of Cárdenas and Lluís. The centered recurrence is a classical Meixner–Pollaczek specialization [3]. The fixed-offset phase theorem uses the stated local limit theorem of Auld and Neammanee [12]. None of these analogies is substituted for a missing noncancellation theorem.

The arithmetic verification checks the prime-length identity symbolically for five primes, 874 exact instances of the prime-unit and valuation family, and 300 full jet block identities. A separate verifier closes the central-strip proof using two exact coefficient formulas on 258 cells and checks 1791 logarithm-free prime-block instances. The geometric obstruction is an exact polynomial identity. These checks supplement the all-parameter algebra and analysis; they do not create a universal proof by extrapolation.

The central strips hold at every depth in their specified widths. The phase theorem holds eventually for each fixed logarithmic degree and fixed offset; it gives no uniform threshold when either parameter grows with the depth. The modular families cover specified arithmetic loci with unbounded depth and logarithmic degree. The full support conjecture and universal integer nonvanishing away from symmetry remain open. No universal claim is made from the diagnostic scan, root location, or a zero residue modulo a prime.

32 Questions

The manuscripts pose no numbered questions. The following are the open directions their scope statements name, re-scoped against one another and against the sibling report. The source’s Question 6 ($a(N) \sim N^2/2$) is answered in [2] and is not repeated.

Question 1 (The transition to the fold). *Fix b . Describe $H_{d,k,2d+b}$, and prove eventual nonvanishing, in the intermediate range $d^{-2/3} \ll \kappa_c - k/d \ll 1$. This range lies between the compact intervals of Theorem 6.2 and the critical window of Theorem 18.1. The merge recorded this gap; no manuscript poses or claims it.*

Question 2 (Exceptional rays and strips). *Decide the finitely many candidate cells per depth in the bounded strips $|k - \kappa^*d| \leq C_{b,I}$ of Theorem 11.1(1). These lie around the transcendental phase-crossing ratios, the first being $\kappa_9 = K(8\pi/9)$. Manuscripts 58 and 45 make no claim there.*

Question 3 (Beyond the regular branch). *Extend the phase laws past the fold κ_c to later saddle branches, and past the established dominance boundaries of the cosine laws of Part III. Manuscripts 58, 40 and 54 exclude these regions explicitly.*

Question 4 (Effective thresholds). *Make any of the eventual statements of Parts I–III effective, with a numerical depth threshold. Every manuscript disclaims one. An effective fixed-offset threshold would bear on the source’s Question 4, which asks for uniformity in the depth, and its Question 5, which asks for arithmetic separation of real zeros.*

Question 5 (The stronger evaluation lemma). *Prove or refute manuscript 64’s candidate statement of Section 30: for every integer $h' \geq 2d$, $[u^d]((h' + u))_{h'+q+1} G_k(2u + h' - q, h' + q + 1)$ vanishes only when $q = h'$ and $k + d$ is even. It would imply the support conjecture. Its exact diagnostic scan of 813120 cells found no other zero.*

The full support conjecture of the source (`xxb:conj:support`) remains open.

33 Provenance

This report was written in the write phase of batch 72A of ProveIt’s incoming-reports intake. The 79 archives of batch 72 arrived in commit 1512ef835. Cluster A, twenty manuscripts on the power-tower derivative family, was placed in commit d292c6765. The manuscripts are numbered by archive modification time, as in the batch index. That commit opened this report with manuscript 58 as base. It staged 58’s `article.tex`, `spectral_sections.tex` and `README.md` as delivered, and the members’ code, data and figure under the prefixes listed in the README. Each manuscript is dated 1 October 2026 and has the author line “Research note prepared for Vladimir Reshetnikov with OpenAI”. Each pins ProveIt commit ca81647a9f679e96d49ffdbc54b0c0ab13d10a36, the source report’s text at that time.

Table 3: The nine manuscripts. Archive names omit the prefix `ProveIt_A290268_` and the suffix `.zip`; all pin `ca81647a9`.

MS	Archive; title	Printed as	Contributed
58	<code>Explicit_Finite_Ratio</code> ; <i>An Explicit Finite-Ratio Noncancellation Region for OEIS A290268</i>	base: Sections 2–8	spectral limit, regular branch, saddle coefficients, phase law, algebraic ratios, first eight offsets, first phase crossing; the shared certificate
54	<code>Airy_Fold</code> ; <i>The Airy Transition at the A290268 Critical Ratio</i>	Part II	trace lemma, discrete centering, quartic localization, additive Airy law, nonvanishing at the critical degree
64	<code>Arithmetic_Phase</code> ; <i>Arithmetic Nonvanishing and Central Phase Laws for OEIS A290268</i>	Sections 9, 10; Part IV	block congruences, prime-unit families and valuations, central strips, root-geometry example; the case $\kappa = 0$ by a second route
45	<code>Sparse_Cancellations</code> ; <i>Sparse Cancellations in the A290268 Tail</i>	Sections 11, 21	$O(1/d)$ rate, bounded strips, row curvature, lattice lemma, patch density
53	<code>Pointwise_Three_Halves</code> ; <i>A Pointwise Three-Halves Correction for OEIS A290268</i>	Sections 12, 24	phase law for $ b \leq \sqrt{d}$, count $\beta(N) + CN^{3/2}$
49	<code>Strict_Quadratic_Gain</code> ; <i>A Strict Pointwise Quadratic Gain for OEIS A290268</i>	Section 20	inner and outer saddles, two-saddle cosine law, separated phases, count $(3/8 + c)N^2$
40	<code>Direct_Contour</code> ; <i>A Direct Contour Proof for the A290268 Regular Region</i>	Section 22	single-contour representation, odd–even comparison, cosine law through $\delta = 0$, curvature formula, analytic exceptional sets
37	<code>Global_049_Bound</code> ; <i>A Global 0.49 Lower Bound for OEIS A290268</i>	Section 23	horizontal selection in the wide cone, positivity triangle, $\liminf a(N)/N^2 \geq 28176/57247$, region figure
61	<code>Growing_Logarithmic_Degrees</code> ; <i>Growing Logarithmic Degrees in OEIS A290268</i>	not printed	superseded by 58 (below)

The zip entry times show that the manuscripts were written in the order 64, 61, 58, 54, 53, 49, 45, 40, 37, the reverse of the batch numbering. Manuscript 34 (sibling report) was written after all of them. Read “earlier” and “preceding” in the manuscripts in that order. The members cite each other only partly. Manuscript 45 cites 58 and 49 Theorem 5.1 and shipped their PDFs as inputs. Manuscript 40 cites 49 Theorem 5.1 as a comparison, not a dependency. Manuscript 54 alludes to 58 without citing it, as does 58 to 64, 61 to 64, 64 to manuscript 70 (now in the source report) and 53 to manuscript 68 (likewise). Merge notes mark each of these.

Manuscript 61, superseded. Manuscript 61 proved two theorems at a fixed offset b . The first is a uniform sublinear range: for every $K(d) = o(d)$, $H_{d,k,2d+b} \neq 0$ for all $0 \leq k \leq K(d)$ at large d , apart from the reflection holes, with a phase law. The second is a positive linear range: there is $\eta_b > 0$ with nonvanishing for $0 \leq k \leq \eta_b d$. Its proof used a signed Laurent product with a positive saddle coefficient, root bounds and an absolute-integral comparison for the shift. Its normalization $2^k (-1)^{q+h} \sigma^\varepsilon (2d)! q! C_h m^{-\varepsilon}$ is (6.5), and its constant a with $a \arctan(1/a) = 1/4$ is

a_0 . Both theorems follow from Theorem 6.2 on $I = [0, \kappa_1]$ with κ_1 below the first exceptional ratio. The applicable phase is nonzero and continuous there, by Corollary 7.1 at $\kappa = 0$ and continuity. For $|b| \leq 8$, Corollary 7.2 even gives $\eta_b = 1/36$. Manuscript 61 called itself a continuation of 64 and was written before 58; it cites neither. It disclaimed numerical values of η_b and D_b , offsets growing with d , and a phase limit in the linear band, and it remarked that its amplitude C_h need not be asymptotic to the elementary-symmetric coefficient when k is large. Nothing from 61 is shipped. Its archive `docs/incoming/ProveIt_A290268_Growing_Logarithmic_Degrees.zip` survives in commit 1512ef835.

Where the merge had to choose.

- *Base*. Manuscript 58, the most general fixed-offset theorem and the input of 45 and 54. Its `spectral_sections.tex` is kept as the input file of Sections 2ff.
- *Order*. Parts by regime rather than by manuscript, as described in Section 1.1. Within Part III, 49 precedes 45 because 45 uses its cosine law, and 45 precedes 40 and 37 because the lattice lemma is printed there.
- *Duplicates*. These are listed in Section 1.2. Manuscript 64’s phase theorems became corollaries of 58 with 64’s proofs as second proofs. Manuscript 45’s Proposition 4.1, which restates 49’s cosine law, was not reprinted. Of the closed-walk trace estimate, 54’s full proof and 53’s offset-dependent proof are both printed, since 53 tracks a parameter that 54 holds fixed; 45’s and 49’s sketches are kept with citations.
- *Superseded headlines*. The count bounds of 37, 49, 53, 45 and 40 are kept, with dated notes at each “1/2 remains open” or “3/8 remains” sentence and the table of Section 25; manuscript 34 is cited, not reprinted.
- *Abstracts*. The eight abstracts were replaced by one. Every non-claim in them is also stated in the manuscripts’ bodies, which are kept.
- *Notation*. The renames of Section 1.4.
- *Bibliography*. The items were merged. Manuscripts 37 and 40 cite DLMF §4.36 with different descriptions, and one item covers both. Manuscript 64’s commented-out bibliography entry inside its fixed-offset section is its item `auld-neammanee-2024` and was dropped as a comment. The members’ citations of one another were replaced by cross-references.

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<https://github.com/VladimirReshetnikov/ProveIt/blob/ca81647a9f679e96d49ffdbc54b0c0ab13d10a36/SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-derivative-term-counts/article.tex>. In the repository:
`SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-derivative-term-counts`.
- [2] ProveIt research report *power-tower-exponent-supports* (batch 72A),
`SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-exponent-supports`. It prints manuscript 34's theorem $a(N) = N^2/2 + o(N^2)$ for A290268 as the case $a = 2$ of $A_a(N) = N^2/2 + o(N^2)$ for x^{x^a} .
- [3] NIST Digital Library of Mathematical Functions, *Meixner–Pollaczek generating function*, Eq. 18.23.7.
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- [8] NIST Digital Library of Mathematical Functions, §24.16, equation 24.16.1, higher-order Bernoulli polynomials.
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